Policy Complementarities in Human Capital Formation: Long-Term Evidence from Preschool and School Feeding*

Camila Galindo† Fabio Sánchez‡

October 22, 2026

Abstract

We study whether two large-scale educational policies act as complements in the formation of human capital. We combine the staggered expansion of public preschools in Colombia with the subsequent decentralized rollout of the national school feeding program. The policies are complementary: school feeding reduces dropout and increases grade completion and postsecondary enrollment, with effects 20–30% larger among students previously exposed to preschool. These reinforcing effects are larger when school feeding is introduced shortly after preschool and fade when introduced in secondary. Preschool alone has limited effects, while school feeding alone can partially compensate for the absence of early investments.

JEL Codes: I21, I38, H75, O15

*We are grateful to Diana Pinilla and Alexander Almeida for excellent research assistance. We also thank Rita Ginja, Krzysztof Karbownik, and seminar participants at Universidad del Rosario, Universidad de los Andes, the Experimental Seminar at The University of Chicago, the 5th Human Capital Workshop at the Bank of Italy, LUISS, EAFIT, the RIDGE-Labor Workshop, the CESifo Venice Summer Institute-The Economics of Children in a Changing World, and the NBER Summer Institute 2026 Children and Families for their helpful comments and suggestions, and Jorge Caputo and Sofía Collante-Zárate for their support in accessing and organizing the administrative data.

†Department of Economics, LUISS; email: cgalindo@luiss.it.

‡Department of Economics, Universidad de Los Andes; email: fasanche@uniandes.edu.co.

Early- and later-life interventions have expanded worldwide, with increased preschool enrollment, the scale-up of school feeding programs, and broader access to financial aid, all aimed at promoting human capital accumulation and improving long-term outcomes. As a result, children are often exposed to multiple interventions, potentially at different stages of development (Johnson and Jackson, 2019; Rossin-Slater and Wüst, 2020). Early investments can yield important benefits (Currie, 2001; Currie and Almond, 2011; Hoynes et al., 2016); however, their long-term impacts are often evaluated in isolation, without accounting for their interaction with subsequent later investments impacting the same children. Whether later-life policies can reinforce early-life policies is ex ante ambiguous and likely depends on the timing and nature of investments (Almond et al., 2018).

In this paper, we ask whether two of the most widespread policies targeting vulnerable children—public preschool and school meals—have complementary effects on human capital accumulation. That is, our goal is to determine whether the effect of a subsequent policy—school feeding—on children's educational outcomes is larger in the presence of an early policy—preschool access. Identifying such interactions requires exogenous variation in both programs (Almond and Mazumder, 2013), a key feature of the context we study. We focus on a staggered expansion of free public preschools in Colombia between 2005 and the mid-2010s, using nearly two decades of administrative data that follow the universe of students who entered first-grade between 2006 and 2011. Efforts to expand access to public preschool began in the late 1990s, leading to a 30% increase in supply over the following decade. We combine this variation in preschool exposure with the subsequent roll-out of a second policy: starting in 2012, the national school feeding program underwent a major expansion, largely determined by targeting rules. The school feeding program provides free food rations during the school day, such as breakfast, snacks, or lunch, to improve student retention and health.

The two expansions generate quasi-experimental variation in exposure and timing, resulting in cohorts of students who experienced neither, one, or both interventions at different points in their schooling trajectories. We use this variation to test whether joint exposure yields higher educational returns than exposure to either program alone. Differences in the timing between investments allow us to also test whether earlier exposure to school feeding leads to stronger complementarities with preschool. Moreover, we ask two critical questions. First, can preschool lead to lasting educational gains without subsequent investments? Second, can school meals remediate the absence of early investments? To answer these questions, we estimate the effects of each program alone by comparing students exposed only to preschool

or only to school feeding with those exposed to neither program.

Our central finding is that school feeding reduces dropout and increases secondary completion and post-secondary enrollment, with effects that are 20–30% larger among students previously exposed to preschool. These findings are consistent with complementarity between the two policies. We also estimate larger reinforcing effects when school feeding is introduced no later than seven years after preschool exposure and fade thereafter. While we do not detect average complementarities in performance on the high school exit exam, they emerge and are stronger the earlier school feeding arrives. Complementarities are concentrated among higher-income students, while school feeding alone generates sizable gains particularly for lower-income children. Overall, preschool alone has limited effects without follow-up investments, while the school feeding program alone produces substantial gains (albeit smaller than in the presence of early investments).

Following Hendren and Sprung-Keyser (2020), we value these policies using the marginal value of public funds. Accounting for complementarities raises the welfare gains from the preschool plus school feeding bundle by 22% relative to an additive benchmark that ignores interactions. Timing also matters for welfare: the preschool plus school feeding bundle delivers almost twice the benefits per dollar of net government cost when school feeding arrives in primary school than when it arrives in secondary school.

Preschool. We begin by estimating the long-run effects of each program separately, and then test for complementarities using variation in exposure overlap. First, we use a staggered difference-in-differences design that leverages the expansion of public preschools across municipalities. Although all municipalities had at least one preschool by 2005, when our analysis begins, most students still lived more than 2 kilometers from the nearest one. Thus, we do not study the introduction of preschool, but rather the expansion of access in areas where it was already present. We classify municipalities as treated if a new public preschool opened within 2 kilometers (the median distance at baseline) of the geographic centroid after 2006, while those with a preschool already within that radius in 2005 are classified as always-treated, and the remainder as never-treated.

After the opening of a new preschool nearby, on average, the distance to the nearest preschool falls by 2.5 km and enrollment increases by 7.1 percentage points (pp) (11% relative to the pre-expansion mean of those not exposed). Preschool exposure has sizable and lasting effects on academic outcomes: it increases the probability of reaching 5th, 9th, and 11th grade by 3, 2.1, and 2 pp, respectively; reduces dropout by 2 pp; and raises higher education enrollment by 1.3 pp. We find similar effects across

subgroups (e.g., gender, socioeconomic level, urban/rural). Scaling the intent-to-treat estimates by the first stage enrollment effects implies treatment-on-the-treated (TOT) effects of 25–40 pp, substantially larger than long-term preschool estimates from other developed (Bailey et al., 2021; Gray-Lobe et al., 2023) and middle-income settings (Berlinski et al., 2025). Our implied TOT estimates, however, average over students with and without subsequent exposure to school feeding.

School Feeding. Starting in 2012, nearly half the students in our sample received school meals, with the share of beneficiaries increasing from 20% in our oldest cohort to 80% in our youngest cohort. To identify the separate impact of the feeding program, we leverage variation in the year it was first implemented in each student's first-grade school. We treat the first-grade school as the unit of exposure, since we observe all students in this grade regardless of preschool or school feeding, avoiding contamination from endogenous schooling transitions.

We observe no differential trends in the probability of receiving meals across schools, prior to treatment. Post-treatment, we estimate an increase in the probability of receiving school meals by 14 pp (45%). School feeding reduces dropout by 8.7 pp (14%), increases the probability of reaching grades 5, 9, and 11 by 7.9, 8.3, and 8.1 pp (13%, 24%, and 24%), respectively, and increases higher education enrollment by 4.5 pp (33%). Moreover, it significantly improves academic performance, increasing the probability of scoring in the top of the distribution in both math and reading. Low-income rural children were more likely to benefit from the program, in line with it's targeting rules, while we find no significant differences by gender.

Preschool ∩ School Feeding. We next explore whether preschool can boost the effect of school feeding. Identification relies on both investments being independent. Consistent with this assumption, exposure to the two policies is uncorrelated, and the rollout of either intervention does not predict whether or when the other is introduced. We also show that school meals take-up (i.e., the effect of school feeding exposure on the probability of receiving meals) and that the average observable characteristics of school feeding compliers—students induced to receive meals by the program's rollout—are similar across preschool exposure groups. Hence, we implement a triple difference-in-differences (DiDiD) strategy to identify the medium- and long-term effects of preschool exposure (capturing whether early investments produce lasting gains without reinforcement), school feeding exposure (capturing whether later investments can remediate the absence of earlier ones), and their interaction (capturing complementarities) on educational outcomes.

Preschool and school feeding act as complementary policies for academic progression. The effects of preschool exposure on completion of grade 9 and 11 and higher

education enrollment are 2–2.5 pp larger among students later exposed to school feeding, compared to those without school feeding exposure. However, we do not find evidence of such complementarities for academic performance. Nevertheless, if the second program arrives during sensitive developmental periods or at schooling transitions, it may still reinforce earlier investments. Hence, we further explore whether the size of the complementarity depends on the timing of school feeding exposure relative to preschool exposure.

Indeed, we find that the effect of preschool on grade completion, dropout reduction, and higher education enrollment is larger when school feeding is introduced 2–7 years after preschool (when students are in primary grades), and vanishes when the gap between investments exceeds eight years (when students are in secondary grades). For academic performance, where average interaction effects are not detected, we estimate sizable complementarities when school feeding follows shortly after preschool particularly within two to three years, when students are still in early primary grades. Overall, these findings underscore the importance of aligning later investments with key academic transitions.

Complementarities across all educational outcomes are significantly stronger for students from higher socioeconomic levels. In contrast, we do not observe significant gender differences and, although the interaction effects are somewhat larger for students in urban areas, these estimates are imprecise across the board. Using detailed records on household composition and characteristics at baseline for a subsample, we find that students in higher strata also have more educated mothers who are less likely to report domestic work as their main economic activity, and live in smaller households. We interpret these results as suggestive of reinforcing behaviors at home, if more educated parents with fewer children can invest more in their children. However, we lack more granular measures to confidently attribute our results to parental behaviors.

While we estimate positive effects of preschool alone on completing grade 5, it does not significantly improve academic progression or test performance in the absence of a subsequent investment. If anything, the implied TOTs for academic performance are negative, albeit imprecisely estimated. In contrast, school feeding can partially remediate the absence of preschool. We find large and significant effects on academic progression and test performance from exposure to school feeding alone, including in long-term outcomes such as higher education enrollment.

Related Literature. We contribute to the recent literature providing reduced-form evidence of dynamic complementarities in skill formation between public or private investments. While the theoretical foundations are well established (Cunha and

Heckman, 2007; Heckman and Mosso, 2014), the empirical evidence using experimental (Bjorvatn et al., 2025, 2024) or quasi-experimental variation (Rossin-Slater and Wüst, 2020; Bau et al., 2020; Goff et al., 2023; Bharadwaj et al., 2018; Kinsler, 2016; Jenkins et al., 2024) largely finds no complementarities or even substitution between investments.[1] Most studies focus on high-income countries, while the evidence for middle- and lower-income countries analyzes interactions between inputs such as parental investments, early-life shocks, and the remediation potential of public policies.[2] We provide novel evidence from a middle-income country on complementarities between two large-scale public educational policies, each with distinct oversight and goals. Exploiting variation in their relative timing, we show that shorter intervals between investments amplify complementarities—an approach rarely feasible when investments are delivered contemporaneously, as in most experimental settings.

Within this literature, we add to scant work examining the interaction between early childhood education or environments and subsequent investments. Existing evidence points to limited or negative interactions.[3] Our findings most closely resemble Johnson and Jackson (2019), who find complementarities between Head Start and per-pupil spending in public schools in the U.S. Similar to their case, complementarities in our setting are likely to arise between two education programs that target different parameters of the production of skills: preschool, and early cognitive investment which rarely provided nutritional or health services, and school feeding, a later enabling investment, which directly enhances these inputs.[4]

We also contribute to the literature evaluating the long-term impacts of preschool on educational outcomes. Recent work for the U.S. shows sizable impacts on educational attainment and behavioral outcomes (Bailey et al., 2021; Gray-Lobe et al., 2023). In Latin America, evidence from Argentina (Berlinski et al., 2008, 2009, 2025) and Mexico (Behrman et al., 2024) documents positive medium- and long–term effects on educational attainment, test scores, and noncognitive skills. Similar to these

[1]For exceptions, see Gilraine (2017); Johnson and Jackson (2019); Chan et al. (2024); Bhalotra and Venkataramani (2015); Mbiti et al. (2019); Bütikofer et al. (2024) using experimental or quasi-experimental variation, and Cunha et al. (2010); Attanasio et al. (2020) using dynamic factor models.

[2]For developing countries, the evidence examines interactions between parental investments or family environments (Bharadwaj et al., 2018; Goff et al., 2023), classroom inputs (Carneiro et al., 2022), and natural disasters or rainfall shocks (Bau et al., 2020), and their interactions with cash transfers or health interventions (Adhvaryu et al., 2024; Duque et al., 2023; Gunnsteinsson et al., 2022).

[3]A nurse home visiting program and preschool are substitutes (Rossin-Slater and Wüst, 2020); better early-life environments reduce the payoff to high school quality (Goff et al., 2023); and experimental evidence shows no interaction between free childcare and cash transfers (Bjorvatn et al., 2025, 2024). Novel work on genetic endowments has also analyzed complementarities with investments (Muslimova et al., 2024; Biroli et al., 2025).

[4]Other papers have explored how early programs may interact with other inputs (Chakraborty and Jayaraman, 2019; Bailey et al., 2021). Importantly, these studies do not use exogenous variation in additional/subsequent investments and hence cannot credibly test for interactions.

studies, we examine an expansion relevant for students from diverse socioeconomic backgrounds (rather than targeting the most vulnerable, as programs such as Head Start). Although we find sizable preschool impacts, most of these are explained by interactions with the school feeding program. Moreover, existing studies for Colombia have shown that high-quality early childhood programs, with higher teaching quality, improve child development outcomes in the short-term (Bernal and Ramírez, 2019; Bernal et al., 2019; Andrew et al., 2024).[5] Our contribution lies in showing that, in a context of heterogeneous infrastructure and service quality, preschool contributes little to achievement or test performance during compulsory schooling unless reinforced by later investments.[6]

Although our paper is not the first to evaluate the impacts of school meals in Colombia, we provide novel evidence showing that their effects, while lessened in the absence of preschool, are sizable and significant.[7] In other middle- and low-income countries, the evidence on school meals is mixed. We also find that earlier arrival of school meals, relative to preschool exposure, has the largest complementarities on compulsory schooling and academic performance. These results are consistent with the evidence from high-income countries showing that earlier or longer exposure to school feeding generates stronger gains (Hoynes et al., 2016; Bütikofer et al., 2018; Lundborg et al., 2022).[8] Our paper contributes to this large literature by showing that school meals are more productive if preceded by an earlier investment (preschool, in our case), but nonetheless can on their own improve outcomes for children with initial disadvantages.

[5]Unlike the preschools we study, these articles examine those under the national early childhood strategy ("From Zero to Forever") launched in 2011 for children ages 0–5 from the lowest socioeconomic levels. Under this policy, preschools offered standardized services with pedagogical guidelines. Importantly, children in our sample were above the policy's target age at the time of its implementation in 2011.

[6]The absence of effects from preschool exposure alone is consistent with the fade-out literature, which finds that early cognitive gains often dissipate while noncognitive improvements persist and shape longer-term outcomes (Currie and Thomas, 1995; Deming, 2009; Heckman et al., 2010, 2013; Bailey et al., 2021; Gray-Lobe et al., 2023). We lack measures of socioemotional skills or disciplinary behavior to test this hypothesis in our context.

[7]Collante-Zárate et al. (2024) evaluate the national feeding program, using a different identification strategy for a larger sample, and find similar results to ours.

[8]See, e.g., Maluccio et al. (2009); McEwan (2013) for evidence from middle- and low-income countries. On high-income countries, see Alderman et al. (2024) and Ayllón and Lado (2025) for reviews of the impacts of school meals.

1 Context: Public Education Investments in Colombia

1.1 Preschool

We study the expansion of free public preschools in Colombia between 2005 and 2015. Most of the regulatory framework supporting preschool enrollment predates this period. According to Article 67 of the 1991 Constitution, the final year of preschool education (known as "Transición"; preschool, hereafter) is mandatory for children under the age of six in public institutions. The earlier grades of "Pre-jardín" (age 3) and "Jardín" (age 4) are not required for preschool entry, and attendance in preschool is not a prerequisite for entering primary school (Decree 2247 of 1997). The main goals of preschool education included promoting children's integral development through play-based learning, and although health and nutrition were recommended, they were not mandatory or part of the pedagogical framework. In 2003, the Ministry of Education established guidelines to allocate preschool slots, including prioritizing children enrolled in services managed by the Colombian Institute of Family Welfare (*ICBF*, in Spanish).[9] Despite these regulations, enforcement was weak. As a result, and notwithstanding its mandatory status, the net preschool enrollment rate was around 30% in the early 2000s (Ministerio de Educación Nacional, 2002) and fewer than half of first-grade students in 2006 had attended preschool.

In parallel to policies on enrollment, the legal framework also encouraged the expansion of public preschool providers. Law 115 of 1994 and Decree 2247 of 1997 mandated a gradual rollout of preschool grades across all public institutions, specifically those offering primary education. By 2005, every municipality had at least one preschool, yet many children still lived far from the nearest one. Between 2005 and 2014, the number of preschools grew by 30%, the average distance to the nearest one fell by 2 kilometers, and enrollment increased by more than 20 pp (Appendix Figure A1). By 2015, the net preschool enrollment rate had reached 62%, remaining at similar levels in subsequent years (Ministerio de Educación Nacional, 2025).

Where did public preschools open, and how did they differ from existing ones? Appendix Table A1 shows that most were located in rural areas. Compared to preschools operating in 2005, newly opened preschools were more likely to offer a full school day, have smaller class sizes, and be located within schools offering primary grades. While we lack direct measures of quality, these patterns suggest that the expansion may have occurred through relatively better services. Newly opened

[9] In 2005, most of ICBF services consisted of small, home-based care centers (*Hogares Comunitarios*, in Spanish) serving the most vulnerable children aged 0-5 (Bernal and Ramírez, 2019).

preschools were also less likely to provide services for children 3–4, indicating that the expansion primarily targeted the mandatory grade. Despite these improvements, the public preschool system in the 2000s remained characterized by uneven service quality and limited oversight, in contrast to ICBF services, which operated under clearer national standards, albeit often with low-quality indicators (Bernal, 2014).

1.2 School Feeding

Between 2011 and 2012, the provision of school meals in public schools in Colombia underwent a major institutional transformation, shifting from centralized delivery by ICBF to a decentralized model overseen by the Ministry of Education (MEN, hereafter), with the objective of achieving universal coverage (Law 1450 of 2011). Although it is one of the country's oldest social programs (operating in various forms since the 1920s), by 2010 it mainly served students from the lowest socioeconomic level in urban areas (Departamento Nacional de Planeación, 2013). At the onset of MEN's oversight in 2012, only 10% of students were registered as receiving meals; by 2019, this number had increased to 73% (Collante-Zárate et al., 2024). Our analysis focuses on the impacts of the program in its current form, implemented under MEN since 2012.

The school feeding program has traditionally provided free nutritional supplements to students in public schools in the form of snacks, breakfast, or lunch. Its primary goal is to support enrollment and attendance, while also promoting healthy habits. To implement the program, MEN provides guidelines to Local Education Authorities ("Secretarías de Educación", in Spanish) which are responsible for allocating resources, identifying beneficiaries, and selecting schools to participate in the program. Priority is given to full-day schools in rural areas, and to urban schools serving ethnic minorities, students with disabilities, or low-income students. Within participating schools, implementation must begin with the lower grades and expand to students in upper grades depending on available resources, prioritizing low-income students from minorities, with disabilities, or victims of conflict (Collante-Zárate et al., 2024).

In theory, these guidelines (depending on budgetary constraints) determine which schools receive school meals, when they receive them, and which students benefit. In practice, the share of full-time students, enrolled in primary grades, or victims of conflict are strong predictors of the probability that schools are assigned to receive meals. Within schools, allocation began from the lower grades, as suggested, but did not adhere to other prioritization criteria. Instead, regardless of student characteristics, either all or none of the students in a given grade received meals (Collante-Zárate

et al., 2024). This suggests that whether a student received meals mainly depended on the timing of arrival of the program at their school (determined by Local Education Authorities) and the student's grade at that time, with younger students more likely to benefit earlier.

2 Data

We match nearly two decades of several administrative records to identify students' preschool attendance, school meals, academic progression, performance on the high school exit exam, and higher education enrollment. Our primary data source is MEN's Integrated Enrollment System (SIMAT), which contains individual-level longitudinal records of student enrollment from "Pre-jardín" to 11th grade for the period 2005-2023, annual indicators of whether each student received school meals (2012–2023), and school-level characteristics. Using SIMAT, we identify whether students completed primary school (reaching at least 5th grade), progressed to 9th and 11th grade, or ever dropped out.[10] To measure academic performance, we match SIMAT to data from "Saber 11", Colombia's national high school exit exam administered by the Colombian Institute for the Evaluation of Education (ICFES). These records, available from 2010–2023, include test scores in math and reading. Finally, we identify post secondary enrollment by linking SIMAT to the National Higher Education Information System (SNIES), also maintained by MEN, which covers the years 2013–2023 and records student enrollment in bachelor's, technical, or technological programs.

For a subset of students, we can also match household records from the System for Identifying Potential Beneficiaries of Social Programs (Sisbén, in Spanish), administered by the National Planning Department (DNP, in Spanish). These records include information on household composition, assets, educational attainment, and employment status of household members from interviews conducted between 2003–2010. We also use data from MEN's School Identification System (SISE, in Spanish), which provides the precise geographic coordinates of all schools nationwide. We complement the individual-level records with municipal-level information from the National Department of Statistics (DANE) and the Center for Economic Development Studies (CEDE). Using DANE's 2005 National Geostatistical Framework (MGN 2005), we compute geographic centroids for all municipalities. The Municipal Panel from CEDE provides data for 2004 on education levels, poverty rates, population, homi-

[10] We define ever dropping out as a binary variable equal to one if the student disappears from SIMAT records in t, $t+1$, and $t+2$, and does not reach grade 11th or takes the high school exit exam.

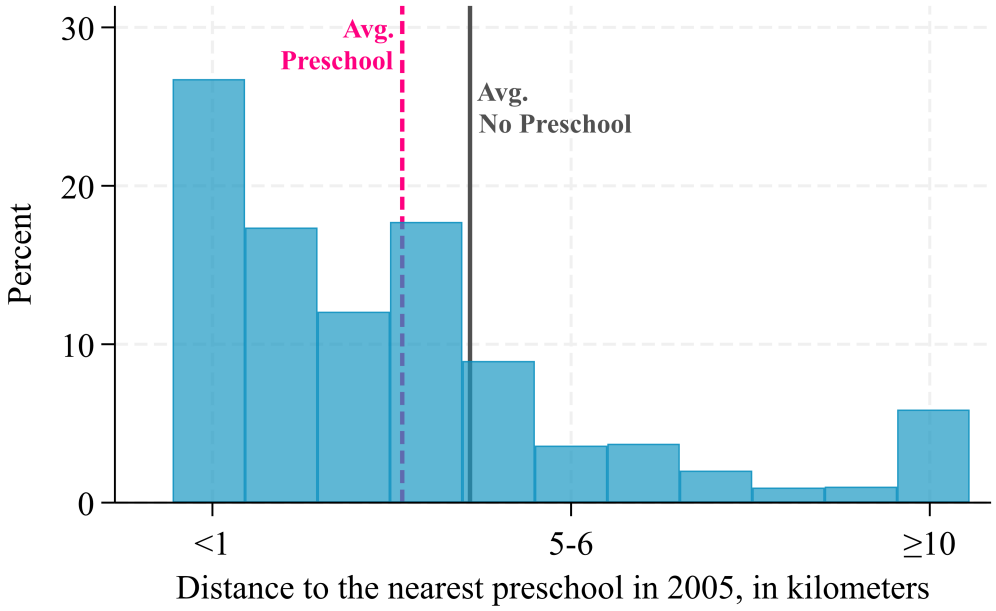
cides, municipal area, and distance to the capital and main markets.

2.1 Sample Definition and Descriptive Statistics

To identify preschool attendance and its long-term impacts, we focus on the universe of students enrolled in 1st grade of primary school in the public system between 2006–2011 (cohorts, hereafter) using SIMAT. By focusing on these cohorts, all students, including those in 1st grade in 2011, have at least two years after on-time high school graduation to enroll in higher education. We restrict the sample to new students in each cohort (i.e., students appearing in 1st grade for the first time). For each cohort, we link students to preschool enrollment records from the preceding year ($t-1$), from 2005–2010. We define a binary indicator for preschool attendance equal to one if the student appears in the preschool records in $t-1$, and zero otherwise. Of roughly 3.3 million students who entered first grade between 2006 and 2011, only 66% had attended preschool.

What explains the preschool enrollment gap? We argue that a key driver is the availability of preschools at the time families make enrollment decisions. To measure preschool availability or exposure, we begin by computing the minimum distance from the geographic centroid of a student's municipality of birth to the nearest preschool in 2005, our baseline year. Figure 1 shows the distribution of this distance: the median distance is 2 km, and on average, the nearest preschool was 1 km closer for students who enrolled (dashed pink line) compared to those who did not enroll (solid gray line). To leverage the expansion of preschools between 2005 to 2015, described in Section 1.1, we restrict the sample to students whose baseline distance was greater than 2 km. We classify students within 2 km at baseline as always-treated (namely, those who were already relatively close to preschool and thus likely unaffected by subsequent openings). This restriction results in a sample of 1,704,991 students across 483 municipalities, representing 44% of all municipalities in the country.

Figure 1. Distribution of Average Distance to the Nearest Preschool, at Baseline



Note: The figure displays the distribution of the average distance from the geographic centroid of a student's municipality of birth to the nearest preschool in 2005 (the baseline year). The dotted pink line denotes the average distance at baseline for students who went to preschool, while the solid gray line denotes the average distance for students who did not go to preschool. Source: SIMAT and DANE.

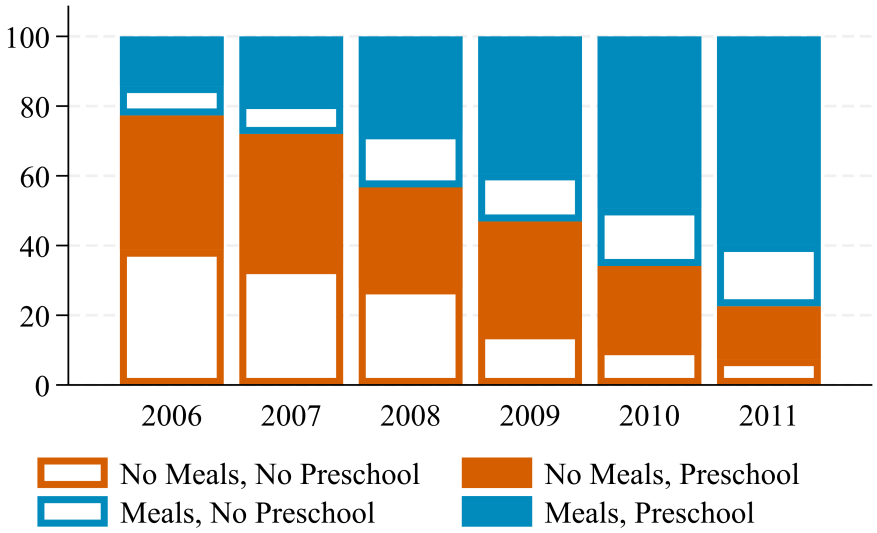
Importantly, Appendix Table A2, Panel A, shows that students in the restricted sample (column "Sample") are, on average, similar to those in the full sample (column "All Students") in terms of observable characteristics, with only small differences in income distribution and baseline math and reading scores. Meanwhile, the share of low-income rural students is higher among those who did not enroll in preschool (column "No Preschool") compared to those who did (column "Preschool"). We also observe higher dropout rates, slower grade progression, and worse performance on the high school exit exam among students who did not attend preschool.

We next examine how preschool openings evolved across municipalities over time for students in our estimating sample. Because we do not observe students' residential addresses, we define exposure at the level of the student's municipality of birth, which predates preschool enrollment. Specifically, we compute the distance from the geographic centroid of each municipality to the nearest preschool using annual data on preschool supply from 2006 to 2015 from SIMAT. We then define the first year in which a preschool was available within a 2 km (the median distance at baseline) radius of the centroid as the timing of exposure. Among the 483 municipalities in our sample, 21% experienced such preschool openings between 2006 and 2015. Appendix Figure A2 illustrates the spatial and temporal variation in preschool exposure

according to this definition.

To explore what predicts preschool exposure, we regress an indicator for whether a municipality was ever exposed, or the year of exposure, on municipal characteristics measured in 2004 (e.g., population, municipal area, distance to the department capital). We also include characteristics of the female population such as educational attainment, labor force participation, and health service use constructed from the 2005 Census data. The share of women with secondary education significantly predicts ever being exposed, though the estimated magnitude is small. In addition, only municipal area, invariant across municipalities, is a statistically significant predictor (Appendix Table A3).

Figure 2. Evolution of Preschool Enrollment and Share of School Meals Beneficiaries, by Cohort



Note: The figure shows the share of students who went to preschool, got school meals, both, or neither by cohort (i.e., the year of entry into first grade). Source: SIMAT.

With the scale-up of the national feeding program beginning in 2012, how did school meal coverage evolve across our cohorts? We identify school meal beneficiaries using SIMAT records and define a binary indicator equal to one if a student received school meals in any year. Appendix Table A2 shows that 48% of students in our sample received school meals. This share is higher among students who attended preschool compared to those who did not (54% versus 38%, respectively). Reflecting the staggered rollout of the program, the share of students receiving meals increased from roughly 20% in the 2006 cohort to nearly 80% in the 2011 cohort (Figure 2). Notably, the figure also shows that preschool enrollment rose over the same period,

from under 50% in the 2006 cohort to nearly 80% in the 2011 cohort. In terms of student characteristics, we do not observe differences by school meals participation in age, sex, or income level; yet, students who received meals are less likely to drop-out, and more likely to complete primary and secondary, and enroll in higher education (Appendix Table B1).

The national school feeding program expanded gradually across schools after 2012. By that time, some students in our cohorts had already repeated grades or dropped out, making grade-based measures of program uptake potentially endogenous to prior student characteristics, including preschool attendance. To address these concerns, we use the first year in which each student's first-grade school received the program as a proxy, since first-grade enrollment predates both the national scale-up and later schooling transitions. Of 32,270 first-grade schools, the vast majority received school meals at some point between 2012–2019. Appendix Figure B1 shows the distribution of program rollout across first-grade schools, with most entering between 2012–2014. As expected from the prioritization rules, Appendix Table B2 shows that rollout timing correlates with observable school characteristics, including full-time school days, rural status, the share of preschool and primary students, and the share of low income students in 2011.

3 Identification and Estimation

In this section, we begin by describing how we identify and estimate the medium- and long-run effects of each intervention separately, before testing for complementarities using variation in exposure overlap. For both interventions, we use a staggered DiD design comparing students affected by each expansion to those unaffected. Given the identification and estimation challenges associated with complementarities, we adopt a parsimonious strategy and estimate our main results using two-way fixed effects (TWFE). Nonetheless, we assess the robustness of our findings using alternative estimators (Cengiz et al., 2019; Sun and Abraham, 2021).

Throughout this section, Y denotes one of the following student-level outcomes of interest: the probability of ever dropping out; the probability of reaching grade 5, 9, or 11; standardized math and reading test scores on the high school exit exam; and the probability of enrolling in higher education.

3.1 Preschool

To identify the effects of preschool on medium- to long-term academic outcomes, we use a staggered DiD design that leverages the expansion of public preschools across municipalities. We classify students as treated if they lived more than 2 kilometers from the nearest preschool in 2005 and a new preschool opened within 2 kilometers of the centroid of their municipality of birth between 2006 and 2015.[11] We assume that exposure is an absorbing state, such that once a preschool opens the municipality remains treated thereafter. Overall, our approach captures variation along the intensive margin of access—specifically, improvements in geographic proximity—rather than the introduction of preschool itself.

Our design compares outcomes across students who gained access to preschool earlier and those who had not yet gained access, or never did, by the time they entered preschool. That is, our control group includes both not-yet-treated and never-treated students. Identification relies on the following assumptions: (i) parallel trends (i.e., in the absence of preschool openings, potential outcomes would have evolved in parallel for all exposed and unexposed groups), (ii) no spillovers (i.e., outcomes in unexposed areas are not affected by nearby preschool openings), and (iii) no anticipation (i.e., outcomes in municipalities where preschools opened were not affected by the upcoming expansion). Under these assumptions, our design recovers the intent-to-treat (ITT) effect of preschool exposure. Moreover, we can also recover implied treatment-on-the-treated (TOT) effects as the ratio of each outcome ITT to the (first stage) ITT on preschool enrollment.

First, to determine whether preschool exposure affects preschool enrollment (i.e., the first stage), we use the following event-study model:

$$D_i = \alpha_{m(i)} + \alpha_{t(i)} + \sum_{j=-5, j\neq -1}^{4} \beta_j \mathbb{1}[\text{Exp}_{m(i),t(i)} = j] + \alpha_{d(i)} \times \alpha_{t(i)} + \epsilon_i, \tag{1}$$

where D_i is a binary variable equal to one if student i (in municipality m and cohort t) enrolled in preschool, and zero otherwise. $\text{Exp}_{m(i),t(i)} = t - \text{Open}_{m(i)}$, where t denotes the student's preschool year (cohort year-1), and $\text{Open}_{m(i)}$ is the year a preschool first became available within 2 km of the centroid of the student's municipality of birth. We restrict the exposure window to 5 years for each cohort ($j \in \{-5, 4\}$, omitting $j = -1$). $\alpha_{m(i)}$ are municipal fixed effects, which capture m

[11] We set the 2 km threshold based on both data and prior evidence. First, 2 km corresponds to the median distance to the nearest preschool at baseline in 2005. Second, low travel time or distance is a well-known determinant of childcare use (Attanasio et al., 2013; Bernal and Fernández, 2013). In Section 4.1 we assess the robustness of our results to employing different radius.

characteristics invariant overtime, and $\alpha_{t(i)}$ are cohort fixed effects invariant across space. We also include a department d trend, $\alpha_{d(i)} \times \alpha_{t(i)}$, which captures common time-varying factors between municipalities in the same department. ϵ_i is the error term.

To assess the effects of preschool exposure on academic outcomes, we follow a similar event-study specification where the dependent variable is replaced by an outcome of interest for student i,

$$Y_i = \alpha_{m(i)} + \alpha_{t(i)} + \sum_{j=-5, j\neq -1}^{4} \beta_j \mathbb{1}[\text{Exp}_{m(i),t(i)} = j] + \alpha_{d(i)} \times \alpha_{t(i)} + \epsilon_i. \tag{2}$$

We also estimate pooled versions of Equations (1) and (2) with TWFE as follows,

$$Y_i = \alpha_{m(i)} + \alpha_{t(i)} + \beta \mathbb{1}[\text{Exp}_{m(i),t(i)} \geq 0] + \alpha_{d(i)} \times \alpha_{t(i)} + \epsilon_i, \tag{3}$$

where β identifies the overall effect of preschool exposure. For inference, we cluster standard errors at the municipality level.

Because our ITTs rely on the parallel trends assumption, we use the leads in Equations (1) and (2), β_j for $j < -1$, to examine whether exposed and unexposed groups were trending similarly prior to preschool openings. That is, we test whether the pre-exposure coefficients are statistically indistinguishable from zero. A second threat to identification is spillovers: families in unexposed municipalities may change their enrollment choices in response to preschool openings in neighboring municipalities. However, given the importance of proximity for preschool enrollment decisions, it is unlikely that families would regularly cross municipal boundaries to access newly opened schools.

Last, TWFE suffers from "forbidden comparisons" (Borusyak et al., 2024). Hence, we contrast our results with alternative estimators robust to treatment effect heterogeneity and dynamic effects (Sun and Abraham, 2021; Cengiz et al., 2019).[12]

3.2 School Feeding Program

We next focus on the medium-and long-term effects of the scale-up of the feeding program from 2012 onward. There are two important considerations for identification.

[12]We use the estimators from Sun and Abraham (2021) and the stacked event-study design of Cengiz et al. (2019) to assess robustness for two reasons. First, our focus on complementarities makes TWFE the most straightforward estimator to implement and interpret, as most recent multiple-treatment estimators do not accommodate staggered adoption. Second, both alternatives are computationally more efficient, given our sample size, and produce dynamic estimates that treat pre-treatment and post-treatment coefficients symmetrically (Roth, 2024; Wooldridge, 2025) and are more directly comparable to TWFE.

First, neither receiving school meals nor the grade at which students receive them is orthogonal to student characteristics. Second, the treatment year is inherently unobserved for students who never received school meals. Restricting the analysis to meal recipients would introduce bias, particularly when estimating complementarities.

To address the first concern, we define treatment as the first year the student's first-grade school received school meals. To address the second, we assign a *placebo* treatment year to non-recipients, defined as the average year of meals receipt among students of the same cohort, sex, and municipality. Using a staggered DiD design, we compare outcomes between students whose first-grade school s had entered the program by the year they received school meals (or the assigned *placebo* year for non-recipients) and those whose first-grade school had not yet received the program in that year.[13] Identification relies on the parallel trends assumption, such that in the absence of school meals, potential outcomes would have evolved in parallel for all groups of first-grade schools that received the program in different years. We assume no spillovers, the rollout of school meals in one school does not affect outcomes in schools that have not yet received the program, and no anticipation. Under these assumptions, we can identify the ITT of school feeding exposure and recover the implied TOTs, dividing outcome ITTs by the first stage ITT.

For estimation, we use an event-study model,

$$F_i = \alpha_{m(s)} + \alpha_{t_\text{sf}(i)} + \alpha_{t(i)} + \sum_{j=-4, j\neq -1}^{5} \gamma_j \mathbb{1}[\text{Exp}^{\text{sf}}_{s,t_\text{sf}(i)} = j] + \alpha_{d(s)} \times \alpha_{t(i)} + \mu_i, \quad (4)$$

where F_i is a binary variable equal to one if student i (of school s) ever received school meals, and zero otherwise. $\text{Exp}^{\text{sf}}_{s,t_\text{sf}(i)} = t_\text{sf} - \text{SF}_s$, where SF_s is the first year in which school s (the student's first-grade school) received the school feeding program (SF), and t_sf is the year the student received meals for the first time or the *placebo* year (for non-beneficiaries). The model includes municipality fixed effects, preschool cohort fixed effects, and department trends. Plus, timing of school feeding fixed effects, $\alpha_{t_\text{sf}(i)}$, to capture common trends across students first exposed to the program in the same year. We control for school-day type (full-day vs. half-day). Last, to account for the "imputation" of treatment year for non-beneficiaries, we additionally control for sex. μ is the idiosyncratic error term. Equation (4) denotes the first stage of school feeding exposure; that is, whether our treatment definition predicts receipt of school meals.

[13] In the context of school feeding and our treatment definition, 5% of students fall into the never-treated group.

For outcomes, we estimate:

$$Y_i = \alpha_{m(s)} + \alpha_{t_sf(i)} + \alpha_{t(i)} + \sum_{j=-4, j\neq -1}^{5} \gamma_j \mathbb{1}[\text{Exp}^{sf}_{s,t_sf(i)} = j] + \alpha_{d(s)} \times \alpha_{t(i)} + \mu_i, \tag{5}$$

and pooled versions of Equations (4) and (5) with TWFE,

$$Y_i = \alpha_{m(s)} + \alpha_{t_sf(i)} + \alpha_{t(i)} + \gamma \mathbb{1}[\text{Exp}^{sf}_{s,t_sf(i)} \geq 0] + \alpha_{d(s)} \times \alpha_{t(i)} + \mu_i, \tag{6}$$

where γ identifies the overall dynamic effect of school feeding exposure. We cluster standard errors at the first-grade school (s) level in Equations (4) to (6).[14]

To assess the validity of the parallel trends assumption, we test whether the lead terms in Equations (4) and (5), γ_j for $j < -1$, are statistically different from zero. For robustness, we use alternative estimators robust to heterogeneous effects and variation in treatment timing (Sun and Abraham, 2021).[15]

3.3 Interaction between Investments

We combine variation in exposure to preschool and to school feeding to estimate complementarities. Our design compares schools exposed to the feeding program with those not yet exposed, in municipalities with and without preschool expansions.

Identification relies on the assumption that the two shocks—preschool and school feeding—are exogenous and independent. That is, preschool exposure should not predict the timing of school feeding, and vice versa. In our setting, preschool expansion largely precedes the scale-up of the national school feeding program. However, given its focus on early grades, the introduction of school meals may have incentivized local authorities to expand preschool provision. This seems unlikely in practice, as opening a preschool typically requires substantial infrastructure investments and staffing beyond what would be justified by the arrival of meals alone. Conversely, it is possible that municipalities with preschool access were prioritized for earlier rollout of school meals, for instance due to existing infrastructure. Yet, since preschool exposure is defined based on whether a new preschool opened within 2 km of the

[14] We exclude school fixed effects from Equations (4)-(6) to preserve the cross-municipality variation in preschool exposure, which is defined at the municipality level and is central to identifying complementarities between preschool and school feeding. Since each school belongs to a single municipality, including school fixed effects would absorb all variation needed to estimate the preschool effect.

[15] Unlike the case of preschool, we refrain from using stacked designs in this case since the share of never-treated school feeding students is only 5% leading to noisy estimates when estimating cohort(stacks)-specific effects.

municipality centroid, it is not clear that such localized expansions would affect the timing of school meals rollout.

To address these concerns, we assess the plausibility of the independence assumption in two ways. First, we test whether the timing of one shock is predicted by exposure to the other. Specifically, we re-estimate Equation (1) replacing the dependent variable with a set of year-specific indicators for school feeding rollout, $\mathbb{1}[\text{SFP}_s = y]$, for $y \in [2012, 2022]$. Similarly, we re-estimate Equation (4) using indicators for preschool opening year, $\mathbb{1}[\text{Open}_{m(s)} = y]$, for $y \in [2006, 2015]$, as the dependent variable. Under the independence assumption, the event-time coefficients (leads and lags) should be statistically indistinguishable from zero and display no systematic pattern.

Second, we examine whether trends in outcomes prior to the rollout of school feeding differ by preschool exposure status. To do so, we re-estimate Equations (4) and (5) saturated by preschool exposure (*i.e.*, we include each $\mathbb{1}[\text{Exp}^{\text{sfp}}_{s,t_\text{sfp}(i)} = j]$ and its interaction with $\mathbb{1}[\text{Exp}_{m(s),t(i)} \geq 0]$). Pre-treatment coefficients should be statistically indistinguishable from zero in both groups, suggesting that trends were parallel before the arrival of school feeding, regardless of preschool exposure. Under independence, we can recover ITTs of preschool exposure, school feeding exposure, and their interaction.

For estimation, we use the following pooled TWFE model,

$$\begin{aligned} Y_i = \alpha_{m(s)} + \alpha_{t_\text{sf}(i)} + \alpha_{t(i)} & \\ + \beta \mathbb{1}[\text{Exp}_{m(i),t(i)} \geq 0] & \\ + \gamma \mathbb{1}[\text{Exp}^{\text{sf}}_{s,t_\text{sf}(i)} \geq 0] & \\ + \theta \mathbb{1}[\text{Exp}_{m(i),t(i)} \geq 0] \times \mathbb{1}[\text{Exp}^{\text{sf}}_{s,t_\text{sf}(i)} \geq 0] + \alpha_{d(s)} \times \alpha_{t(i)} + u_i, & \end{aligned} \tag{7}$$

where β captures the effect of exposure to preschool in the absence of shool feeding, while γ captures the effect of exposure to school feeding without preschool. θ is our parameter of interest: it identifies the additional effect of exposure to school feeding when preceded by preschool. For inference, we cluster standard errors at the first-grade school, s, level.

Last, we recover implied TOTs of preschool with and without school feeding, by estimating their corresponding first stages. In specific, we re-estimate Equation (7) with $D_{i,s} \times F_{i,s}$ and $D_{i,s} \times (1 - F_{i,s})$ as dependent variables.

4 Results

4.1 Preschool

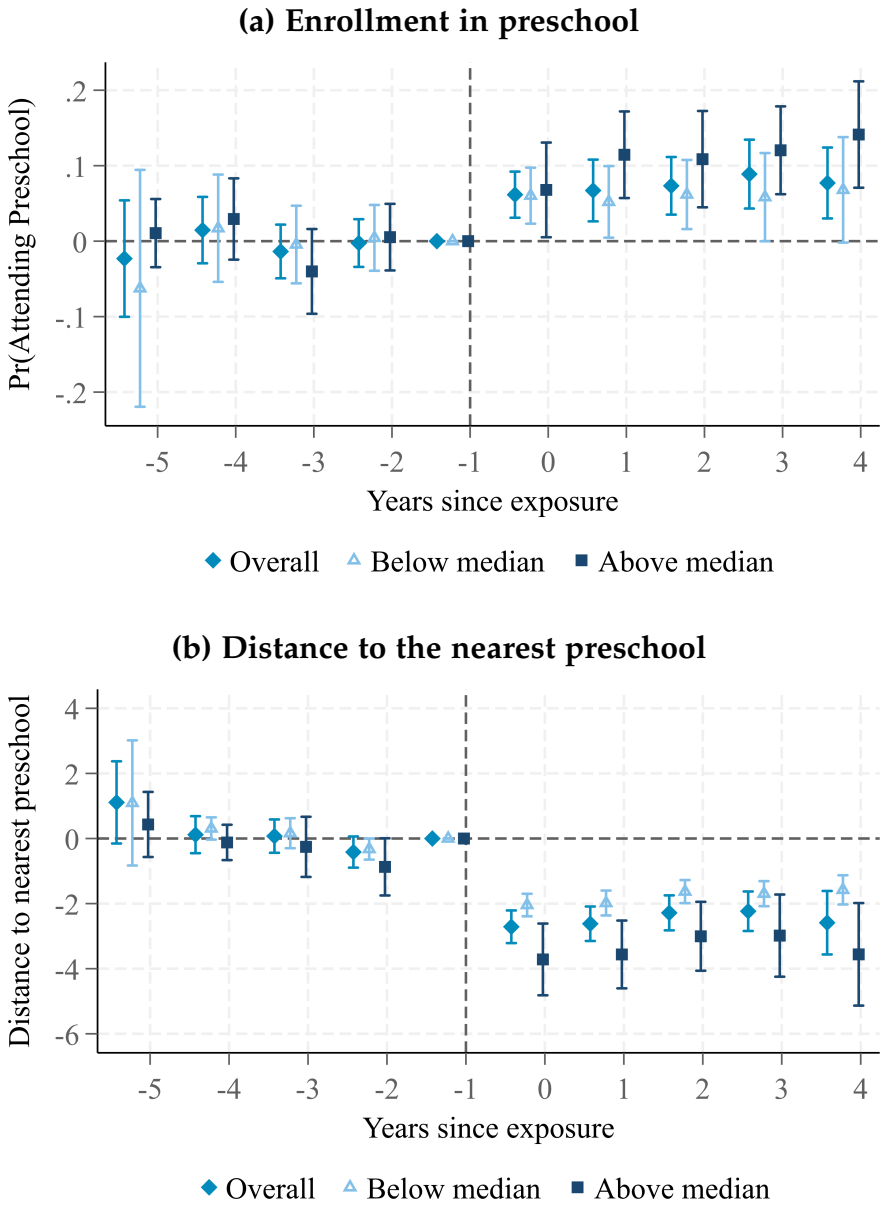
Our measure of preschool exposure determines both enrollment and proximity. Figure 3 presents event-study estimates of Equation (1) for the probability of preschool enrollment in panel (a), and the distance to the nearest preschool in panel (b). We show results for the full estimation sample (diamonds) and separately by baseline proximity in 2005, splitting children above (squares) and below (triangles) the sample median distance of 4 km. Across both sets of outcomes, the pre-treatment coefficients are close to zero, indicating that enrollment and availability were evolving in parallel prior to the opening of a new preschool within 2km. This pre-treatment pattern provides support for the validity of the assumption of parallel trends. After preschool openings, attendance increases by 7.1 pp (11%) on average, along with a reduction of 2.5 km in the distance to the nearest preschool. Children who were initially farther away (above the 2005 median distance) experience larger gains: enrollment rises by 10 pp (17%) and distance falls by 3.2 km on average (Appendix Table A4).

The empirical evidence supports our choice of a 2 km radius to measure exposure to preschool openings. First, the average dynamic effect of openings on attendance declines with distance: at 0.5km the effect is 10.5 pp, while at 3km it falls to 3.3 pp (Appendix Table A5 and Figure A3). Second, although the 2km estimates are somewhat smaller than those at 1.5km, they are more persistent and precise. We therefore interpret the 2km results as conservative estimates of preschool exposure.[16] Moreover, our findings remain largely unchanged when we implement a stacked design (Cengiz et al., 2019) or apply the Sun and Abraham (2021) estimator (Appendix Table A7 and Figure A5).

We find persistent effects of preschool exposure on academic trajectories. Table 1 presents pooled estimates from Equation 3 for the probability of reaching different grades, dropping out, and enrolling in higher education. Outcome-specific event-studies estimates do not show violations of parallel trends (Appendix Figure A6, first row). Preschool exposure increases the probability of reaching 5th, 9th, and 11th grade by 3, 2.1, and 2 pp, respectively. The probability of ever dropping out de-

[16] Preschool exposure does not crowd out private preschool enrollment or induce migration (e.g., making families more likely to move to another municipality for their child's education). First, the vast majority of students in our sample enrolled in a public preschool, and the effect of preschool exposure on private preschool enrollment is negligible (Figure A4, Panel A; Table A6, Panel (a)). Second, only 15% of students in our sample migrate between birth and first grade (i.e., reporting different municipalities at birth and in first grade). Exposure does not increase the probability of migrating, and its effects on preschool enrollment are fully driven by non-migrants (Figure A4, Panel B; Table A6, Panel (b)).

Figure 3. Effect of Preschool Exposure on Enrollment and Distance to Nearest School



Note: Panel (a) shows event-study estimates for the probability of enrolling in preschool, using Equation (1). Panel (b) shows estimates using the distance to the nearest preschool from the centroid of the student's municipality of birth as dependent variable. Results shown for the full estimating sample ("Overall") and split by below ("Below median") and above ("Above median") the median distance of 4 kilometers (in the estimating sample). Confidence intervals at the 95% level, calculated with clustered standard errors at the municipality level.

creases by 2 pp (4%), while the probability of enrolling in higher education increases by 6%. These results are robust to alternative estimators (Appendix Table A9). Combining these estimates with the first stage (Appendix Table A4, Panel A, Column 1), results in implied treatment-on-the-treated effects (TOTs) of 25-40 pp, well above estimates found in other contexts. However, we argue that these sizable implied TOTs reflect not only the effect of preschool but also the fact that exposed children might

have benefited from additional interventions and programs. Thus, we interpret these results as capturing the overall long-term effects of preschool exposure, combining children who were and were not subject to subsequent investments.

Table 1. Effect of Preschool Exposure on Academic Trajectories

	(1)	(2)	(3)	(4)	(5)
	Probability of reaching grade			**Pr(Ever**	**Pr(Higher**
	5th	**9th**	**11th**	**Drop-out)**	**Education)**
Exposed×Post	0.029**	0.021**	0.020**	-0.019*	0.013**
	(0.013)	(0.009)	(0.008)	(0.010)	(0.007)
Number of obs.	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991
Number of clusters	483	483	483	483	483
Mean	0.745	0.546	0.479	0.483	0.228

Note: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Preschool exposure improves student performance in the high school exit exam. Because preschool increases the probability of taking the test by 2 pp (Table 2, Panel A, Column 1), restricting our analysis to test takers, for whom we observe scores, would bias our estimates. Hence, we perform two exercises following the literature. First, we adopt the parametric approach in Angrist et al. (2006) and impute the lowest percentile score for non-test takers and for students falling below that threshold. Under this approach, preschool exposure increases math and reading scores by about 0.03 standard deviations (Table 2, Panel A, Columns 2–3), equivalent to 2–3 months of business-as-usual schooling (Evans and Yuan, 2019). The implied TOTs are 0.3–0.4 standard deviations, large effects likely reflecting a combination of impacts of preschool and later investments.

Second, test score gains could arise from improvements at different points in the distribution. Following Gray-Lobe et al. (2023), we construct binary variables taking the value of one if a student takes the test and scores above a quartile threshold, and zero otherwise (including those who did not take the test). Panel B of Table 2 shows that preschool exposure increases the probability of scoring above the 25th percentile in math by 1.5 pp (4.4%) and above the median by 0.7 pp (3.2%). In reading, preschool exposure can boost students above the bottom quartile, but has no effect at the median or higher thresholds (Table 2, Panel C). Taken together, these results suggest that test score gains are concentrated among lower-performing students and are partly driven by the extensive-margin effect of increased test taking. Event-study

estimates across all performance outcomes show no evidence of pre-trend violations (Appendix Figure A6, second row) and our pooled results remain largely unchanged under alternative estimators (Appendix Table A10).[17]

Table 2. Effect of Preschool Exposure on Performance in the High School Exit Exam

	(1)	(2)	(3)
Panel A: Test Taking and Parametric Scores			
	Took Test	**Math**	**Reading**
Exposed×Post	0.020***	0.030**	0.026**
	(0.008)	(0.012)	(0.012)
Number of observations	1,704,991	1,704,991	1,704,991
Number of clusters	483	483	483
Mean	0.457	0.006	0.006
Mean (test takers)		0.789	0.806
Panel B: Probability of Math Score above quartile			
	Bottom	**Median**	**Top**
Exposed×Post	0.015***	0.007**	0.001
	(0.005)	(0.004)	(0.003)
Number of observations	1,704,991	1,704,991	1,704,991
Number of clusters	483	483	483
Mean	0.336	0.219	0.107
Panel C: Probability of Reading Score above quartile			
	Bottom	**Median**	**Top**
Exposed×Post	0.012**	0.004	0.000
	(0.005)	(0.004)	(0.002)
Number of observations	1,704,991	1,704,991	1,704,991
Number of clusters	483	483	483
Mean	0.336	0.223	0.108

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Last, we find similar effects of preschool expansions across subgroups: there are no significant differences in academic outcomes by gender or urban–rural status (Appendix Figures A7 and A8). We observe somewhat larger effects for children from higher socioeconomic backgrounds, consistent with a stronger first stage in this group (Appendix Figure A9), rendering differences in implied TOTs negligible. Among urban children, the implied TOTs are larger compared to rural children as a result of a

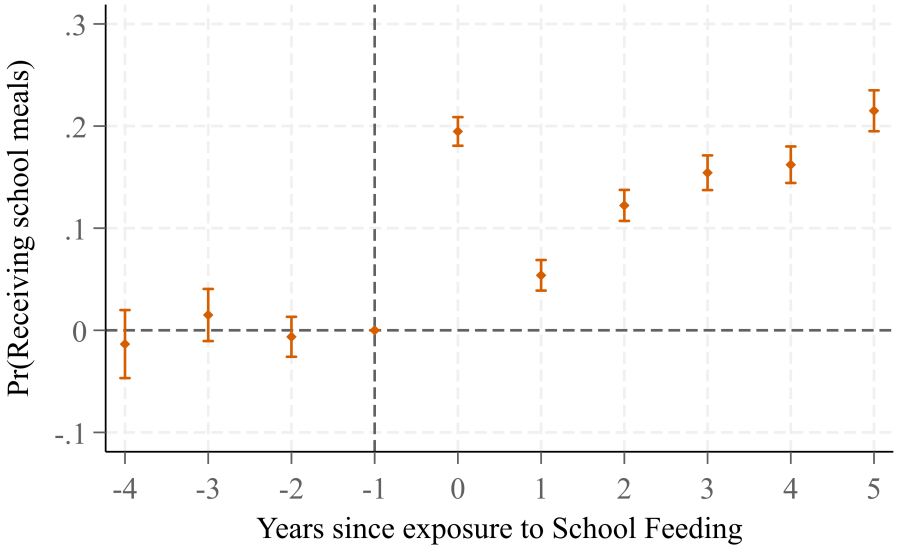
[17]To assess whether additional infrastructure around preschool openings drives our results, we restrict treated municipalities to those where preschools opened within existing primary schools (86% of openings), excluding openings that preceded primary schools or were standalone (14%). Appendix Table A8 shows an 8.2 pp increase in preschool enrollment, compared with 7.1 pp in the full sample, and somewhat larger ITT effects on academic outcomes. Hence, the implied TOT estimates remain similar in magnitude.

weaker first stage among the former (Appendix Figure A9).

4.2 School Feeding

In this section, we examine the aggregate effects of school feeding on academic outcomes irrespective of whether students attended preschool. Although our sample, identifying variation, and estimation method differ, the results we present below are largely similar to the evaluation of the school feeding program by Collante-Zárate et al. (2024).

Figure 4. Effect of School Feeding Exposure on the Probability of Receiving School Meals



Note: Event-study estimates for the probability of receiving school meals, using Equation (4). Confidence intervals at the 95% level, calculated with clustered standard errors at the school level.

First, our measure of school feeding exposure determines the receipt of school meals. Figure 4 shows an immediate increase in the probability of receiving school meals after treatment, followed by a drop in the next period and a sustained increased afterwards. The pre-treatment coefficients are centered around zero, providing support of no differential trends in receiving meals across schools prior to treatment. The post-treatment pattern results from our use of an "indirect" measure of exposure to school meals. By 2012 children from the 2005-2006 cohorts were less likely to be in primary or lower grades, making them less likely to receive meals rightaway. In contrast, students in the 2009-2010 cohorts, who were in earlier grades in 2012 when the school feeding program began its expansion, were more likely to receive meals right after the program's arrival given the priorization rules (see Section 1.2). Nonetheless, our measure of exposure to school feeding increases the probability of receiving

meals by 14 pp, corresponding to a large increase of 45% with respect to the mean (Table 3, Panel A, Column 1).[18,19]

Consistent with its overarching objective of keeping children in school, exposure to the feeding program has sizable effects on academic progression, in reducing drop-out, and in increasing higher education enrollment. The probability of reaching grade 5th increases by 8 pp (13%), grade 9th by 9.3 pp (24%), and grade 11th by 8.1 pp (24%) (see Table 3, Panel A, Columns 2–4). We estimate that the probability of ever dropping-out reduces by 9 pp (13%) (see Table 3, Panel B, Column 1). Last, the probability of enrolling in higher education increases in 4.5 pp (33%) (see Table 3, Panel B, Column 2).[20]

In addition, school feeding improves academic performance. Nutrition and food security have been associated with improved learning outcomes (Maluccio et al., 2009; Anderson et al., 2018; Belot and James, 2011). Panel B, Columns (3) and (4), of Table 3 show that exposure to school feeding increases the average parametric scores in math and reading in 0.13 standard deviations, similar to previous findings. Unlike preschool, which solely improves scores for low-achieving students, the feeding program also boosts the likelihood of achieving test scores in the top of the distribution (Appendix Table B3). The probability of scoring above the 75th percentile in math and reading increases by about 2 pp. As in the case of preschool, the implied TOTs of academic outcomes are considerably large but might reflect complementarities between early and late investments. We will explore this hypothesis in the next section.[21,22]

[18] Appendix Figure B2 splits the dynamics of school meal receipt by the grade in which students were first exposed to the program. We observe that the pattern of Figure 4 is driven by an immediate increase in the probability of receiving meals in grades 4–5, which decreases in the following periods, while the probability of receiving meals in secondary rises gradually overtime.

[19] Appendix Figure B5 shows that the first stage of school feeding is larger for low-income children from rural areas, consistent with the program's prioritization rules.

[20] Our estimates of the effects of school feeding on academic progression and dropout are broadly consistent with Collante-Zárate et al. (2024), who report effects of 20–29% on grade 11 completion and 11–30% on dropout. We estimate a larger effect on higher education enrollment (35% vs. 10–15%), likely due to a lower baseline enrollment rate in our control group (8.4% vs. 30–35% in Collante-Zárate et al. (2024)), resulting from the authors restricting their analysis to students in 6th grade in 2012-2013.

[21] Outcome-specific event-studies for the school feeding program show no evidence of pre-trends (Appendix Figure B3). Our results for the school feeding program are largely unchanged when implementing the Sun and Abraham (2021) estimator using never-treated and the last-treated cohort as control. See Figure B4 Tables B4-B6 in the Appendix.

[22] We do not observe heterogeneous effects of school feeding by gender (Appendix Figures B6 and B7). Consistent with the first stage results, ITT estimates are higher among low-income, rural children for outcomes such as reaching grades 5, 9, and 11, and reducing the probability of dropping out. Lastly, while the effects on the math and reading test scores are also stronger for these groups, we do not find significant differences in the probability of scoring above the median or top of the distribution.

Table 3. Effect of School Feeding Exposure on Receiving School Meals and Academic Outcomes

	(1)	(2)	(3)	(4)
Panel A: First stage and Academic Progression				
	Pr(Received School Meals)	**Probability of reaching grade**		
		5th	**9th**	**11th**
Exposed to Feeding×Post	0.136***	0.079***	0.093***	0.081***
	(0.005)	(0.005)	(0.005)	(0.005)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270
Mean	0.296	0.614	0.381	0.328
Panel B: Drop-out, Enrollment in Higher Education, and Parametric Scores				
	Pr(Ever Drop-out)	**Pr(Higher Education)**	**Math**	**Reading**
Exposed to Feeding×Post	-0.087***	0.045***	0.138***	0.131***
	(0.005)	(0.003)	(0.011)	(0.011)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270
Mean	0.649	0.136	-0.295	-0.289

Notes: Each column reports the effect of school feeding exposure on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

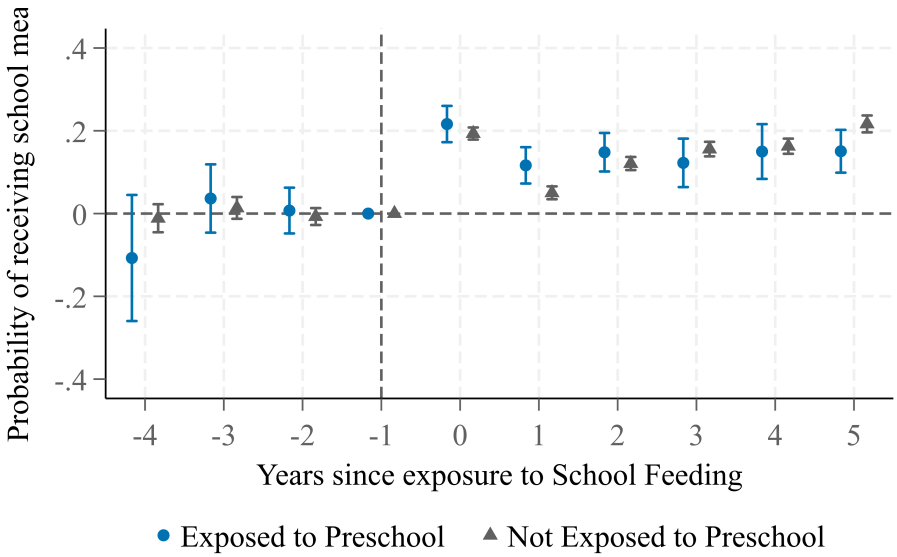
4.3 Interaction between Preschool and School Feeding

We have shown that preschool and school feeding each have meaningful effects on academic outcomes. In this section, we examine whether the returns to school feeding are higher for children exposed to preschool versus those who were not exposed. That is, motivated by the theoretical concept of dynamic complementarity (Cunha and Heckman, 2007), we use a reduced-form approach to test for interaction effects between preschool and school feeding.

First, we test whether the rollout of one intervention predicts the timing of the other, as outlined in Section 3.3. Appendix Figure C1 shows no systematic patterns in leads or lags, and most coefficients are statistically indistinguishable from zero. Second, Figure 5 displays event-study estimates of exposure to *school feeding*, by exposure to *preschool*. That is, we estimate a saturated version of Equation (4), as detailed in Section 3.3. The pre-treatment coefficients are not significant and are centered around zero for both groups, implying that trends in receiving meals were parallel prior to the program's arrival regardless of whether students were exposed to preschool.[23] Taken together, these results support the validity of our research de-

[23] Appendix Figures C2 and C3 show event-study estimates of exposure to school feeding by expo-

Figure 5. Effect of School Feeding Exposure on the Probability of Receiving School Meals, by Exposure to Preschool



Note: Event-study estimates for the probability of receiving school meals, using a saturated Equation (4) where each lead and lag is interacted by a binary indicator of exposure to preschool. Confidence intervals at the 95% level, calculated with clustered standard errors at the school level.

sign to identify complementarities between policies.[24] Post-treatment, school feeding exposure increases the probability of receiving meals in a similar pattern and magnitude regardless of preschool exposure, although the effects are more noisy among those exposed due to a smaller sample size.

On average, we find evidence of policy complementarities between preschool and school feeding during compulsory schooling, with effects that extend into pursuing post-secondary degrees. Table 4 reports estimates from Equation (7), showing that preschool exposure amplifies the effects of school feeding. The effect of school feeding on completing grades 9 and 11 is 2–2.5 pp larger for students previously exposed to preschool, relative to those who were not. We observe interaction effects of similar magnitudes on the probability of enrolling in higher education. The latter is consistent with the concept of dynamic complementarity, such that earlier investments increase the returns to later ones. We also observe that, without the boost from

sure to preschool for academic outcomes. The pre-treatment coefficients are centered around zero and largely not significant across all academic outcomes.

[24]To further validate the independence assumption, we compute the average characteristics of students who would react to the preschool (school feeding) expansion (namely, the compliers) conditional on exposure to school feeding (preschool). Without independence, the composition of compliers might differ depending on whether they were exposed to the other treatment. Appendix Table C1 shows that regardless of preschool exposure, school feeding compliers are similar on individual and household characteristics. Appendix Table C2 shows differences that are imprecisely estimated, but preschool compliers are, in both groups (regardless of school feeding exposure), overwhelmingly from low-income rural households.

school feeding which targets academic permanence, preschool exposure has no effect on dropout or academic progression.[25] In contrast, we estimate a positive standalone effect of preschool on grade 5 completion, but no interaction effect at this margin. This likely reflects the timing of the feeding program's rollout: most cohorts in our sample had already exited primary school before school feeding expanded.

We recover implied TOTs by dividing the estimates in rows 1 and 4 of Table 4 by the corresponding first stage effects in Appendix Table C3, row 1, Column (3), and row 4, Column (5), respectively. Among compliers (i.e., students who enrolled in preschool due to improved access and later received school meals) the probability of reaching 9th and 11th grade increased by 28 pp, with similarly large effects on higher education enrollment.

We do not find robust evidence of complementarity between preschool and school feeding on test performance. Interaction effects are absent for standardized scores in math and reading, with only modest gains observed in the upper tail of the reading distribution (Appendix Table C4). Although test scores improve significantly among children exposed to both programs, these effects are not statistically different from those among children exposed to preschool alone. In addition, school feeding on its own generates substantial gains, while preschool alone shows little effect.[26] When combined with school feeding, we estimate implied TOT effects of preschool of approximately 14–18 pp in increasing the probability that students score above the bottom quartile in math and reading; standardized test scores increase by roughly 0.3 standard deviations. These patterns suggest that school meals are the primary driver of gains in academic performance, while preschool contributes little additional value on its own. However, given the size of the point estimates, we may be under-powered to detect moderate interaction effects.

4.4 Mechanisms and Discussion

4.4.1 Timing between investments

While the results of the previous section suggest complementarities in academic permanence and progression rather than in performance, the timing of the second investment may also shape the extent of these interactions. Two considerations motivate

[25] Results for the probability of taking the high school exit exam are qualitatively the same to those of the probability of reaching grade 11th. In Colombia, the high school exit exam is a requirement to graduate high school resulting in negligible differences between these two outcomes.

[26] These results are consistent with the finding that early childhood interventions may not lead to cognitive gains in the long-term, commonly known as test score fade-out (see Cascio, 2021). In our case, we find that preschool exposure can have long-term impacts in academic performance only in the presence of additional investments.

Table 4. Effect of Preschool Exposure, School Feeding Exposure, and their Interaction on Academic Trajectories

	(1)	(2)	(3)	(4)	(5)
	Probability of reaching grade			**Pr(Ever**	**Pr(Higher**
	5th	**9th**	**11th**	**Drop-out)**	**Education)**
Preschool	0.027*	-0.005	-0.001	0.009	-0.008
	(0.014)	(0.013)	(0.012)	(0.008)	(0.006)
Feeding	0.079***	0.091***	0.080***	-0.086***	0.044***
	(0.005)	(0.006)	(0.006)	(0.006)	(0.005)
Preschool×School Feeding	0.002	0.025**	0.020*	-0.027**	0.020**
	(0.014)	(0.013)	(0.011)	(0.012)	(0.008)
Total Preschool	0.029***	0.020***	0.019***	-0.018***	0.012***
	(0.005)	(0.005)	(0.004)	(0.004)	(0.003)
Implied TOT	0.574	-0.106	-0.002	0.191	-0.166
(Preschool without Feeding)	[0.044]	[0.686]	[0.908]	[0.458]	[0.118]
Implied TOT	0.403	0.278	0.264	-0.25	0.166
(Preschool and Feeding)	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270	32,270
Mean	0.6108	0.3805	0.3270	0.6494	0.0861

Note: Each column reports estimates from Equation (7) on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01. p-values for implied TOTs in brackets, computed using bootstrap.

our focus on the timing of exposure to school feeding. First, the theory of dynamic complementarity posits the logic of sensitive periods, when investments are more productive (Cunha et al., 2010), and implies that the effectiveness of later investments depends on the stock of early acquired skills and may lose effectiveness when subsequent inputs arrive too late in the life cycle (Heckman and Mosso, 2014). Second, empirical studies of school feeding programs find that earlier (Hoynes et al., 2016; Bütikofer et al., 2018) or longer (Chakraborty and Jayaraman, 2019; Lundborg et al., 2022) exposure results in larger gains. Hence, in what follows we explore whether school feeding is more complementary to preschool when implemented earlier.

A natural candidate for measuring the timing of school feeding exposure is the grade in which students first received meals. In fact, half of the students in our sample received school meals for the first time during primary school, with the average grade of first exposure equal to sixth grade. However, using grade of exposure as the timing variable would introduce bias, since grade attainment is itself affected by preschool. Instead, we use the difference between the year of exposure to school feeding and the year of exposure to preschool. This measure captures the timing of the feeding program arrival relative to preschool arrival, and is not determined by endogenous school progression. Among students exposed to preschool and school feeding, only 22% were exposed to the latter two to five years after preschool, while 42% were exposed six to seven years later (see Appendix Figure C4).[27]

We estimate the following equation:

$$Y_{i,s} = \alpha_{m(s)} + \alpha_{t(i)} + \beta \mathbb{1}[\text{Exp}_{m(s),t(i)} \geq 0] + \gamma \mathbb{1}[\text{Exp}^{\text{sfp}}_{s,t_\text{sfp}} \geq 0] + \sum_{g=1}^{4} \theta_g \mathbb{1}[\text{Window}_s = g] + \alpha_{d(s)} \times \alpha_{t(i)} + \epsilon_{imt}, \tag{8}$$

where $\text{Window}_s = \text{SFP}_s - \text{Open}_{m(s)}$ if $\mathbb{1}[\text{Exp}_{m(s),t(i)} \geq 0] \times \mathbb{1}[\text{Exp}^{\text{sfp}}_{s,t_\text{sfp}(i)} \geq 0] = 1$, and 0 otherwise. It denotes the difference in years between school feeding and preschool exposure, with 1 denoting 2–3 years, 2 for 4–5 years, 3 for 6–7 years, and 4 more than 8 years. If $\theta_1 \geq \theta_2 \geq \theta_3 \geq \theta_4 > 0$, then the return to preschool is higher in the presence of other early investments. To assess the validity of our identification strategy, we estimate the same specification using the probability of reaching grade 5 as the outcome. This serves as a placebo test: since most students in our cohorts would

[27] Similar to Lundborg et al. (2022), earlier arrival also means longer exposure in our context. Appendix Figure C5, Panel (a), shows a negative slope between total years with meals and timing of arrival. That is, earlier arrival (on the x-axis) is associated with longer receipt of school meals (on the y-axis). Hence, we cannot disentangle the effects of timing from the effects of length exposure in our context.

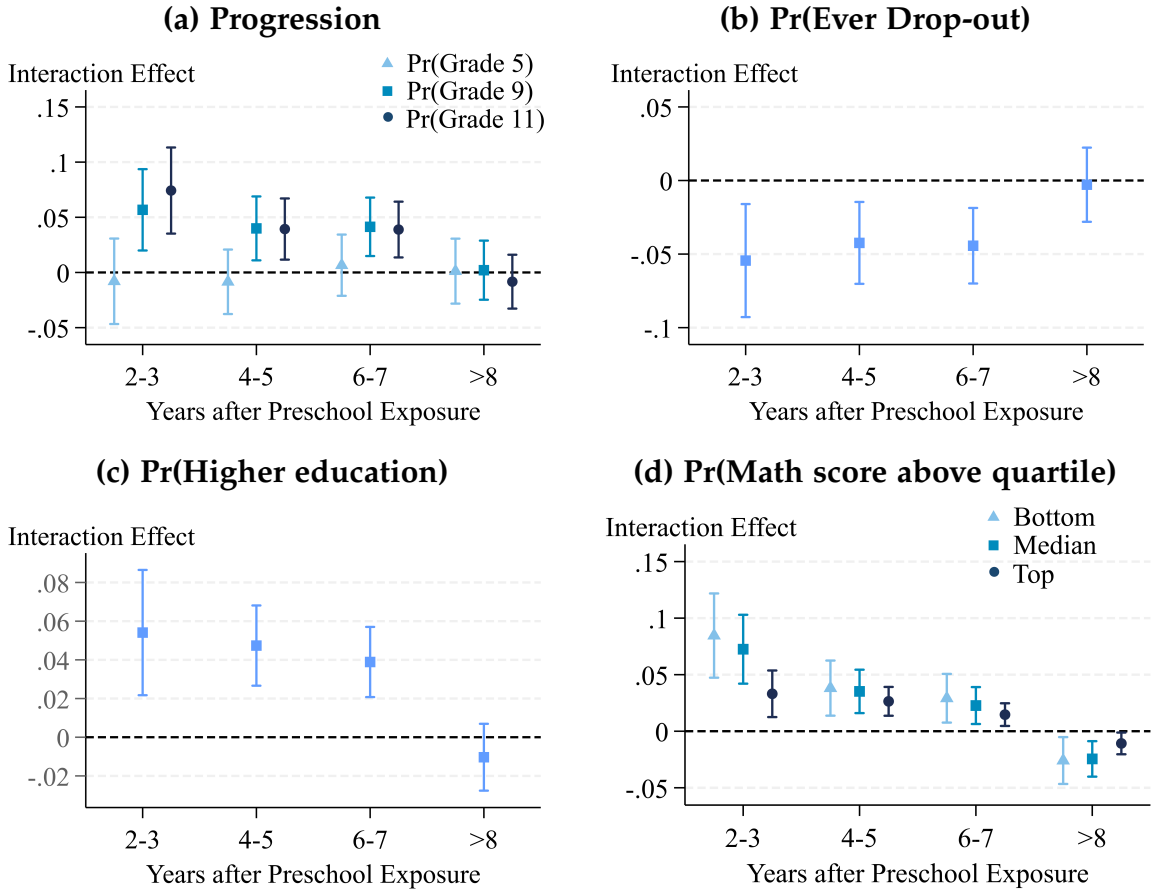
have completed primary school before the introduction of school feeding (post-2012), and few had exposure windows as short as two to three years, we should not expect to find any differential effects by timing for this outcome.

We find that the effect of preschool on grade completion and dropout reduction is greater when school feeding is introduced earlier (see Figure 6). Panel (a) shows that the interaction effect between preschool and school feeding exposure on the probability of reaching grades 9 and 11 declines as the time between the two investments increases. For students with two to three years between preschool and school feeding exposure, the interaction effect on reaching grades 9 and 11 exceeds 5 pp, roughly twice the average interaction effect reported in Table 4, and largely disappears when the gap between investments exceeds eight years. Moreover, consistent with the placebo test outlined above, we find no differential effect by timing on reaching grade 5. Panel (b) shows a similar gradient for the probability of ever dropping out: it falls by up to 5 pp when the gap between investments is two to three years, but there is no interaction effect when the gap exceeds eight years.

We also find that preschool and school feeding can be complementary for higher education enrollment when the latter investment reaches students at the right time. Panel (c) of Figure 6 shows a positive interaction effect of 2–6 pp between preschool and school feeding when the gap between investments is two to seven years, with no detectable interaction effect for longer gaps. To unpack these differential effects, Panel (b) Appendix Figure C5 shows the average grade at which students received school meals for the first time versus the exposure window between investments. Shorter (longer) gaps between preschool and feeding are generally associated with earlier (later) grades of feeding receipt. Notably, students below the 7 year window typically receive meals during primary school. This pattern suggests that complementarities with preschool for higher education enrollment are strongest when subsequent investments align with schooling transitions, rather than arriving too late (more than eight years after preschool).[28]

[28]Some students in our cohorts may have been exposed to financial aid programs. *Ser Pilo Paga*, the country's largest need-based and merit-based financial aid initiative (Londoño-Vélez et al., 2020) operated from 2014 to 2018, overlapping with the timing of higher education decisions for part of our sample. In 2018, it was replaced by *Generación E*, which maintained similar eligibility criteria and expanded coverage for low-income students (Londoño-Vélez et al., 2025). To assess whether our results for higher education enrollment are confounded by the expansion of financial aid, we estimate Equation (8) using the probability of being eligible for aid as the outcome. Exposure to school feeding between two and seven years after preschool increases eligibility by at most 0.2 pp, suggesting that financial aid programs are unlikely to drive the higher education patterns we observe (see Appendix Figure C6, Panel (d)).

Figure 6. Interaction Effect of Preschool and School Feeding, by time elapsed between preschool and school feeding exposure



Note: The figures plot estimates from Equation (8) of the interaction effects between preschool and school feeding exposure, by relative arrival between the two investments. Confidence intervals at the 95% level, calculated with clustered standard errors at the school level.

Finally, we estimate sizable interaction effects between preschool and school feeding on academic performance when the latter is introduced two to three years after preschool exposure (see Panel (d) Figure 6, and Appendix Figure C6). Within that window, the interaction effect on the probability of scoring above the bottom quartile in Math is approximately 8 pp, falling by half when the gap between investments is four to seven years, and turning negative when the gap exceeds eight years. We observe a similar pattern for the interaction effect on the probability of scoring above the median. Moreover, although smaller in magnitude (about 2.5 pp), the interaction effect on scoring above the top quartile is also positive when school feeding follows preschool by two to five years, suggesting that early exposure can also improve performance at the upper end of the distribution. We find qualitatively similar patterns for the probability of scoring above each quartile in Reading, as well as for standardized Math and Reading test scores (see Appendix Figure C6).

4.4.2 Complementarities by student characteristics

Although we do not find significant heterogeneity in the ITT effects of each investment, it is plausible that their complementarity varies across subgroups. Most existing reduced-form studies on dynamic complementarities find limited evidence of interaction effects, particularly among low-income populations (Rossin-Slater and Wüst, 2020), or show that families may compensate by either reinforcing or offsetting investments (Bharadwaj et al., 2018; Goff et al., 2023). In contrast, some evidence suggests that cash transfers to low-income families can remediate the negative effects of early-life shocks (Adhvaryu et al., 2024; Duque et al., 2023). Motivated by these findings, we estimate Equation (7) separately by gender, socioeconomic stratum, Sisbén status, and whether the student attended a rural or urban school in first grade.

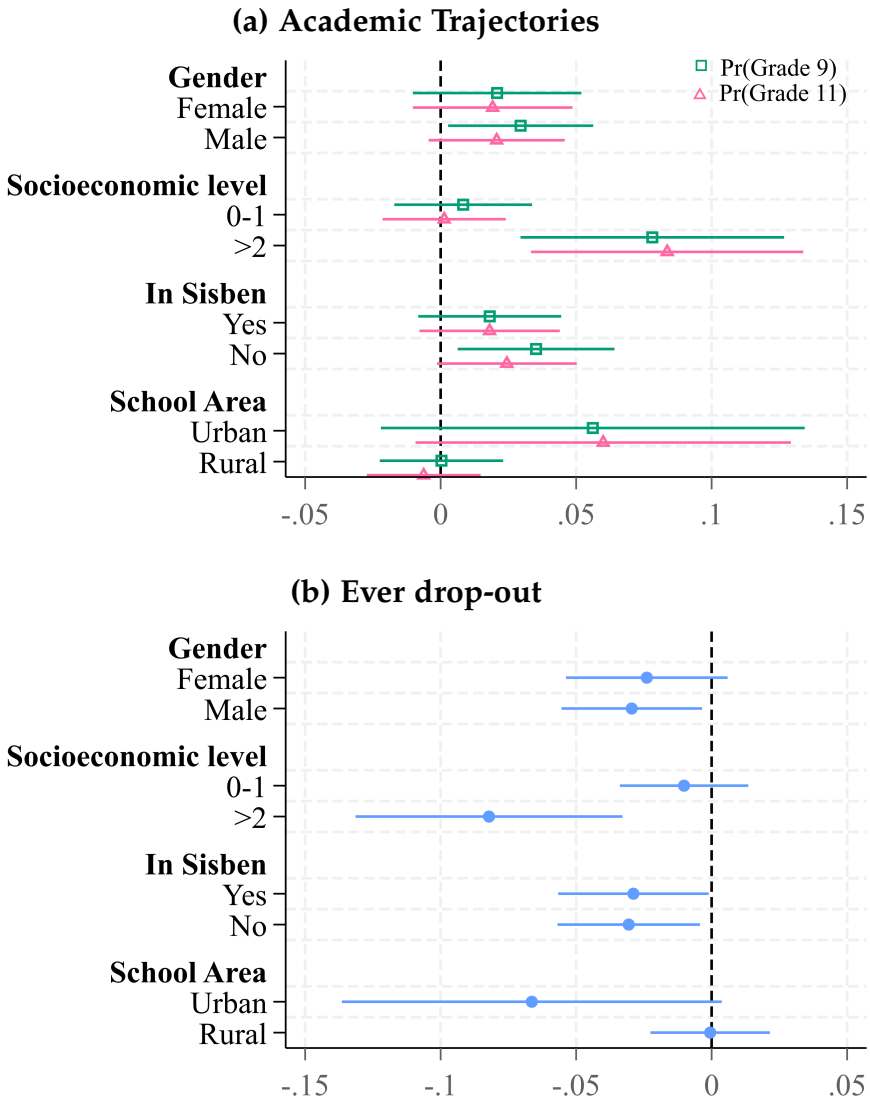
Consistent with previous studies, we find no evidence of complementarities among low-income students on any outcome (see Figure 7 and Appendix Figure C7). In addition, we do not observe significant gender differences in the interaction effect between preschool and school feeding. The interaction effects are somewhat larger for students in urban schools, but these estimates are consistently imprecise. In contrast, complementarities are significantly stronger among children from relatively higher socioeconomic strata. For instance, Panel (a) of Figure 7 shows that the interaction effect on the probability of reaching grades 9 and 11 is nearly 10 pp among students in strata 2–4, and statistically different from the null effect observed among students in the lowest stratum. Moreover, the absence of differences by Sisbén status indicates that complementarities vary along more granular socioeconomic levels than broad poverty classifications reveal. We observe a similar pattern across all outcomes, suggesting that complementarities are concentrated among relatively better-off students.

To unpack the sizable interaction effects among students from higher socioeconomic strata, we use a subsample matched to detailed household records from Sisbén. These records include information on household composition, members' labor market participation and education, housing conditions, and other socioeconomic indicators. About 47% (55%) of students in strata 0–1 (2–4) can be linked to Sisbén (see Appendix Table C5). Students in Sisbén are more likely to enroll in preschool and receive school meals than those not in Sisbén, but are otherwise similar in age and gender composition. We focus the analysis on this matched subsample. As expected, students from higher socioeconomic levels tend to live in smaller households, have more educated mothers who are more likely to work, and are less likely to report domestic work as their main economic activity (Appendix Table C6).[29] These patterns

[29]For a smaller subsample, we also observe paternal education and employment status. Appendix Table C7 shows negligible differences in the share of fathers who work, although fathers of students

are suggestive of complementarities that may depend on reinforcing environments at home, although we cannot directly observe parental investments to credibly test this argument.

Figure 7. Interaction Effect of Preschool and School Feeding on Academic Trajectories, by student attributes



Note: The figures plot estimates of interaction effects between preschool and school feeding exposure, by selected student characteristics. Confidence intervals at the 95% level, calculated with clustered standard errors at the school level.

Complementarities among relatively better-off students may also stem from the relative quality of public preschool compared to other available options. For these students, public provision may represent a lower-quality substitute for higher-quality private preschools. In the absence of the expansion of public preschools, higher-income children may have attended private preschools of arguably better quality,

in strata 2–4 are more educated.

while their lower-income peers may not have enrolled in any early education at all. These differences in fallback options are likely important drivers of differential program effects (Kline and Walters, 2016; Cornelissen et al., 2018). We observe that among students in strata 2–4, preschool exposure alone has negative albeit imprecise effects on all academic outcomes, while among lower-income students, effects are negligible but not negative (Appendix Figure C9). These patterns are unlikely to be driven by differential take-up of preschool or school meals (Appendix Figure C10). Notably, school feeding has sizable positive effects across the board, with the largest gains among lower-income students. Taken together, these findings suggest that complementarities among better-off students may arise because public preschool is a lower-quality alternative to their outside options, offering limited or even negative returns unless followed by subsequent investments. In contrast, for lower-income students, school feeding alone yields substantial gains, regardless of prior preschool exposure.

5 Quantifying social welfare

We now translate the estimates of exposure to preschool, school feeding, and both interventions into the Marginal Value of Public Funds (MVPF). The MVPF is computed as the ratio of beneficiaries willingness to pay (WTP) over the net government costs of a policy of interest. Appendix D describes how we combine our causal estimates with information on unitary costs of preschool and school meals to form: (i) the WTP^{P} and $Cost^{\text{P}}$, or the benefits and costs associated to the preschool expansion, (ii) the WTP^{SFP} and $Cost^{\text{SFP}}$, or the benefits and costs associated to the school feeding program expansion, and (iii) the $WTP^{\cap}$ and $Cost^{\cap}$, or the additional earning gains and costs due to the interaction effects between the two programs.

We use the WTP and net government costs estimates in Appendix D to compute the MVPF of the expansion of preschool and school feeding (i) allowing for complementarities and (ii) under an additive benchmark that sets complementarities to zero. Relative to the additive benchmark, complementarities can raise WTP (via higher implied earnings) but also increase net costs through induced public spending on additional years of schooling; the net effect on the MVPF is therefore a priori ambiguous. The bundle accounting for complementarities yields:

$$MVPF^{\text{DC}} = \frac{WTP^{\text{P}} + WTP^{\text{SFP}} + WTP^{\cap}}{Cost^{\text{P}} + Cost^{\text{SFP}} + Cost^{\cap}} = \frac{28,019,926}{5,756,795} = 4.87,$$

whereas the additive benchmark yields:

$$MVPF^{\text{NC}} = \frac{WTP^{\text{P}} + WTP^{\text{SFP}}}{Cost^{\text{P}} + Cost^{\text{SFP}}} = \frac{22,418,964}{5,622,420} = 3.99.$$

Thus, allowing for complementarities raises the estimated MVPF by $4.87/3.99 - 1 \approx 22\%$ relative to the additive benchmark. Hence, the increase in WTP more than compensates for the higher costs associated to additional schooling.

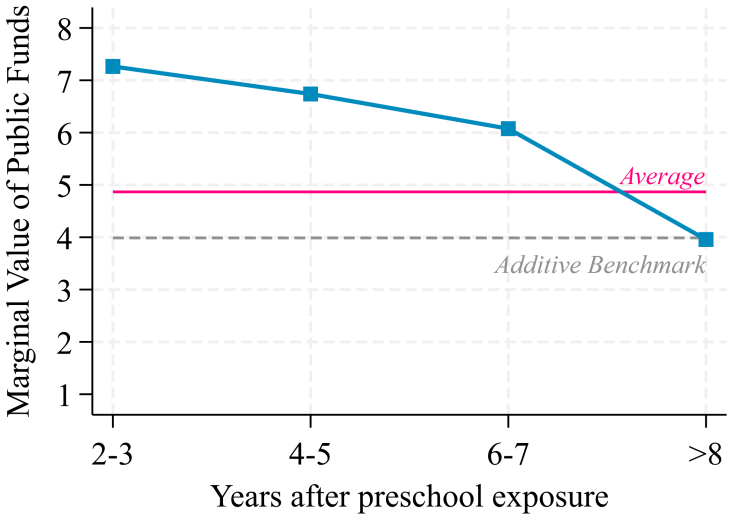
Timing Between Investments. Our estimates in Section 4.4.1 allow us to show how the MVPF of the policy bundle varies with the time elapsed between investments. Because earlier exposure to school meals relative to preschool has larger effects on educational attainment (see Figure 6), the WTP increases as the lag between investments shrinks. Earlier exposure to school meals also affects the government budget through two channels. First, it increases direct program spending by extending the duration of meal provision: Appendix Table D2 shows that meals are provided for an average of 4.2 years when they arrive 2–3 years after preschool, compared to 3.0 years when they arrive 8+ years later. Second, higher educational attainment gains at shorter windows increase indirect public costs through additional years of schooling. We quantify these trade-offs by computing window-specific MVPFs,

$$MVPF^{\text{DC}}(g) = \frac{WTP^{\text{P}} + WTP^{\text{SFP}} + WTP^{\cap}(g)}{Cost^{\text{P}} + Cost^{\text{SFP}}(g) + Cost^{\cap}(g)},$$

where $g \in \{2\text{–}3, 4\text{–}5, 6\text{–}7, > 8\}$ denotes the number of years between school feeding and preschool exposure. As in our bundle MVPF, all upfront preschool spending is included in $Cost^{\text{P}}$, while $Cost^{\text{SFP}}(g)$ and $Cost^{\cap}(g)$ capture, respectively, window-specific direct spending on school meals (through the average number of years of meal provision in window g) and induced public education spending from additional years of schooling (e.g., lower dropout and higher post-secondary enrollment in window g).

The timing of subsequent investments matters for welfare. Figure 8 shows that the MVPF of the preschool plus school feeding bundle decreases with the lag between investments. Window-specific MVPFs range from 7.3 when feeding arrives 2–3 years after preschool to 3.9 when feeding arrives more than 8 years later. When feeding arrives within six years of preschool exposure, the MVPF lies well above the additive benchmark (gray line) and also exceeds the pooled bundle MVPF of 4.87 (pink line, "Average"). In contrast, when feeding arrives in secondary school (8+ years after preschool), the MVPF essentially coincides with the additive benchmark and is

Figure 8. Marginal Value of Public Funds, by time elapsed between preschool and school feeding exposure



Note: The figure plots window-specific marginal values of public funds (MVPFs) for the preschool and school feeding policy bundle as a function of the lag g (in years) between preschool exposure and the arrival of school feeding. The pink horizontal line shows the pooled bundle MVPF ($MVPF^{DC} = 4.87$), and the gray dashed line shows the additive benchmark MVPF obtained by setting complementarities to zero ($MVPF^{NC} = 3.99$). MVPFs use present values of beneficiaries' willingness to pay and net government costs, which include direct program spending, induced public education spending from additional schooling, and fiscal externalities from changes in earnings.

roughly half the value in the earliest arrival window.

Given that our causal estimates mainly pertain to educational attainment, the reported MVPFs are likely lower bounds. Recent work by Humphries et al. (2025) and Montpetit et al. (2025) shows that early investments can generate benefits beyond educational attainment (e.g., parental labor supply), and the true MVPF of the policies could well exceed our estimates.

6 Conclusion

We study policy complementarities in human capital formation between two large-scale public investments: preschool and school meals. Our empirical strategy exploits the staggered expansion of public preschools in Colombia (2005–2015) and the subsequent scale-up of the national school feeding program starting in 2012. The two expansions resulted in some students exposed to both, one, or neither intervention, with the timing of school feeding determined by program targeting rules. Under the assumption that exposure to each intervention and their overlap are plausibly exogenous, we test: (i) whether investments are more productive when preceded by

early ones, and (ii) whether the timing of follow-up investments matters. We further examine whether preschool alone (i.e., an early investment without follow-ups) and school meals alone (i.e., later remediation for early disadvantage) can yield sustained impacts.

Using 12-18 years of administrative records, we follow cohorts of primary school entrants from 2006-2011 through their academic trajectories. Consistent with complementarity, we find that students exposed to both interventions (in particular when school feeding was introduced shortly after preschool), experience lower dropout, higher secondary completion and post-secondary enrollment, and improved test scores. In contrast, preschool alone has limited effects without follow-up investments, while the school feeding program alone produces substantial gains (though smaller than in the presence of preschool).

Complementarities across all educational outcomes are substantially stronger for students from relatively higher socioeconomic levels. If anything, complementarities seem to be negligible among low-income students. This pattern aligns with prior evidence from Rossin-Slater and Wüst (2020) and Bjorvatn et al. (2025), who document limited or even negative interaction between preschool and other investments for vulnerable children. Additional analyses using a subsample of students with detailed household records suggest that family environments, with more and better resources, are a potential channel behind these heterogeneous results. Although we lack direct measures of parental investments, our findings are consistent with the hypothesis that compensatory (or offsetting) behaviors at home among low-income households may undo school-based complementarities (Bau et al., 2020; Goff et al., 2023). In contrast, we speculate that among relatively better-off families, reinforcement at home (or even the absence of interference) may allow these complementarities to emerge.

Our findings also show that preschool alone has limited long-term effects. A *naive* analysis of the preschool expansion suggests sizable impacts across all outcomes, with large implied TOT estimates. However, once we disentangle direct effects from interaction effects, most of the observed gains are concentrated among students who were also exposed to subsequent investments. Accounting for these complementarities increases the estimated welfare impact of jointly expanding preschool and school feeding by 22% relative to an additive benchmark.

From a policy perspective, this highlights the importance of follow-up interventions to sustain and amplify early gains. From an empirical perspective, our results show that the evaluation of early interventions may overstate their effectiveness if later *complementary* investments are not taken into account. This concern is not unique to our setting, as children around the world are often exposed to multiple and over-

lapping investments. But it is also a limitation to our own analysis: we acknowledge that many children in our sample likely benefited from additional programs, as our study period coincides with a broader expansion of educational policies in the country. Whether other educational investments, such as the expansion of financial aid programs in the late 2010s, also act as complements in the production of human capital remains an important and open question for future research.

References

Adhvaryu, Achyuta, Teresa Molina, Anant Nyshadham, and Jorge Tamayo, "Helping Children Catch Up: Early Life Shocks and the PROGRESA Experiment," *The Economic Journal*, January 2024, *134* (657), 1–22.

Alderman, Harold, Donald Bundy, and Aulo Gelli, "School Meals Are Evolving: Has the Evidence Kept Up?," *The World Bank Research Observer*, August 2024, *39* (2), 159–176.

Almond, Douglas and Bhashkar Mazumder, "Fetal Origins and Parental Responses," *Annual Review of Economics*, August 2013, *5* (Volume 5, 2013), 37–56.

_ , Janet Currie, and Valentina Duque, "Childhood Circumstances and Adult Outcomes: Act II," *Journal of Economic Literature*, December 2018, *56* (4), 1360–1446.

Anderson, Michael L., Justin Gallagher, and Elizabeth Ramirez Ritchie, "School meal quality and academic performance," *Journal of Public Economics*, December 2018, *168*, 81–93.

Andrew, Alison, Orazio P. Attanasio, Raquel Bernal, Lina Cardona Sosa, Sonya Krutikova, and Marta Rubio-Codina, "Preschool Quality and Child Development," *Journal of Political Economy*, July 2024, *132* (7), 2304–2345.

Angrist, Joshua, Eric Bettinger, and Michael Kremer, "Long-Term Educational Consequences of Secondary School Vouchers: Evidence from Administrative Records in Colombia," *American Economic Review*, June 2006, *96* (3), 847–862.

Attanasio, Orazio, Costas Meghir, and Emily Nix, "Human Capital Development and Parental Investment in India," *The Review of Economic Studies*, November 2020, *87* (6), 2511–2541.

Attanasio, Orazio P., Vincenzo Di Maro, and Marcos Vera-Hernández, "Community Nurseries and the Nutritional Status of Poor Children. Evidence from Colombia," *The Economic Journal*, 2013, *123* (571), 1025–1058. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ecoj.12020.

Ayllón, Sara and Samuel Lado, "The Causal Impact of School-Meal Programmes on Children in Developed Economies: A Meta-Analysis," August 2025.

Bailey, Martha J., Shuqiao Sun, and Brenden Timpe, "Prep School for Poor Kids: The Long-Run Impacts of Head Start on Human Capital and Economic Self-Sufficiency," *American Economic Review*, December 2021, *111* (12), 3963–4001.

Bau, Natalie, Martin Rotemberg, Manisha Shah, and Bryce Steinberg, "Human Capital Investment in the Presence of Child Labor," May 2020.

Behrman, Jere R., Ricardo Gomez-Carrera, Susan Parker, Petra Todd, and Weilong Zhang, "Starting Strong: Medium-and Longer-run Benefits of Mexico's Universal Preschool Mandate," 2024.

Belot, Michèle and Jonathan James, "Healthy school meals and educational outcomes," *Journal of Health Economics*, May 2011, *30* (3), 489–504.

Berlinski, Samuel, Guillermo Cruces, Sebastian Galiani, Paul Gertler, and Fabian Gonzalez, "Long-run Effects of Universal Pre-Primary Education Expansion: Evidence from Argentina," December 2025.

_ , Sebastian Galiani, and Marco Manacorda, "Giving children a better start: Preschool attendance and school-age profiles," *Journal of Public Economics*, June 2008, *92* (5), 1416–1440.

_ , _ , and Paul Gertler, "The effect of pre-primary education on primary school performance," *Journal of Public Economics*, February 2009, *93* (1), 219–234.

Bernal, Raquel, "Diagnóstico y recomendaciones para la atención de calidad a la primera infancia en Colombia," Cuadernos de Fedesarrollo 11568, Fedesarrollo 2014.

_ and Camila Fernández, "Subsidized childcare and child development in Colombia: Effects of Hogares Comunitarios de Bienestar as a function of timing and length of exposure," *Social Science & Medicine*, November 2013, *97*, 241–249.

_ and Sara María Ramírez, "Improving the quality of early childhood care at scale: The effects of "*From Zero to Forever*"," *World Development*, June 2019, *118*, 91–105.

_ , Orazio Attanasio, Ximena Peña, and Marcos Vera-Hernández, "The effects of the transition from home-based childcare to childcare centers on children's health and development in Colombia," *Early Childhood Research Quaterly*, April 2019, *47*, 418–431.

Bhalotra, Sonia R. and Atheendar Venkataramani, "Shadows of the Captain of the Men of Death: Early Life Health Interventions, Human Capital Investments, and Institutions," 2015. SSRN Working Paper.

Bharadwaj, Prashant, Juan Pedro Eberhard, and Christopher A. Neilson, "Health at Birth, Parental Investments, and Academic Outcomes," *Journal of Labor Economics*, April 2018, *36* (2), 349–394. Publisher: The University of Chicago Press.

Biroli, Pietro, Titus Galama, Stephanie von Hinke, Hans van Kippersluis, Cornelius A Rietveld, and Kevin Thom, "The Economics and Econometrics of Gene–Environment Interplay," *The Review of Economic Studies*, June 2025, p. rdaf034.

Bjorvatn, Kjetil, Denise Ferris, Selim Gulesci, Arne Nasgowitz, Vincent Somville, and Lore Vandewalle, "Long-Term Effects of Preschool Subsidies and Cash Transfers on Child Development: Evidence from Uganda," *AEA Papers and Proceedings*, May 2024, *114*, 459–462.

_ , _ , _ , _ , _ , and _ , "Childcare, Labor Supply, and Business Development: Experimental Evidence from Uganda," *American Economic Journal: Applied Economics*, 2025.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting Event-Study Designs: Robust and Efficient Estimation," *The Review of Economic Studies*, February 2024, *91* (6), 3253–3285. _eprint: https://academic.oup.com/restud/article-pdf/91/6/3253/60441633/rdae007.pdf.

Bütikofer, Aline, Krzysztof Karbownik, and Ingrid Mikkelsen Semb, "Getting a Second Chance? Prenatal Health Shocks and Infant Health Care," in "Essays in Empirical Labor Economics," University of Oslo, 2024. PhD dissertation by Ingrid Mikkelsen Semb.

Bütikofer, Aline, Eirin Mølland, and Kjell G. Salvanes, "Childhood nutrition and labor market outcomes: Evidence from a school breakfast program," *Journal of Public Economics*, December 2018, *168*, 62–80.

Carneiro, Pedro, Yyannu Cruz-Aguayo, Rafael Hernandez-Pachon, and Norbert Schady, "Dynamic complementarity in elementary schools: Experimental estimates from Ecuador," 2022.

Cascio, Elizabeth U, "Early childhood education in the United States: What, when, where, who, how, and why 1," in "The Routledge handbook of the economics of education," Routledge, 2021, pp. 30–72.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, "The Effect of Minimum Wages on Low-Wage Jobs*," *The Quarterly Journal of Economics*, August 2019, *134* (3), 1405–1454.

Chakraborty, Tanika and Rajshri Jayaraman, "School feeding and learning achievement: Evidence from India's midday meal program," *Journal of Development Economics*, June 2019, *139*, 249–265.

Chan, Marc K., Antonio Dalla-Zuanna, and Kai Liu, "Understanding Program Complementarities: Estimating the Dynamic Effects of Head Start with Multiple Alternatives," IZA Discussion Paper 17297, IZA Institute of Labor Economics September 2024.

Collante-Zárate, Sofía, Catherine Rodríguez, and Fabio Sánchez, "The Power of a Meal. School Feeding and its Educational Effects: Evidence from Colombia," Documento CEDE No.24 June 2024.

Cornelissen, Thomas, Christian Dustmann, Anna Raute, and Uta Schönberg, "Who Benefits from Universal Child Care? Estimating Marginal Returns to Early Child Care Attendance," *Journal of Political Economy*, December 2018, *126* (6), 2356–2409. Publisher: The University of Chicago Press.

Cunha, Flavio and James Heckman, "The Technology of Skill Formation," *American Economic Review*, 2007, *97* (2), 31–47.

_ , James J. Heckman, and Susanne M. Schennach, "Estimating the Technology of Cognitive and Noncognitive Skill Formation," *Econometrica*, 2010, *78* (3), 883–931. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA6551.

Currie, Janet, "Early Childhood Education Programs," *Journal of Economic Perspectives*, June 2001, *15* (2), 213–238.

_ and Douglas Almond, "Chapter 15 - Human capital development before age five," in David Card and Orley Ashenfelter, eds., *Handbook of Labor Economics*, Vol. 4, Elsevier, 2011, pp. 1315–1486.

_ **and Duncan Thomas**, "Does Head Start Make a Difference?," *The American Economic Review*, 1995, *85* (3), 341–364. Publisher: American Economic Association.

Deming, David, "Early Childhood Intervention and Life-Cycle Skill Development: Evidence from Head Start," *American Economic Journal: Applied Economics*, July 2009, *1* (3), 111–134.

Departamento Nacional de Planeación, "Evaluación de Operaciones y Resultados para Determinar el Grado de Efectividad del Programa de Alimentación Escolar – PAE," Technical Report, Departamento Nacional de Planeación 2013. Accessed: 2025-10-12.

Duque, Valentina, Maria Rosales-Rueda, and Fabio Sanchez, "How Do Early-Life Shocks Interact with Subsequent Human Capital Investments? Evidence from Administrative Data," 2023. Working Paper.

Evans, David K and Fei Yuan, "Equivalent Years of Schooling: A Metric to Communicate Learning Gains in Concrete Terms. Policy Research Working Paper 8752.," *World Bank*, 2019.

Gilraine, Michael, "School accountability and the dynamics of human capital formation," 2017. Working Paper.

Goff, Leonard, Ofer Malamud, Cristian Pop-Eleches, and Miguel Urquiola, "Interactions Between Family and School Environments: Access to Abortion and Selective Schools," *Journal of Human Resources*, April 2023. Publisher: University of Wisconsin Press Section: Articles.

Gray-Lobe, Guthrie, Parag A Pathak, and Christopher R Walters, "The Long-Term Effects of Universal Preschool in Boston*," *The Quarterly Journal of Economics*, February 2023, *138* (1), 363–411.

Gunnsteinsson, Snaebjorn, Teresa Molina, Achyuta Adhvaryu, Parul Christian, Alain Labrique, Jonathan Sugimoto, Abu Ahmed Shamim, and Keith P. West, "Protecting infants from natural disasters: The case of vitamin A supplementation and a tornado in Bangladesh," *Journal of Development Economics*, September 2022, *158*, 102914.

Heckman, James J. and Stefano Mosso, "The Economics of Human Development and Social Mobility," *Annual Review of Economics*, August 2014, *6* (Volume 6, 2014), 689–733.

Heckman, James, Rodrigo Pinto, and Peter Savelyev, "Understanding the Mechanisms through Which an Influential Early Childhood Program Boosted Adult Outcomes," *American Economic Review*, October 2013, *103* (6), 2052–2086.

_ **, Seong Hyeok Moon, Rodrigo Pinto, Peter Savelyev, and Adam Yavitz**, "Analyzing social experiments as implemented: A reexamination of the evidence from the HighScope Perry Preschool Program," *Quantitative Economics*, 2010, *1* (1), 1–46. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/QE8.

Hendren, Nathaniel and Ben Sprung-Keyser, "A Unified Welfare Analysis of Government Policies*," *The Quarterly Journal of Economics*, August 2020, *135* (3), 1209–1318.

Hoynes, Hilary, Diane Whitmore Schanzenbach, and Douglas Almond, "Long-Run Impacts of Childhood Access to the Safety Net," *American Economic Review*, April 2016, *106* (4), 903–934.

Humphries, John Eric, Christopher Neilson, Xiaoyang Ye, and Seth D. Zimmerman, "Parents' Earnings and the Returns to Universal Pre-Kindergarten," Working Paper 33038, National Bureau of Economic Research. October 2025.

Jenkins, Jade M., Tyler W. Watts, and Kenneth A. Dodge, "In Search of Dynamic Complementarities between Early and Later Education: Evidence from North Carolina's Pre-K and K–12 School Funding Reforms," *AEA Papers and Proceedings*, May 2024, *114*, 474–479.

Johnson, Rucker C. and C. Kirabo Jackson, "Reducing Inequality through Dynamic Complementarity: Evidence from Head Start and Public School Spending," *American Economic Journal: Economic Policy*, November 2019, *11* (4), 310–49.

Kinsler, Josh, "Teacher Complementarities in Test Score Production: Evidence from Primary School," *Journal of Labor Economics*, January 2016, *34* (1), 29–61. Publisher: The University of Chicago Press.

Kline, Patrick and Christopher R. Walters, "Evaluating Public Programs with Close Substitutes: The Case of Head Start," *The Quarterly Journal of Economics*, November 2016, *131* (4), 1795–1848.

Londoño-Vélez, Juliana, Catherine Rodríguez, and Fabio Sánchez, "Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia," *American Economic Journal: Economic Policy*, May 2020, *12* (2), 193–227.

_ , _ , _ , and Luis Esteban Álvarez Arango, "Targeting Social Assistance: The Evolution of College Financial Aid in Colombia," *AEA Papers and Proceedings*, May 2025, *115*, 340–44.

Lundborg, Petter, Dan-Olof Rooth, and Jesper Alex-Petersen, "Long-Term Effects of Childhood Nutrition: Evidence from a School Lunch Reform," *The Review of Economic Studies*, March 2022, *89* (2), 876–908.

Maluccio, John A., John Hoddinott, Jere R. Behrman, Reynaldo Martorell, Agnes R. Quisumbing, and Aryeh D. Stein, "The Impact of Improving Nutrition During Early Childhood on Education among Guatemalan Adults," *The Economic Journal*, April 2009, *119* (537), 734–763.

Mbiti, Isaac, Karthik Muralidharan, Mauricio Romero, Youdi Schipper, Constantine Manda, and Rakesh Rajani, "Inputs, Incentives, and Complementarities in Education: Experimental Evidence from Tanzania*," *The Quarterly Journal of Economics*, August 2019, *134* (3), 1627–1673.

McEwan, Patrick J., "The impact of Chile's school feeding program on education outcomes," *Economics of Education Review*, February 2013, *32*, 122–139.

Ministerio de Educación Nacional, "Plan Sectorial 2002–2006: La Revolución Educativa," 2002. Accessed: 2025-10-12.

_ , "13. Ficha - Tasa de cobertura neta," Technical Report, Sistema de Información Nacional de Educación Básica y Media (SINEB), Ministerio de Educación Nacional, Bogotá, Colombia 2025. Accessed: 2025-10-12.

Montpetit, Sébastien, Pierre-Loup Beauregard, and Luisa Carrer, "A Welfare Analysis of Universal Childcare: Lessons From a Canadian Reform," CHILD Working Paper Series 128, Centre for Household, Income, Labour and Demographic Economics (CHILD), Collegio Carlo Alberto July 2025.

Muslimova, Dilnoza, Hans Van Kippersluis, Cornelius A. Rietveld, Stephanie Von Hinke, and S. Fleur W. Meddens, "Gene-Environment Complementarity in Educational Attainment," *Journal of Labor Economics*, November 2024. Publisher: The University of Chicago Press.

Rossin-Slater, Maya and Miriam Wüst, "What Is the Added Value of Preschool for Poor Children? Long-Term and Intergenerational Impacts and Interactions with an Infant Health Intervention," *American Economic Journal: Applied Economics*, July 2020, *12* (3), 255–86.

Roth, Jonathan, "Interpreting event-studies from recent difference-in-differences methods," *arXiv preprint arXiv:2401.12309*, 2024.

Sun, Liyang and Sarah Abraham, "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects," *Journal of Econometrics*, December 2021, *225* (2), 175–199.

Wooldridge, Jeffrey M, "Two-way fixed effects, the two-way mundlak regression, and difference-in-differences estimators: JM Wooldridre," *Empirical Economics*, 2025, pp. 1–43.

APPENDIX

A Preschool **ii**

A.1 Additional Tables ii

A.2 Additional Figures x

B School Feeding **xviii**

B.1 Additional Tables xviii

B.2 Additional Figures xxiii

C Complementarities **xxvii**

C.1 Additional Tables xxvii

C.2 Additional Figures xxxiv

D Social Welfare **xliii**

A Preschool

A.1 Additional Tables

Table A1. Characteristics of Preschools, by Opening Period (Existing versus New, 2006–2015)

	Preschools		
	All	Existing	New
Rural schools (%)	0.74	0.70	0.81
Full-day schools (%)	0.26	0.24	0.28
Size and grades offered (2005 or at opening)			
Average preschool size	18.09	23.96	9.47
Schools with grades -1 and -2 (%)	0.04	0.05	0.02
Schools with primary grades (%)	0.81	0.77	0.88
Schools with primary and secondary grades (%)	0.17	0.21	0.11
Size and grades offered (2005-2015)			
Average preschool size	16.60	21.72	9.09
Schools with grades -1 and -2 (%)	0.02	0.03	0.01
Schools with primary grades (%)	0.81	0.76	0.88
Schools with primary and secondary grades (%)	0.18	0.23	0.11
Number of preschools	43,548	25,893	17,655

Notes: Authors' calculations using data from SIMAT. "All" includes preschools operating at any point between 2005 and 2015. "Existing" refers to preschools that opened before or during 2005 (the baseline year). "New" includes preschools that opened between 2006 and 2015 (the period of observed preschool expansion). Grades -1 and -2 correspond to "Jardín" and "Pre-jardín", the two years preceding transition to first grade.

Table A2. Descriptive Statistics by Sample and Preschool Enrollment

	All Students	Sample	Preschool	No Preschool
Panel A: Socioeconomic Characteristics				
Age at 1st grade	6.77	6.77	6.74	6.83
Female	0.49	0.48	0.49	0.48
0, 1	0.65	0.71	0.67	0.78
2, 3 y 4	0.35	0.29	0.33	0.22
Urban	0.64	0.61	0.67	0.51
Panel B: Academic Trajectories and Performance				
School dropout	0.47	0.48	0.41	0.61
Primary Completion (5th grade)	0.76	0.75	0.82	0.62
9th grade	0.56	0.55	0.62	0.42
Secondary Completion (11th grade)	0.49	0.48	0.55	0.36
Enrolled in Higher Education	0.14	0.13	0.15	0.10
Standardized values of Math	0	-0.04	-0.02	-0.11
Standardized values of Reading	0	-0.05	-0.03	-0.10
Received School Meals	0.49	0.48	0.54	0.38
Number of observations	3,298,805	1,704,991	1,092,755	612,236
Number of municipalities	952	483	483	483

Notes: "All Students" refers to the unrestricted sample of first-grade entrants from 2006-2011. Column "Sample" restricts to students who were above 2 kilometers from the nearest school in 2005. The last two columns restrict to students who enrolled and did not enroll in preschool.

Table A3. Predictors of Exposure to Preschool

	Probability Ever Treated		Covariate on Rollout
	All	In sample	(Among Treated)
Variable	(1)	(2)	(3)
Total Population (in millions)	-0.087	-0.682	0.001
	(0.156)	(1.022)	(0.001)
Area (per 10,000 squared km)	-0.056*	-0.062	0.005*
	(0.031)	(0.045)	(0.003)
Distance to Capital (per 100 km)	0.008	-0.014	0.056
	(0.021)	(0.033)	(0.034)
Log of Total GDP	0.003	0.022	0.005
	(0.010)	(0.016)	(0.042)
Ratio of Urban/Rural Population	-0.002	-0.005	-0.001
	(0.002)	(0.005)	(0.140)
Total Public Schools	-0.000	-0.000	1.438
	(0.001)	(0.001)	(1.379)
Preschool Students (per 10,000)	-0.062	0.074	0.000
	(0.185)	(0.504)	(0.001)
Primary Students (per 10,000)	0.020	0.046	0.009
	(0.032)	(0.158)	(0.008)
Homicide Rate (per 1,000)	0.000	0.000	10884.564
	(0.000)	(0.000)	(29279.513)
ICBF Beneficiaries (per 1,000)	-0.000	0.016	0.004
	(0.017)	(0.040)	(0.006)
Characteristics of Female Population (%), Census 2005			
Primary Education	0.001	0.000	-0.190
	(0.002)	(0.003)	(0.378)
Secondary Education	-0.008***	-0.015***	0.175
	(0.002)	(0.004)	(0.310)
Labor Force Participation	0.001	-0.002	0.050
	(0.002)	(0.003)	(0.273)
Uses formal health services	0.001	0.004	-0.115
	(0.002)	(0.003)	(0.505)
Uses informal health services	-0.002	-0.003	0.132
	(0.003)	(0.004)	(0.203)
Number of municipalities	952	483	101

Notes: Columns (1)-(2) show regression estimates for the probability of preschool exposure; the dependent variable is an indicator equal to one if a new preschool opened in a radius of 2 km from the municipality centroid between 2006-2015, zero otherwise. Column (3) present estimates for rollout balance among treated municipalities, using each covariate as dependent variable and the year of preschool arrival as explanatory variable. Characteristics of the female population are measured as percentages (0–100), computed from the 2005 Population Census (DANE). Use of health services omits the share of women reporting no service use. All regressions include department fixed-effects. Robust standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A4. Effect of Preschool Exposure on Preschool Attendance and Distance to Nearest School - Pooled Estimates

	(1)	(2)	(3)
Panel A: Probability of Attending Preschool			
	Overall	**Below Median**	**Above Median**
Exposed× Post	0.071***	0.058***	0.098***
	(0.015)	(0.019)	(0.027)
Number of observations	1,704,991	1,039,092	665,899
Number of clusters	483	236	247
Mean	0.639	0.681	0.573
Panel B: Distance to nearest preschool in km			
	Overall	**Below Median**	**Above Median**
Exposed× Post	-2.450***	-1.839***	-3.122***
	(0.243)	(0.145)	(0.489)
Number of observations	1,704,991	1,039,092	665,899
Number of clusters	483	236	247
Mean	5.234	3.136	8.522

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A5. Effect of Preschool Exposure at different radii on Preschool Attendance - Aggregate Estimates

Variable	0.5km	1km	1.5km	2km	2.5km	3km
Exposed×Post	0.105**	0.054*	0.089***	0.074***	0.060***	0.033*
	(0.043)	(0.030)	(0.022)	(0.018)	(0.020)	(0.020)
Number of observations	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A6. Effect of Preschool Exposure on Preschool Attendance by Preschool Type and Migration Status - Pooled Estimates

	(1)	(2)	(3)
Panel A: Type of Preschool			
	Overall	Private	Public
Exposed× Post	0.071***	0.001	0.070***
	(0.015)	(0.002)	(0.016)
Number of observations	1,704,991	1,704,991	1,704,991
Number of clusters	483	483	483
Mean	0.639	0.015	0.624
Panel B: Migration Status			
		Pr(Preschool Attendance)	
	Pr(Migrating)	Non-migrants	Migrants
Exposed× Post	0.008	0.082***	0.017
	(0.010)	(0.017)	(0.012)
Number of observations	1,704,991	1,428,767	276,224
Number of clusters	483	483	483
Mean	0.159	0.644	0.613

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A7. Effect of Preschool Exposure on Preschool Attendance - Alternative Estimators

	TWFE	Sun and Abraham	Stacked
Exposed×Post	0.071***	0.065***	0.073***
	(0.015)	(0.017)	(0.015)
Number of observations	1,704,991	1,704,991	15,693,124

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A8. Effect of Preschool Exposure on Academic Outcomes, Preschool Openings in Existing Primary Schools

	Pr(Preschool)	Probability of reaching grade 5th	9th	11th	Pr(Ever Drop-out)	Pr(Higher Education)
Exposed×Post	0.082***	0.032**	0.025**	0.022***	-0.023**	0.012**
	(0.017)	(0.015)	(0.010)	(0.009)	(0.011)	(0.006)
Number of observations	1,681,413	1,681,413	1,681,413	1,681,413	1,681,413	1,681,413
	Pr(Reading Score above quartile)			Pr(Math Score above quartile)		
	Bottom	Median	Top	Bottom	Median	Top
Exposed×Post	0.017***	0.008*	0.002	0.013**	0.005	0.000
	(0.006)	(0.004)	(0.003)	(0.006)	(0.004)	(0.003)
Number of observations	1,681,413	1,681,413	1,681,413	1,681,413	1,681,413	1,681,413

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Subsample of municipalities where preschools opened within existing primary schools (86% of all treated municipalities) and never-treated municipalities; treated municipalities with standalone preschool openings or preschool openings preceding primary school openings are excluded (14% of all treated municipalities). Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A9. Effect of Preschool Exposure on Academic Progression - Alternative Estimators

	Probability of reaching grade			Pr(Ever	Pr(Higher	Pr(Took
	5th	9th	11th	Drop-out)	Education)	Test)
			TWFE			
Exposed×Post	0.029**	0.021**	0.020***	-0.019*	0.013**	0.020***
	(0.013)	(0.009)	(0.008)	(0.010)	(0.005)	(0.008)
Number of observations	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991
			Sun and Abraham (2021)			
Exposed×Post	0.025**	0.017**	0.017**	-0.016**	0.011**	0.017**
	(0.012)	(0.008)	(0.007)	(0.008)	(0.005)	(0.007)
Number of observations	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991
			Stacked			
Exposed×Post	0.032**	0.024**	0.022***	-0.022**	0.014***	0.021***
	(0.015)	(0.010)	(0.008)	(0.005)	(0.005)	(0.008)
Number of observations	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

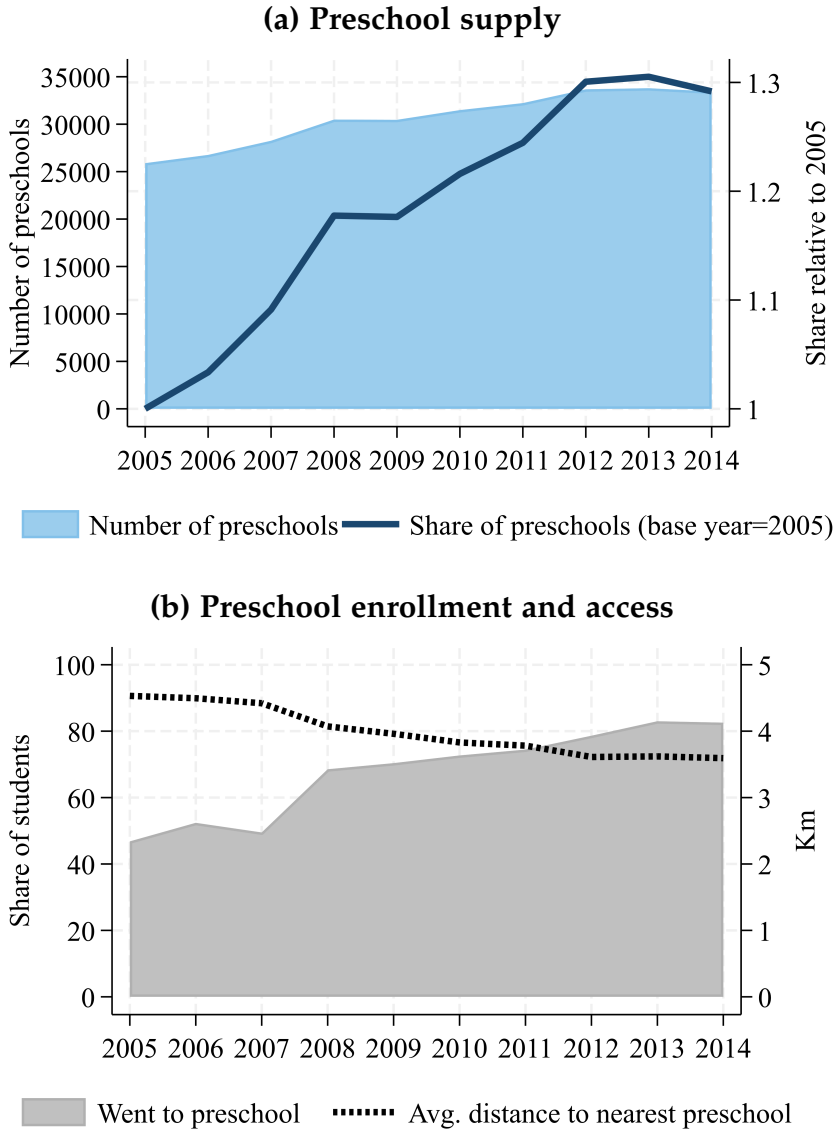
Table A10. Effect of Preschool Exposure on Academic Performance - Alternative Estimators

	Parametric Scores		**Pr(Reading Score above quartile)**			**Pr(Math Score above quartile)**		
	Math	**Reading**	**Bottom**	**Median**	**Top**	**Bottom**	**Median**	**Top**
				TWFE				
Exposed×Post	0.030**	0.026**	0.015***	0.007**	0.001	0.012**	0.004	0.000
	(0.012)	(0.012)	(0.005)	(0.004)	(0.003)	(0.005)	(0.004)	(0.002)
Number of observations	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991
				Sun and Abraham (2021)				
Exposed×Post	0.024**	0.020*	0.012**	0.005	0.001	0.008	0.003	0.001
	(0.012)	(0.012)	(0.005)	(0.004)	(0.002)	(0.005)	(0.004)	(0.002)
Number of observations	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991	1,704,991
				Stacked				
Exposed×Post	0.033***	0.028**	0.016***	0.008**	0.002	0.013***	0.005	0.000
	(0.012)	(0.012)	(0.005)	(0.003)	(0.002)	(0.005)	(0.003)	(0.002)
N	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124	15,693,124

Notes: Each column reports the effect of preschool exposure on the relevant outcome. Clustered standard errors at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

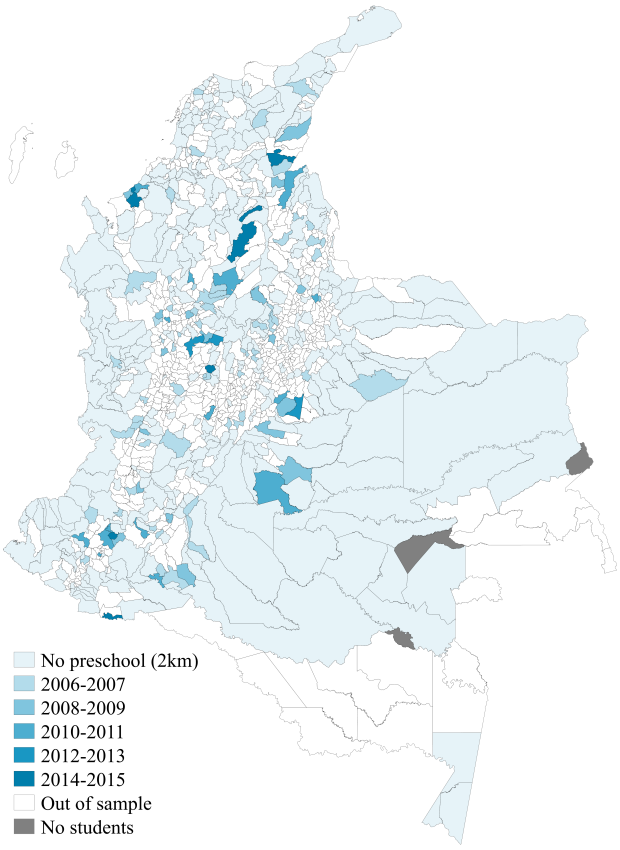
A.2 Additional Figures

Figure A1. Evolution of Preschool Supply and Enrollment (2005–2014)



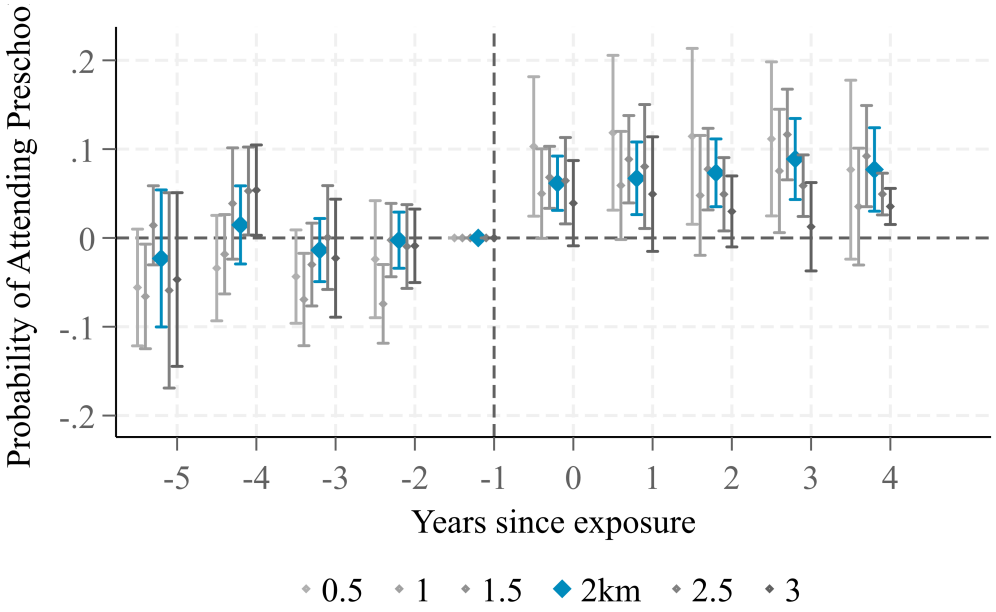
Note: Authors' calculations using data from SIMAT. The share of students who attended preschool is calculated over new entrants to first grade in the following year.

Figure A2. Availability of preschool over space and time



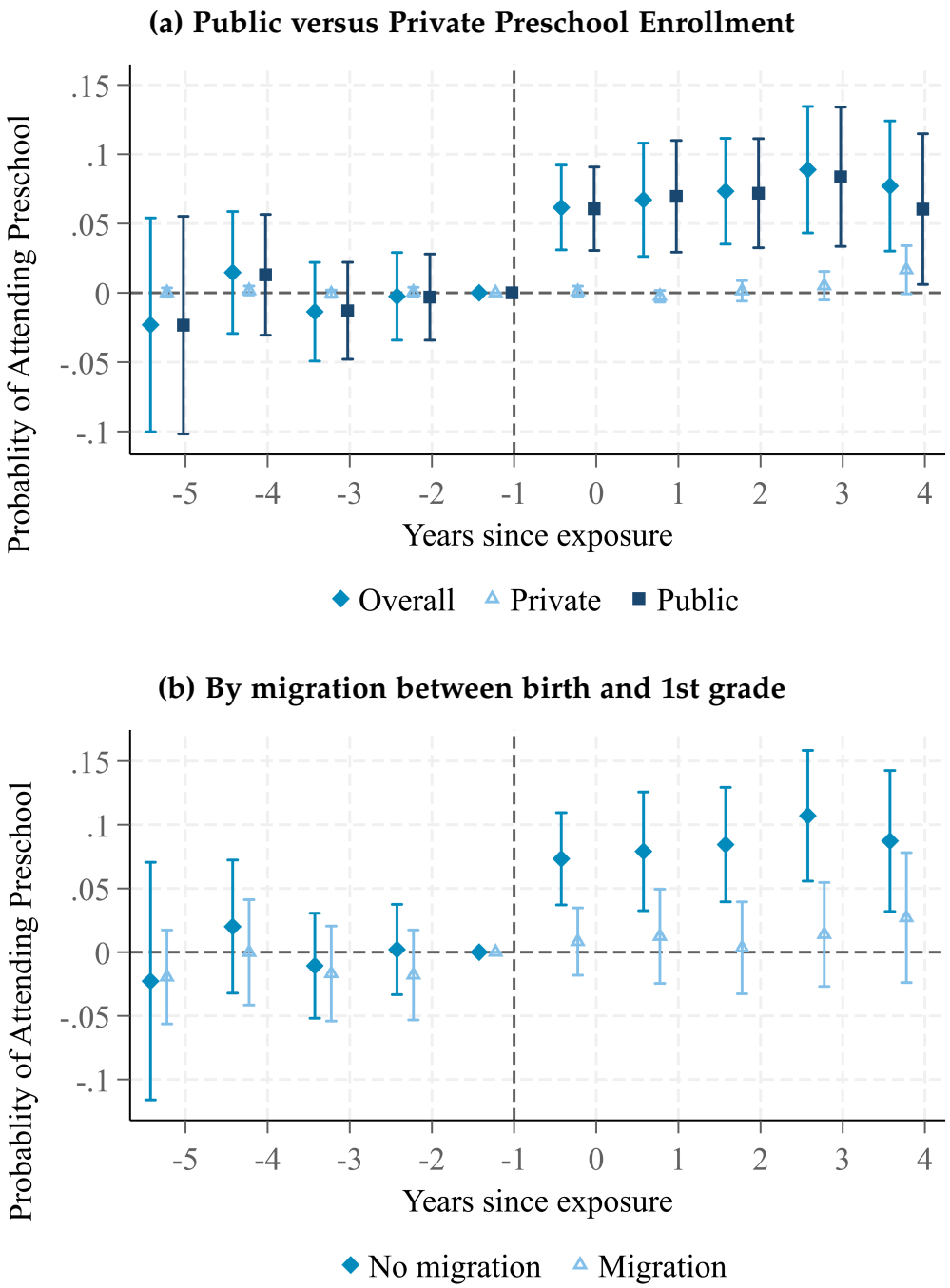
Note: The map displays the geographical distribution of preschool openings, categorized by timing. "Out of sample" refers to students who were below 2 kilometers from the nearest preschool at baseline. "No students" refers to municipalities with no students entering first grade between 2006–2011 in SIMAT.

Figure A3. Effect of Preschool Exposure at Different Radii on the Probability of Attending Preschool



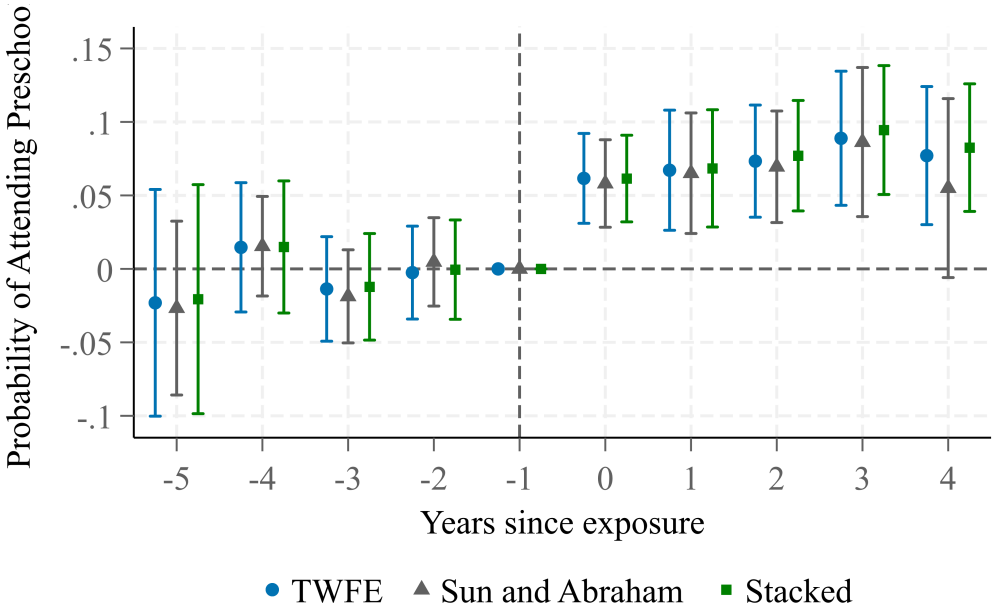
Note: The figure presents event-study estimates by different radii. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A4. Effect of Preschool Exposure on the Probability of Attending Preschool, by Preschool Type and Migration Status



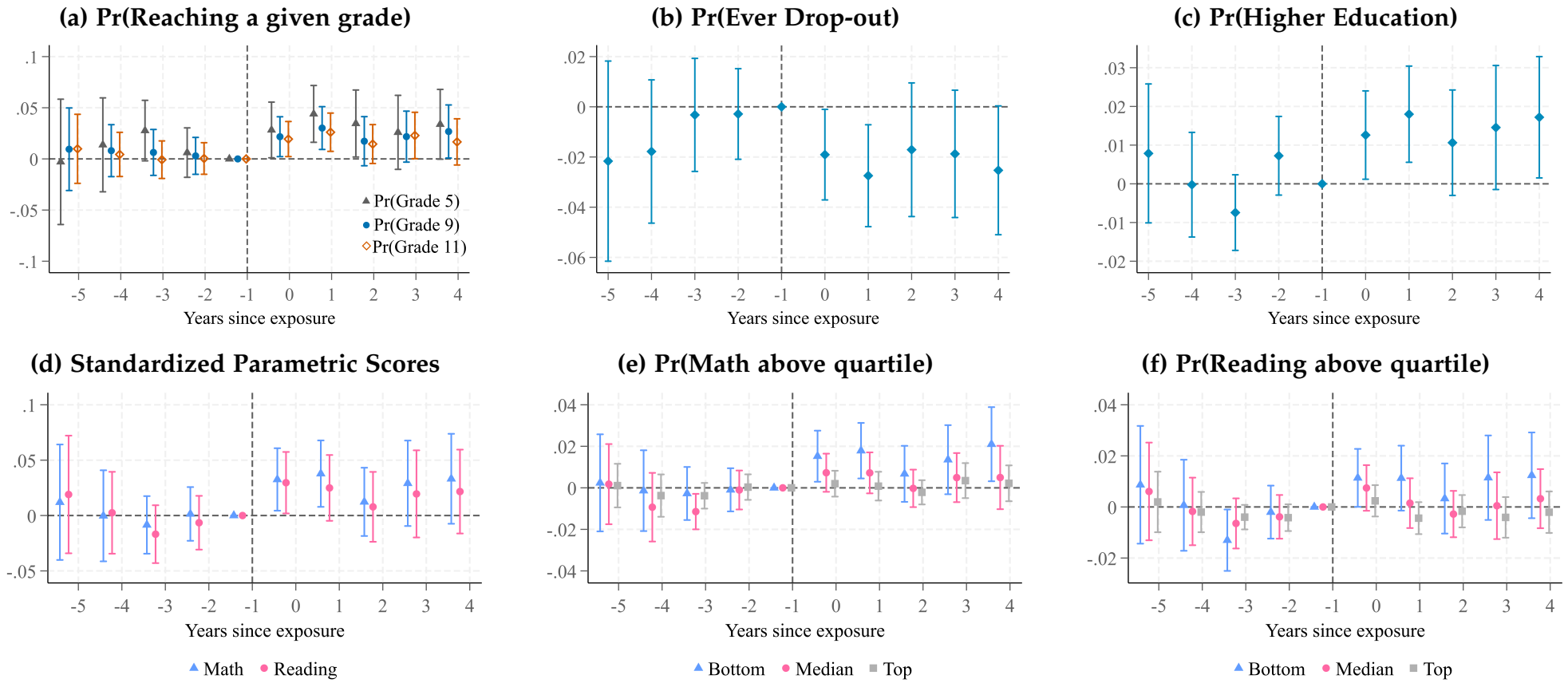
Note: The figure presents dynamic difference-in-difference estimates by type of preschool (public vs. private, in Panel (a)), and by migration defined as reporting a different municipality of residence at 1st grade relative to the municipality of birth (Panel (b)). Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A5. Effect of Preschool Exposure on the Probability of Attending Preschool, by Alternative Estimators



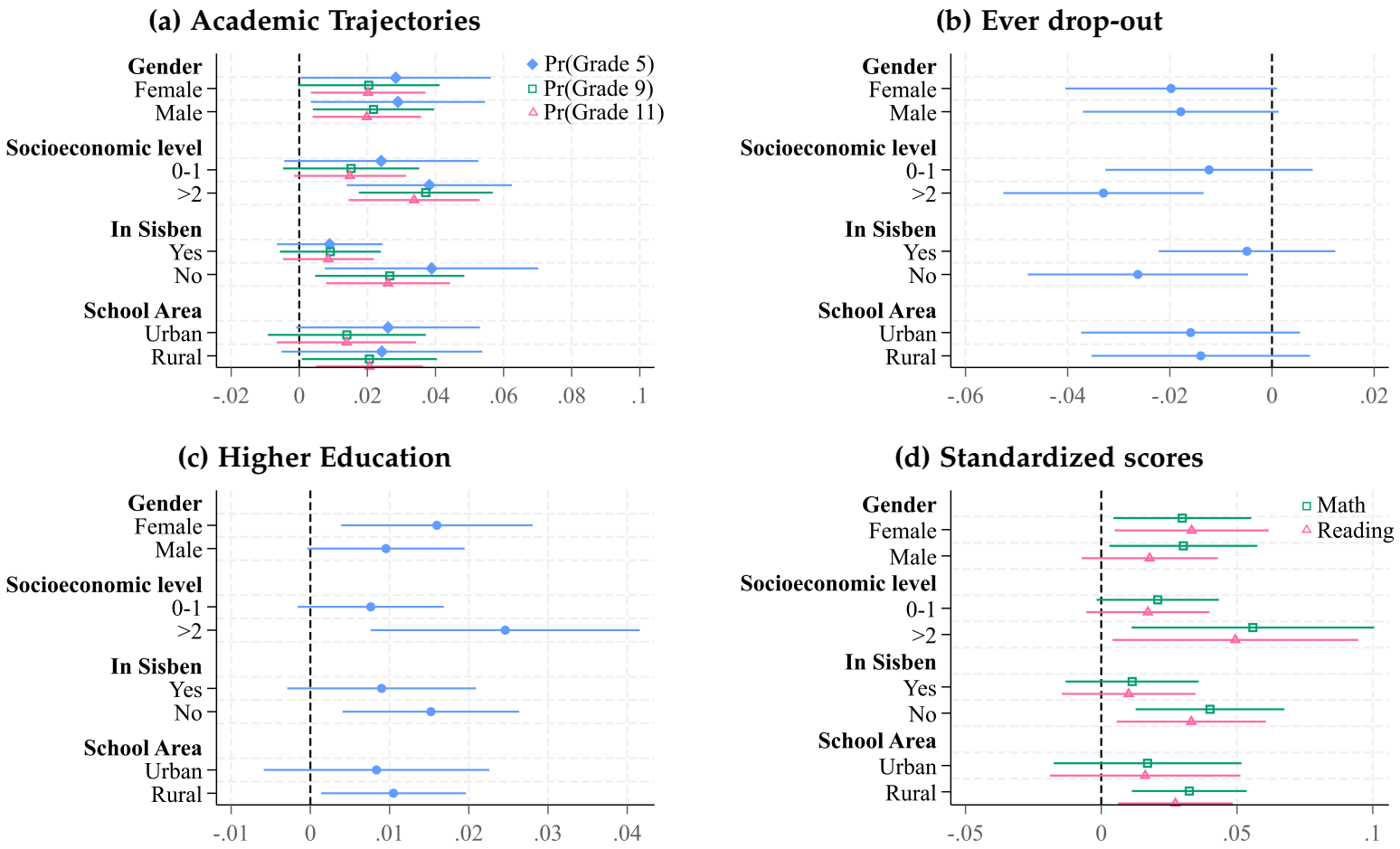
Note: The figure presents dynamic difference-in-difference estimates using TWFE and alternative estimators. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A6. Event-study estimates of Preschool Exposure on Academic Outcomes



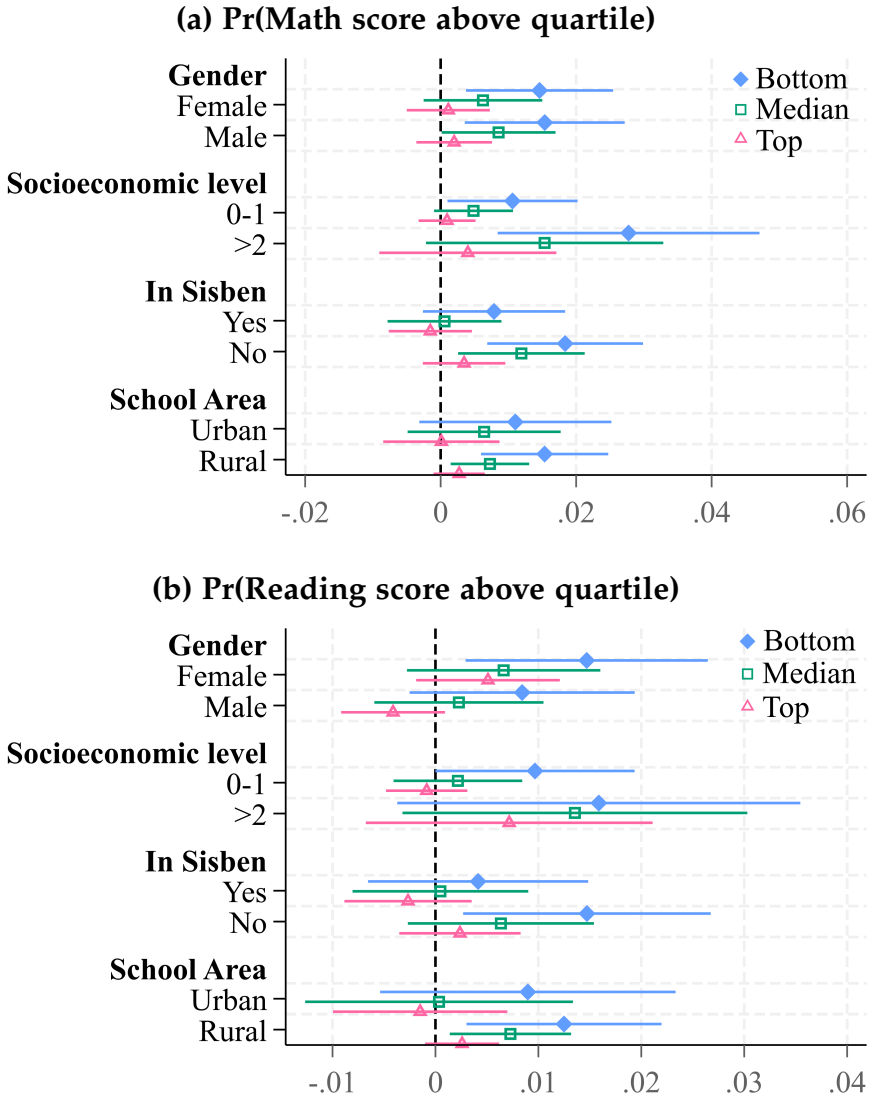
Note: The figures present event-study estimates for the relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A7. Heterogeneous Effects of Preschool Exposure on Academic Trajectories and Parametric Test Scores, by Student Characteristics



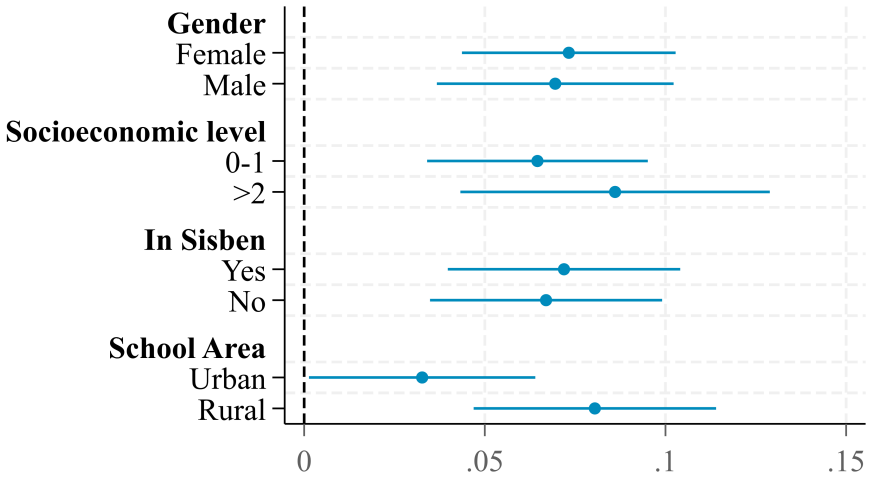
Notes: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A8. Heterogeneous Effects of Preschool Exposure on Academic Performance, by Student Characteristics



Notes: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure A9. Heterogeneous Effects of Preschool Exposure on Preschool Enrollment, by Student Characteristics



Note: Note: The figures present event-study estimates for the relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level in panel (a) and at the school level in panel (b).

B School Feeding

B.1 Additional Tables

Table B1. Descriptive Statistics by School Meals Participation

	School Meals	No School Meals
Panel A: Socioeconomic Characteristics		
Age at 1st grade	6.74	6.80
Female	0.48	0.49
0, 1	0.72	0.70
2, 3 y 4	0.28	0.30
Urban	0.61	0.62
Panel B: Academic Trajectories and Performance		
School dropout	0.30	0.65
Primary completion (5th grade)	0.96	0.55
9th grade	0.74	0.37
Secondary completion (11th grade)	0.63	0.34
Enrolled in Higher education	0.14	0.13
Standardized values of Math	0.19	-0.18
Standardized values of Reading	0.19	-0.17
Number of observations	820,535	884,456
Number of schools	29,435	29,397

Notes: Summary statistics, by school meal participation.

Table B2. Predictors of Exposure to School Feeding

Variable (2011)	(1) Year	(2) 2012-2013	(3) 2014	(4) 2015	(5) 2016	(6) 2017-2019
Full-time school	0.147***	-0.011*	-0.082***	0.033***	0.070***	-0.011**
	(0.020)	(0.006)	(0.006)	(0.007)	(0.006)	(0.005)
Rural school	0.893***	-0.135***	-0.161***	0.063***	0.098***	0.132***
	(0.018)	(0.006)	(0.007)	(0.006)	(0.005)	(0.004)
Share of preschool students	1.695*	-0.138	-0.169	0.197	-0.211	0.253
	(0.874)	(0.208)	(0.272)	(0.285)	(0.266)	(0.271)
Share of primary students	0.408***	-0.074***	0.029	-0.082***	0.069***	0.051**
	(0.077)	(0.020)	(0.021)	(0.023)	(0.022)	(0.020)
Share of secondary students	-0.435***	0.050**	0.135***	-0.083***	-0.030	-0.074***
	(0.074)	(0.020)	(0.021)	(0.023)	(0.021)	(0.018)
Share of low income students	0.287**	-0.003	-0.109**	-0.073*	0.210***	-0.017
	(0.121)	(0.040)	(0.045)	(0.040)	(0.035)	(0.022)
Number of schools	32,270	32,270	32,270	32,270	32,270	32,270
Mean	2013.9	0.242	0.200	0.245	0.184	0.126

Notes: Regression estimates for the probability of school feeding exposure. The outcome variable in column (1) is the year when the school feeding program first arrived. Columns (2)-(6) present estimates for the probability of receiving school meals in the corresponding years. All regressions include department fixed-effects. Robust standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table B3. Effect of School Feeding Exposure on Performance in the High School Exit Exam

	(1)	(2)	(3)
Panel A: Test Taking and Parametric Scores			
	Took Test	**Math**	**Reading**
Exposed to Feeding×Post	0.083***	0.138***	0.131***
	(0.005)	(0.011)	(0.011)
Number of observations	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270
Mean	0.302	-0.295	-0.289
Panel B: Probability of Math Score above quartile			
	Bottom	**Median**	**Top**
Exposed to Feeding×Post	0.058***	0.037***	0.019***
	(0.005)	(0.004)	(0.002)
Number of observations	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270
Mean	0.202	0.120	0.052
Panel C: Probability of Reading Score above quartile			
	Bottom	**Median**	**Top**
Exposed to Feeding×Post	0.055***	0.034***	0.016***
	(0.005)	(0.004)	(0.002)
Number of observations	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270
Mean	0.204	0.125	0.056

Notes: Each column reports the effect of school feeding exposure on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table B4. Effect of School Feeding Exposure on Receiving School Meals - Alternative Estimators

		Sun and Abraham	
	TWFE	Never-treated	Last-treated
Exposed to Feeding×Post	0.136***	0.137***	0.136***
	(0.005)	(0.005)	(0.005)
Number of observations	1,704,854	1,704,854	1,682,012

Notes: Each column reports the effect of school feeding exposure on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table B5. Effect of School Feeding Exposure on Academic Progression - Alternative Estimators

	Probability of reaching grade			Pr(Ever Drop-out)	Pr(Higher Education)	Pr(Took Test)
	5th	9th	11th			
			TWFE			
Exposed to Feeding×Post	0.079***	0.093***	0.081***	-0.087***	0.045***	0.083***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854
			Sun and Abraham: Never-treated			
Exposed to Feeding×Post	0.081***	0.095***	0.083***	-0.089***	0.047***	0.085***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854
			Sun and Abraham: Last-treated			
Exposed to Feeding×Post	0.080***	0.094***	0.082***	-0.088***	0.047***	0.085***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Number of observations	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012

Notes: Each column reports the effect of school feeding exposure on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

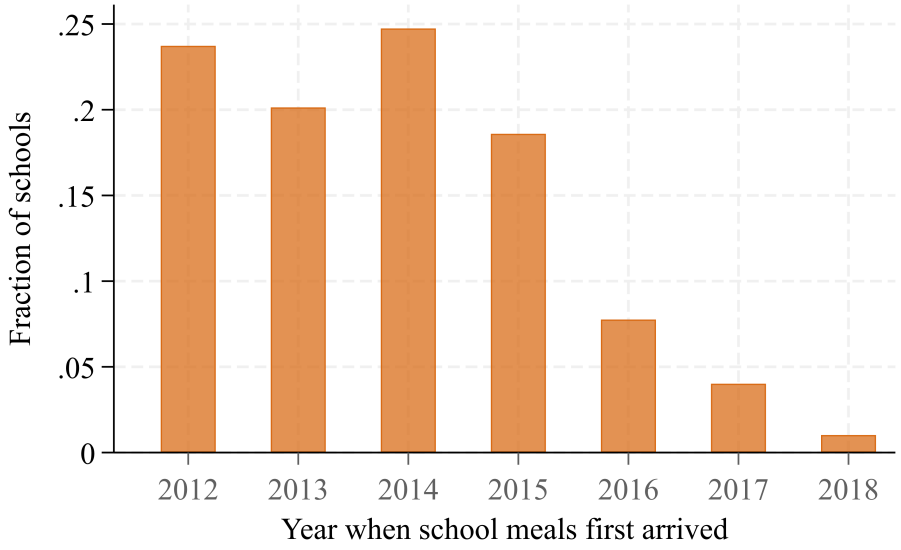
Table B6. Effect of School Feeding Exposure on Academic Performance - Alternative Estimators

	Parametric Scores		**Pr(Reading Score above quartile)**			**Pr(Math Score above quartile)**		
	Math	**Reading**	**Bottom**	**Median**	**Top**	**Bottom**	**Median**	**Top**
				TWFE				
Exposed to Feeding×Post	0.138***	0.131***	0.058***	0.037***	0.019***	0.055***	0.034***	0.016***
	(0.011)	(0.011)	(0.005)	(0.004)	(0.002)	(0.005)	(0.004)	(0.002)
Number of observations	1704854	1704854	1704854	1704854	1704854	1704854	1704854	1704854
				Sun and Abraham: Never-treated				
Exposed to Feeding×Post	0.142***	0.135***	0.060***	0.038***	0.020***	0.057***	0.036***	0.016***
	(0.011)	(0.011)	(0.005)	(0.004)	(0.002)	(0.005)	(0.004)	(0.002)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854
				Sun and Abraham: Last-treated				
Exposed to Feeding×Post	0.142***	0.135***	0.060***	0.039***	0.020***	0.057***	0.036***	0.017***
	(0.011)	(0.011)	(0.005)	(0.004)	(0.002)	(0.005)	(0.004)	(0.002)
Number of observations	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012	1,682,012

Notes: Each column reports the effect of school feeding exposure on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

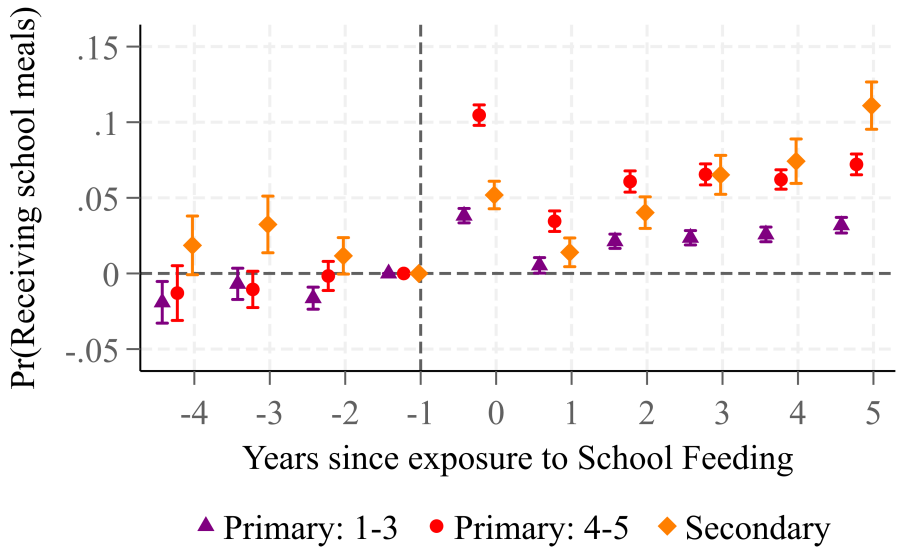
B.2 Additional Figures

Figure B1. Distribution of school meal program arrival across first-grade schools



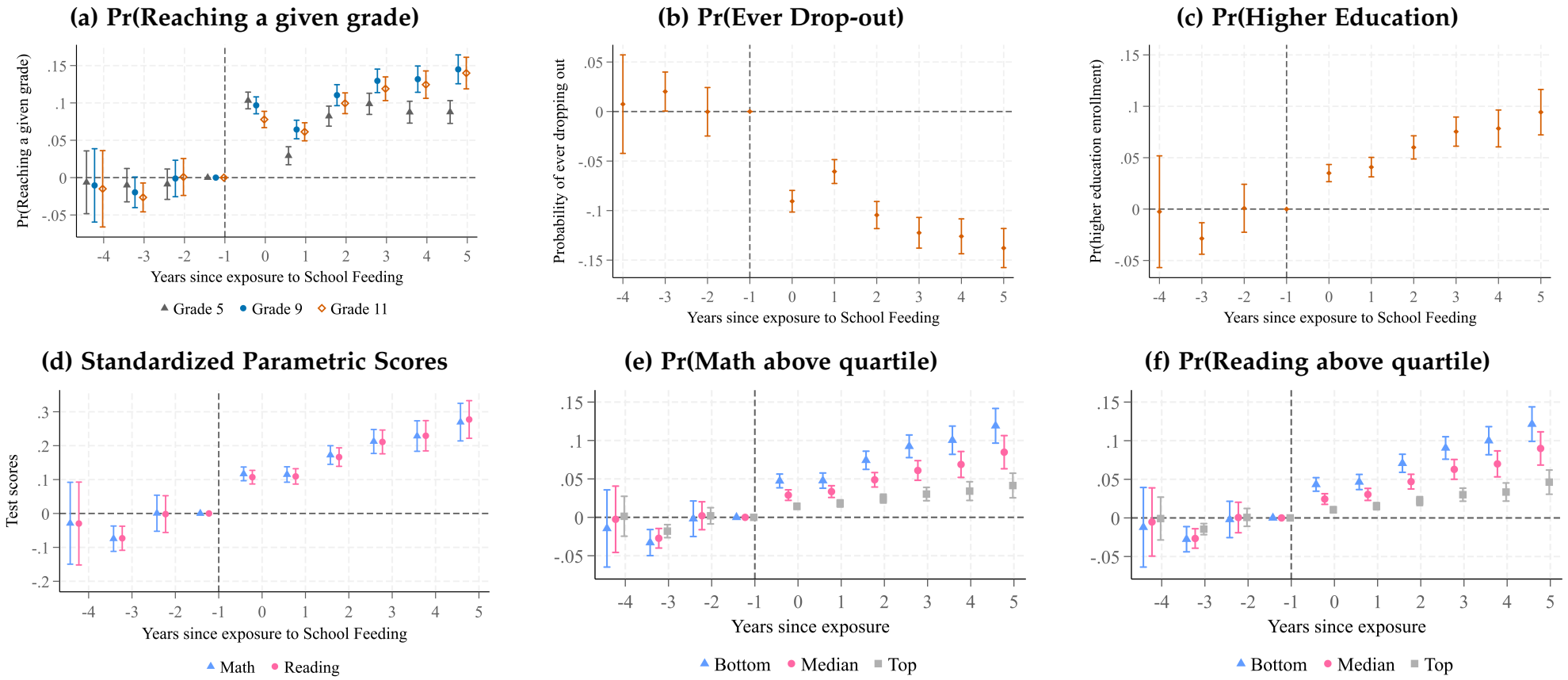
Notes: The figure presents the distribution of arrival of school meals across first-grade schools for students in our sample.

Figure B2. Effect of School Feeding Exposure on the Probability of Receiving School Meals in Primary and Secondary



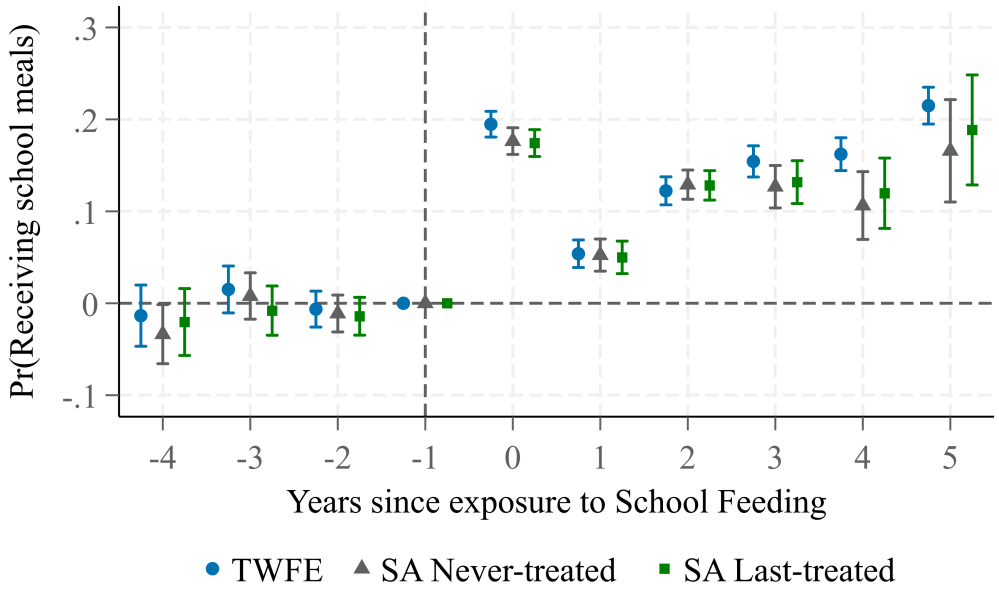
Notes: The figures present event-study estimates for the probability of receiving school meals in different grades. Confidence intervals at the 95% level, computed with clustered standard errors at the municipality level.

Figure B3. Event-study estimates of Exposure to School Feeding on Academic Outcomes



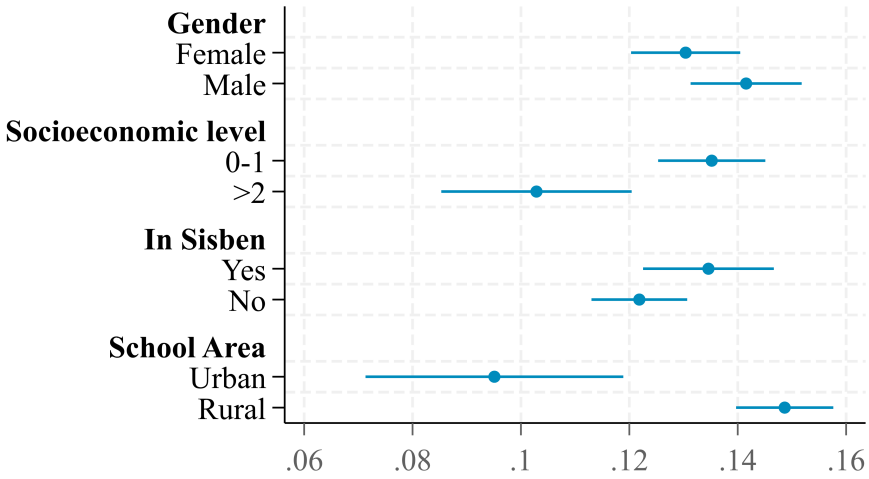
Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure B4. Effect of School Feeding Exposure on the Probability of Receiving School Meals, by Different Estimators



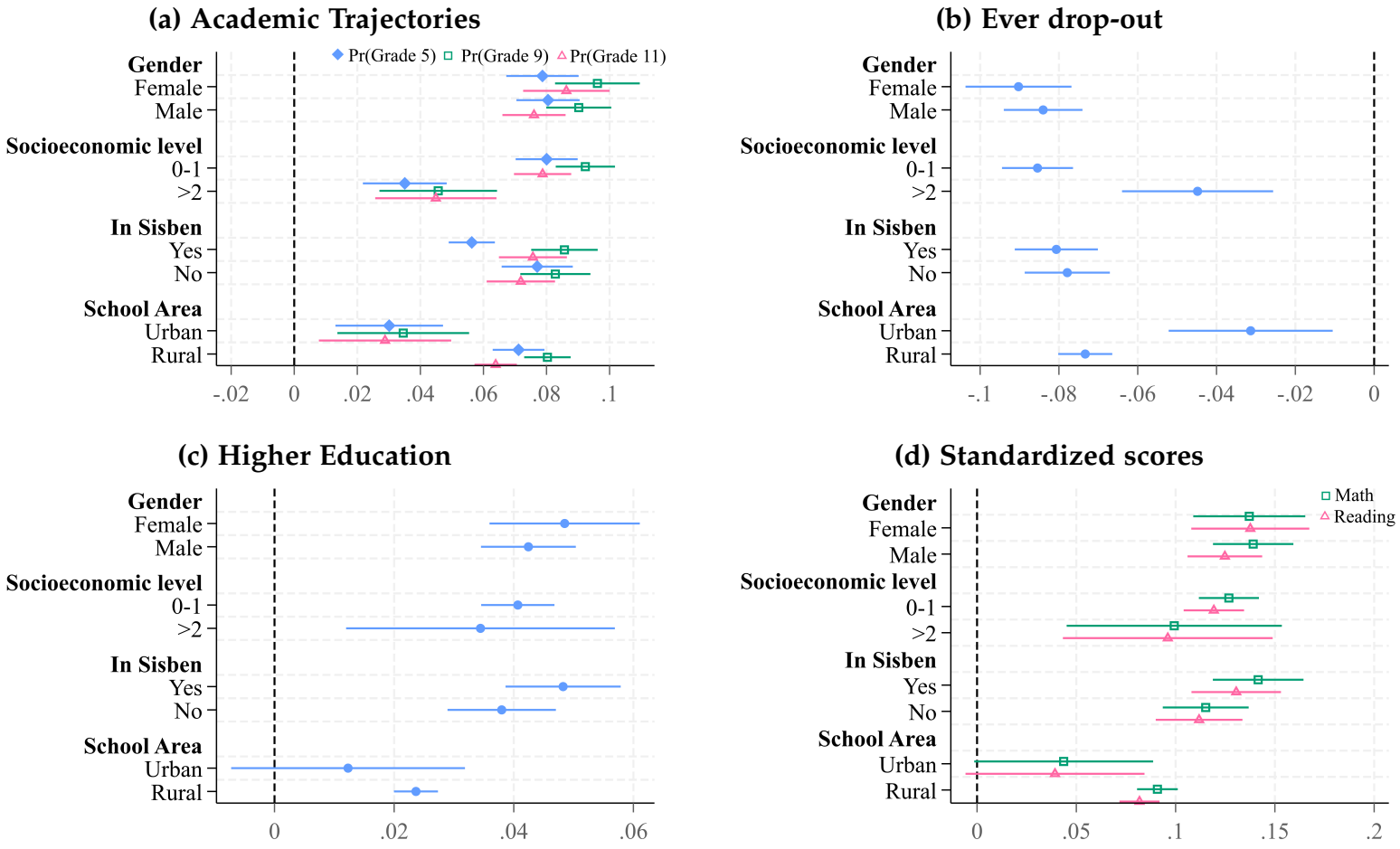
Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure B5. Heterogeneous Effects of School Feeding Exposure on Receiving School Meals, by Student Characteristics



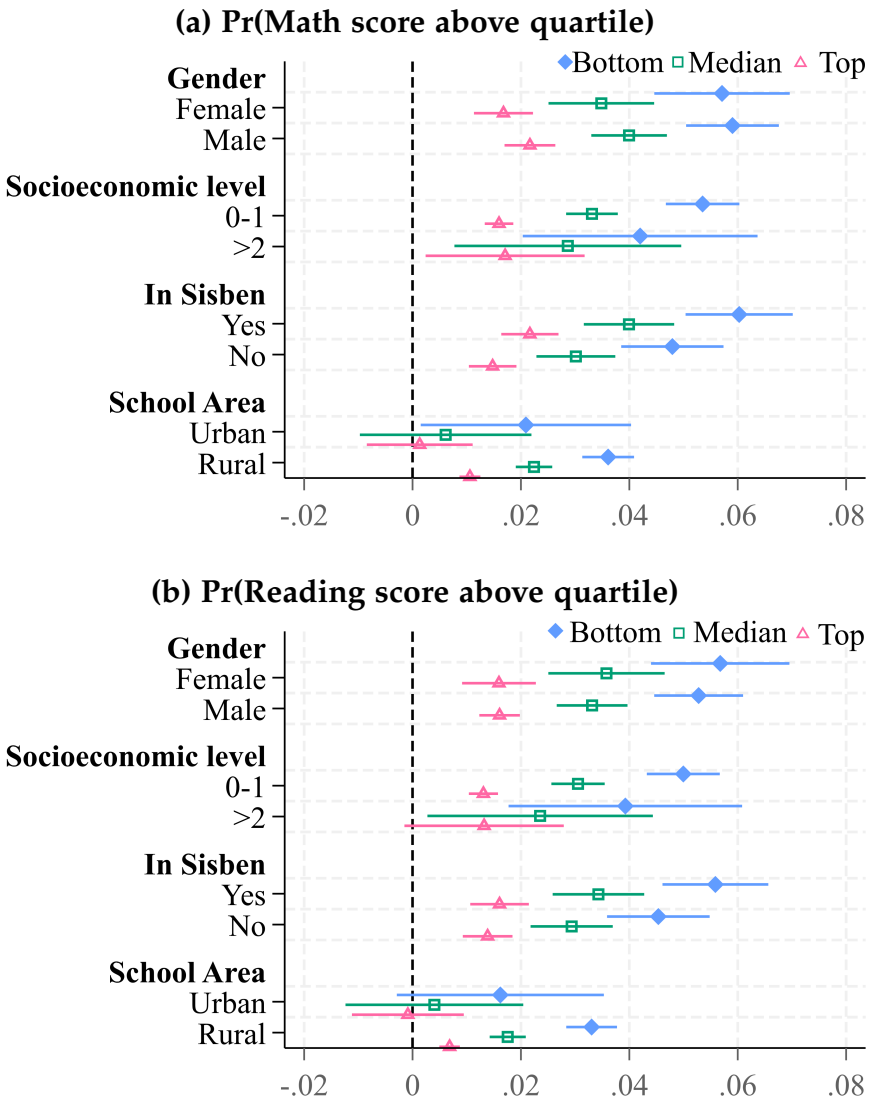
Notes: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure B6. Heterogeneous Effects of School Feeding Exposure on Academic Trajectories and Parametric Test Scores, by Student Characteristics



Notes: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure B7. Heterogeneous Effects of School Feeding Exposure on Academic Performance, by Student Characteristics



Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

C Complementarities

C.1 Additional Tables

Table C1. School Feeding Program Complier composition, by Preschool Exposure

	SF Complier		With Preschool		Without Preschool	
	Avg.	**ΔFull Sample**	**Avg.**	**ΔSF Complier**	**Avg.**	**ΔSF Complier**
Preschool Age	5.770***	-0.002	5.807***	0.037	5.767***	-0.003
	(0.015)	(0.015)	(0.049)	(0.049)	(0.015)	(0.015)
SES=0-1	0.799***	0.092***	0.694***	-0.105**	0.807***	0.008
	(0.017)	(0.017)	(0.042)	(0.042)	(0.017)	(0.017)
Urban	0.796***	0.181***	0.872***	0.076	0.795***	-0.001
	(0.032)	(0.032)	(0.092)	(0.092)	(0.033)	(0.033)
Sisben Household	0.525***	0.121***	0.562***	0.037	0.520***	-0.005
	(0.012)	(0.012)	(0.032)	(0.032)	(0.012)	(0.012)
Household size	5.923***	0.265***	5.895***	-0.028	5.929***	0.005
	(0.077)	(0.077)	(0.250)	(0.250)	(0.080)	(0.080)
Mother's employment: Working	0.180***	-0.059***	0.199***	0.019	0.180***	0.000
	(0.011)	(0.011)	(0.035)	(0.035)	(0.012)	(0.012)
Mother's employment: Unemployed	0.018***	-0.006*	0.013	-0.006	0.019***	0.000
	(0.004)	(0.004)	(0.012)	(0.012)	(0.004)	(0.004)
Mother's employment: Housework	0.726***	0.052***	0.739***	0.012	0.724***	-0.002
	(0.014)	(0.014)	(0.042)	(0.042)	(0.014)	(0.014)
Mother's education: Primary	0.575***	0.034*	0.650***	0.075	0.565***	-0.010
	(0.019)	(0.019)	(0.054)	(0.054)	(0.020)	(0.020)

Notes: "Avg." reports the estimated mean of each baseline characteristic for the indicated group, obtained from a fixed effects instrumental variables regression based on Equation (7) that instruments school meals receipt F_i with $\mathbb{1}[\text{Exp}^{\text{sfp}}_{s,t_{\text{sfp}(i)}} \geq 0]$ and uses $F \times X$ as the dependent variable. "Δ Full Sample" reports the difference between the SFP complier and the average student. The "Δ SFP Complier" columns report the difference between the subgroup mean and the SFP complier mean. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table C2. Preschool Complier composition, by School Feeding Exposure

	Preschool Complier		**With School Feeding**		**Without School Feeding**	
	Avg.	**ΔFull Sample**	**Avg.**	**Δ Preschool Complier**	**Avg.**	**Δ Preschool Complier**
Preschool Age	5.857***	0.085	5.856***	-0.001	5.893***	0.037
	(0.062)	(0.062)	(0.068)	(0.068)	(0.165)	(0.165)
SES=0-1	0.807***	0.100*	0.805***	-0.002	0.960***	0.152
	(0.055)	(0.055)	(0.059)	(0.059)	(0.111)	(0.111)
Urban	0.074	-0.541***	0.069	-0.005	-0.021	-0.095
	(0.085)	(0.085)	(0.091)	(0.091)	(0.143)	(0.143)
Sisben Household	0.732***	0.328***	0.724***	-0.008	0.769***	0.037
	(0.052)	(0.052)	(0.056)	(0.056)	(0.131)	(0.131)
Household size	5.478***	-0.181	5.495***	0.017	4.533***	-0.945
	(0.293)	(0.293)	(0.328)	(0.328)	(0.840)	(0.840)
Mother's employment: Working	0.055	-0.184***	0.050	-0.005	0.073	0.019
	(0.047)	(0.047)	(0.052)	(0.052)	(0.117)	(0.117)
Mother's employment: Unemployed	-0.003	-0.028*	-0.000	0.003	-0.057	-0.053
	(0.015)	(0.015)	(0.017)	(0.017)	(0.044)	(0.044)
Mother's employment: Housework	0.978***	0.303***	0.989***	0.011	0.960***	-0.018
	(0.056)	(0.056)	(0.062)	(0.062)	(0.136)	(0.136)
Mother's education: Primary	0.974***	0.433***	0.991***	0.017	0.925***	-0.050
	(0.065)	(0.065)	(0.074)	(0.074)	(0.154)	(0.154)

Notes: "Avg." reports the estimated mean of each baseline characteristic for the indicated group, obtained from a fixed effects instrumental variables regression based on Equation (7) that instruments preschool enrollment D_i with $\mathbb{1}[\text{Exp}_{m(i),t(i)} \geq 0]$ and uses $D \times X$ as the dependent variable. "Δ Full Sample" reports the difference between the Preschool complier and the average student. The "Δ Preschool Complier" columns report the difference between the subgroup mean and the Preschool complier mean. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table C3. Policy Complementarity between Preschool and School Feeding Exposure on Treatments

	(1) **Both**	(2) **Only Meals**	(3) **Only Preschool**	(4) **Neither**	(5) **Total Preschool**
Preschool	0.000	0.003	0.047***	-0.051***	0.047***
	(0.012)	(0.010)	(0.014)	(0.014)	(0.015)
Feeding	0.121***	0.014***	-0.040***	-0.094***	0.081***
	(0.004)	(0.002)	(0.006)	(0.005)	(0.005)
Preschool×School Feeding	0.047***	-0.022**	-0.023*	-0.003	0.024*
	(0.012)	(0.010)	(0.014)	(0.014)	(0.014)
Total Preschool	0.048***	-0.018***	0.024***	-0.054***	0.072***
	(0.005)	(0.004)	(0.006)	(0.006)	(0.007)
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270	32,270
Mean	0.1706	0.1180	0.3417	0.3697	0.5123

Notes: Each column reports estimates from Equation (7) on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table C4. Policy Complementarity between Preschool and School Feeding Exposure on Academic Performance

	(1)	(2)	(3)	(4)
	Probability of scoring > quartile			
Panel A: Math	**Bottom**	**Median**	**Top**	**Score**
Preschool	0.004	0.000	-0.005	-0.003
	(0.010)	(0.007)	(0.005)	(0.021)
Feeding	0.058***	0.037***	0.019***	0.136***
	(0.005)	(0.004)	(0.002)	(0.011)
Preschool×School Feeding	0.009	0.006	0.006	0.031
	(0.010)	(0.007)	(0.005)	(0.021)
Total Preschool	0.013***	0.006**	0.001	0.027***
	(0.004)	(0.003)	(0.002)	(0.008)
Implied TOT	0.085	0.000	-0.106	-0.064
(Preschool without Feeding)	[0.669]	[0.984]	[0.367]	[0.864]
Implied TOT	0.181	0.083	0.014	0.375
(Preschool and Feeding)	[0.001]	[0.055]	[0.594]	[0.001]
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270
Mean	0.201	0.120	0.053	-0.294
	Probability of scoring > quartile			
Panel B: Reading	**Bottom**	**Median**	**Top**	**Score**
Preschool	0.001	-0.003	-0.010**	-0.012
	(0.010)	(0.008)	(0.004)	(0.021)
Feeding	0.054***	0.034***	0.015***	0.129***
	(0.005)	(0.004)	(0.002)	(0.011)
Preschool×School Feeding	0.009	0.006	0.011**	0.035*
	(0.010)	(0.007)	(0.004)	(0.021)
Total Preschool	0.010***	0.003	0.000	0.023***
	(0.004)	(0.003)	(0.002)	(0.008)
Implied TOT	0.021	-0.064	-0.213	-0.255
(Preschool without Feeding)	[0.941]	[0.723]	[0.070]	[0.548]
Implied TOT	0.139	0.042	0.000	0.319
(Preschool and Feeding)	[0.012]	[0.342]	[0.891]	[0.006]
Number of observations	1,704,854	1,704,854	1,704,854	1,704,854
Number of clusters	32,270	32,270	32,270	32,270
Mean	0.203	0.125	0.056	-0.288

Notes: Each column reports estimates from Equation (7) on the relevant outcome. Clustered standard errors at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01. p-values for implied TOTs in brackets, computed using bootstrap.

Table C5. Average characteristics by socioeconomic level and Sisben subsamples

	Level=0-1		Level=2-4	
	Sisben=No	Sisben=Yes	Sisben=No	Sisben=Yes
Age at 1st grade	6.85	6.74	6.74	6.69
Female	0.48	0.48	0.49	0.49
Urban	0.46	0.55	0.90	0.89
Preschool enrollment	0.52	0.70	0.65	0.79
School meals	0.39	0.59	0.40	0.52
Number of observations	633,574	571,492	224,417	275,508
Share	0.53	0.47	0.45	0.55

Notes: "Level" refers to the student socioeconomic level reported in SIMAT.

Table C6. Differences in household size and maternal characteristics, by socioeconomic level (Sisben subsample)

	Socioeconomic Level		
	0-1	2-4	Difference
Household size:			
< 4	0.333	0.449	0.116***
	(0.001)	(0.001)	(0.001)
5	0.207	0.219	0.012***
	(0.001)	(0.001)	(0.001)
$6 - 7$	0.257	0.207	-0.049***
	(0.001)	(0.001)	(0.001)
> 8	0.203	0.124	-0.079***
	(0.001)	(0.001)	(0.001)
Mother's employment status:			
Working	0.192	0.336	0.144***
	(0.001)	(0.001)	(0.001)
Unemployed	0.019	0.036	0.017***
	(0.000)	(0.000)	(0.000)
Housework	0.722	0.575	-0.147***
	(0.001)	(0.001)	(0.001)
Mother's education level:			
Primary	0.622	0.375	-0.247***
	(0.001)	(0.001)	(0.001)
Secondary	0.358	0.575	0.218***
	(0.001)	(0.001)	(0.001)
Higher Education	0.021	0.050	0.029***
	(0.000)	(0.000)	(0.000)
Number of observations	571,490	275,508	846,998

Notes: Subsample of students matched to Sisbén records. "Housework" refers to domestic work.

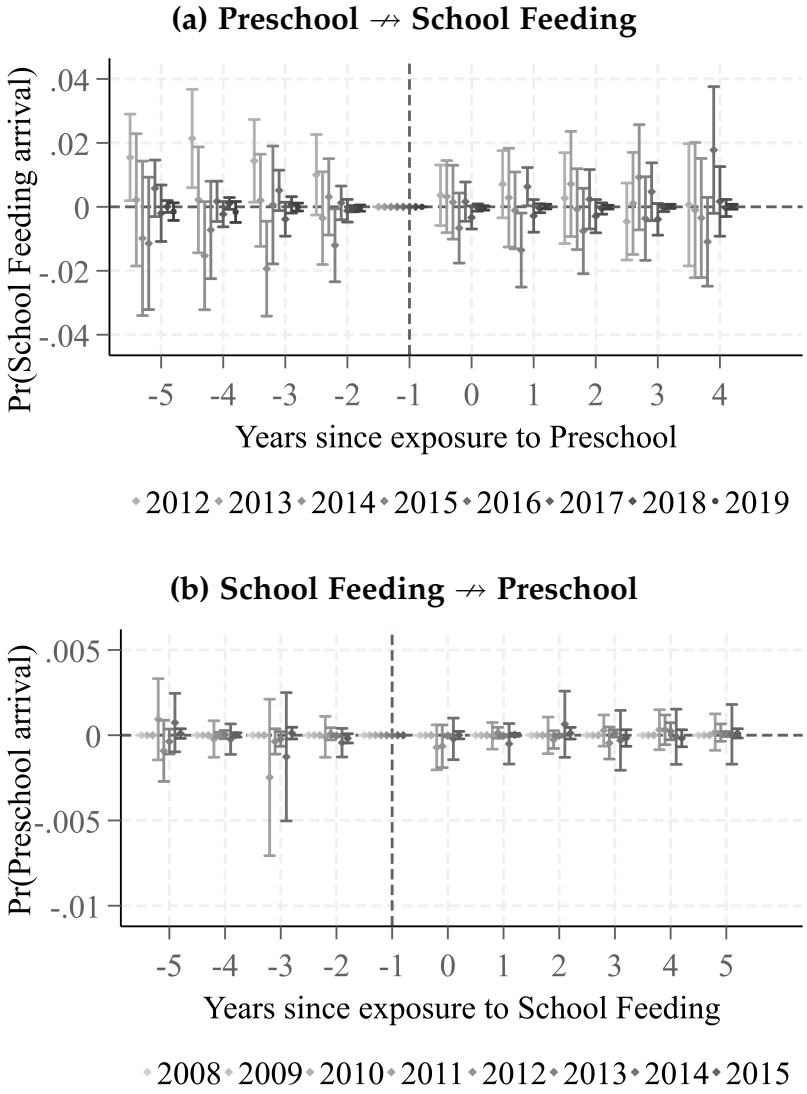
Table C7. Differences in paternal characteristics, by socioeconomic level (Sisben subsample)

	Socioeconomic Level		
	0-1	2-4	Difference
Father's employment status:			
Working	0.897	0.899	0.002**
	(0.000)	(0.001)	(0.001)
Unemployed	0.047	0.060	0.013***
	(0.000)	(0.001)	(0.001)
Housework	0.007	0.006	-0.001***
	(0.000)	(0.000)	(0.000)
Father's education level:			
Primary	0.689	0.451	-0.238***
	(0.001)	(0.001)	(0.001)
Secondary	0.293	0.502	0.209***
	(0.001)	(0.001)	(0.001)
Higher Education	0.018	0.047	0.030***
	(0.000)	(0.000)	(0.000)
Number of observations	424,114	192,120	616,234

Notes: Subsample of students matched to Sisbén records with information on father's employment and educational level. "Housework" refers to domestic work.

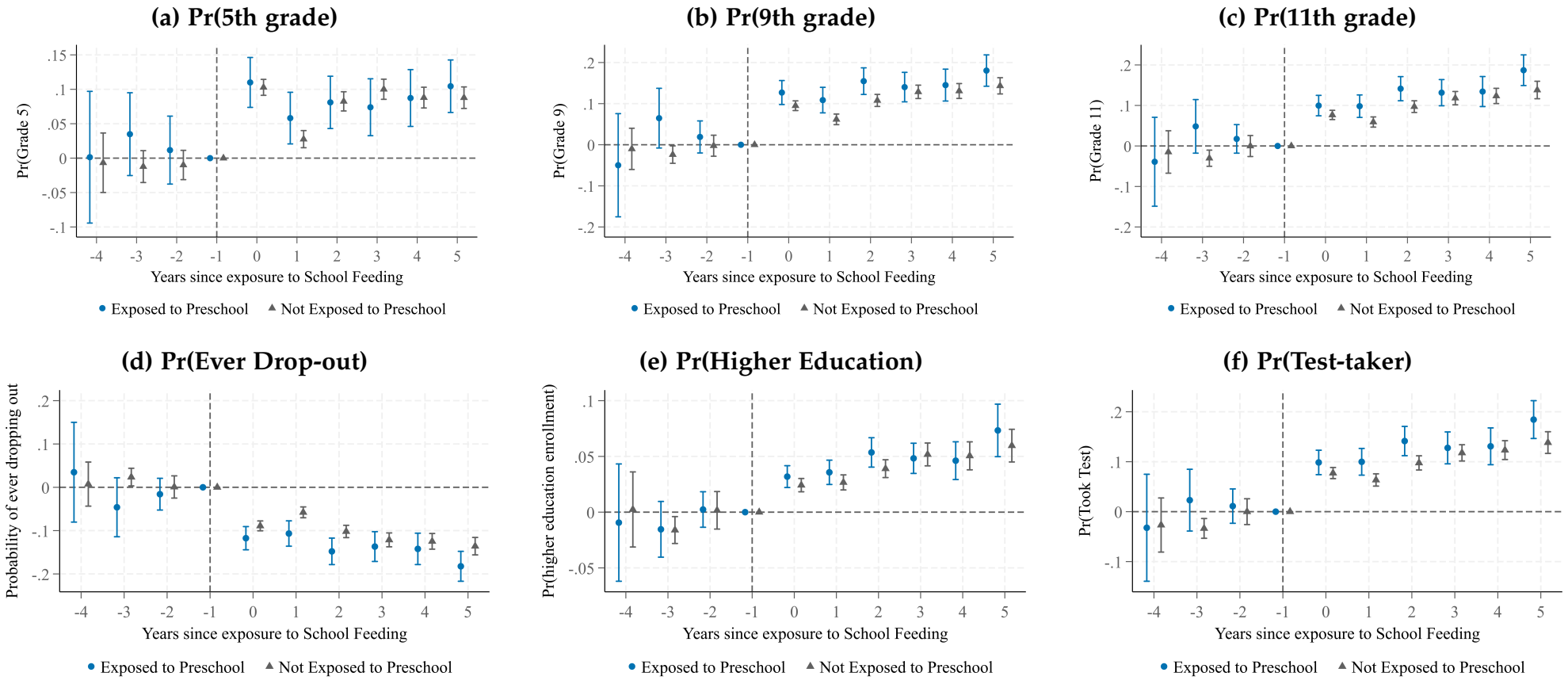
C.2 Additional Figures

Figure C1. Independence between Investments



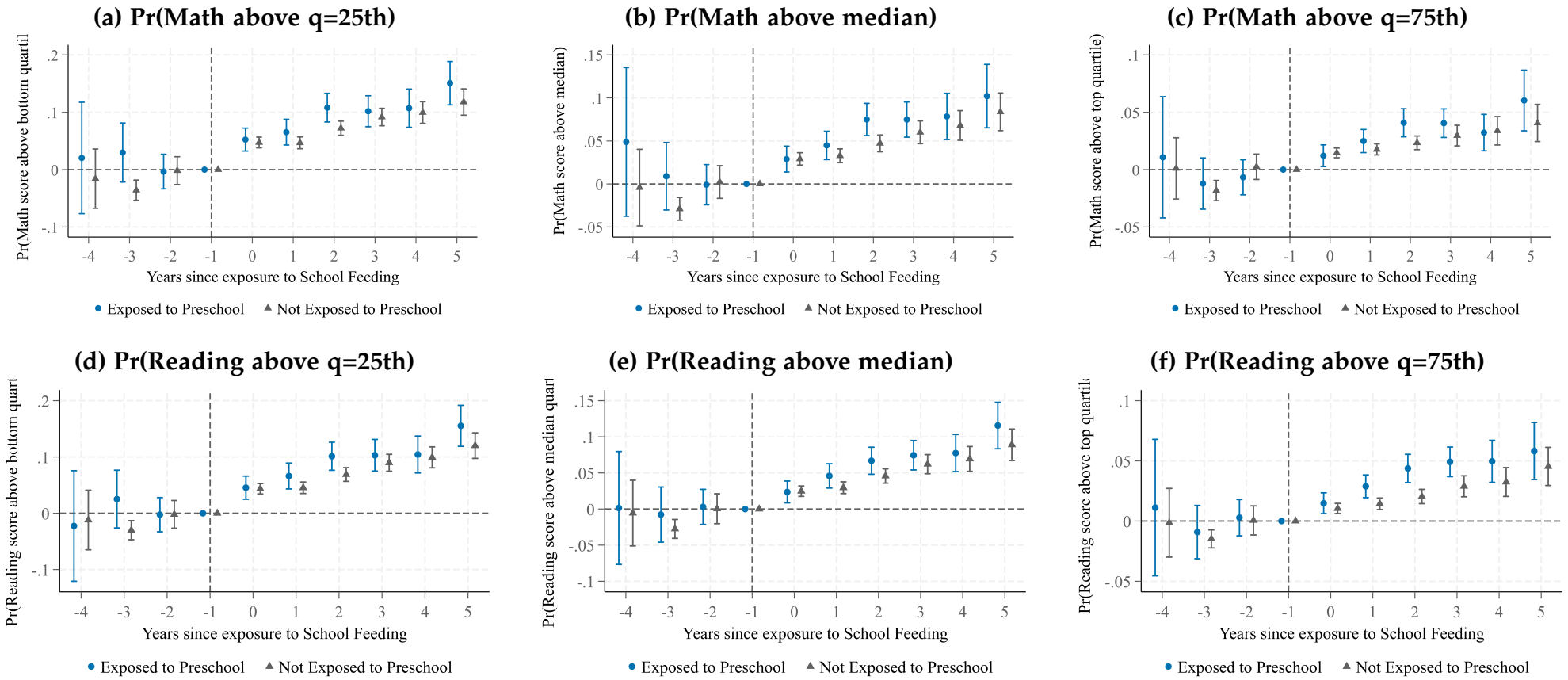
Note: The figure plots event-study coefficients estimating the probability of investment arrival as a function of exposure to the other intervention. Panel (a) shows coefficients from regressions of school feeding arrival (2012–2019) on a set of indicator variables for each year relative to the timing of preschool exposure. Panel (b) shows coefficients from regressions of preschool arrival (2008–2015) on a set of indicator variables for each year relative to the timing of school feeding exposure. Confidence intervals at the 95% level.

Figure C2. Effect of School Feeding Exposure on Academic Outcomes, by Exposure to Preschool



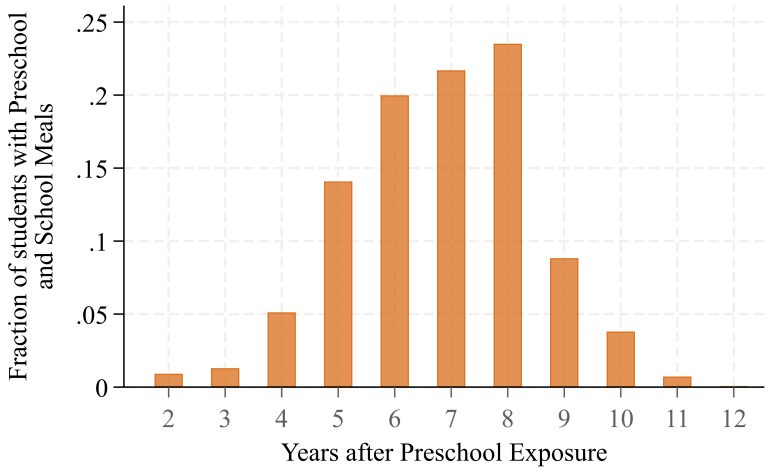
Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure C3. Effect of School Feeding Exposure on Academic Performance, by Exposure to Preschool



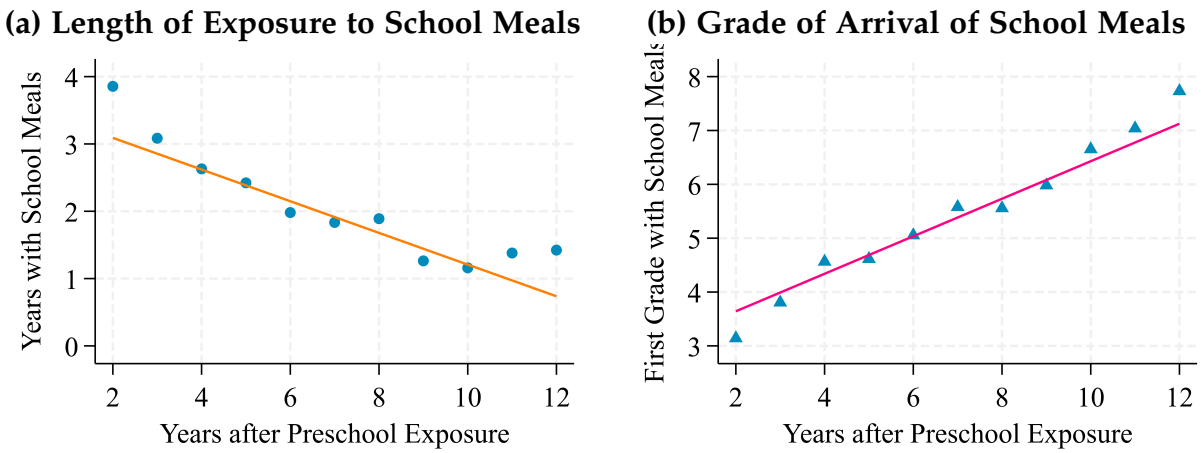
Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure C4. Distribution of years between preschool and school meals exposure



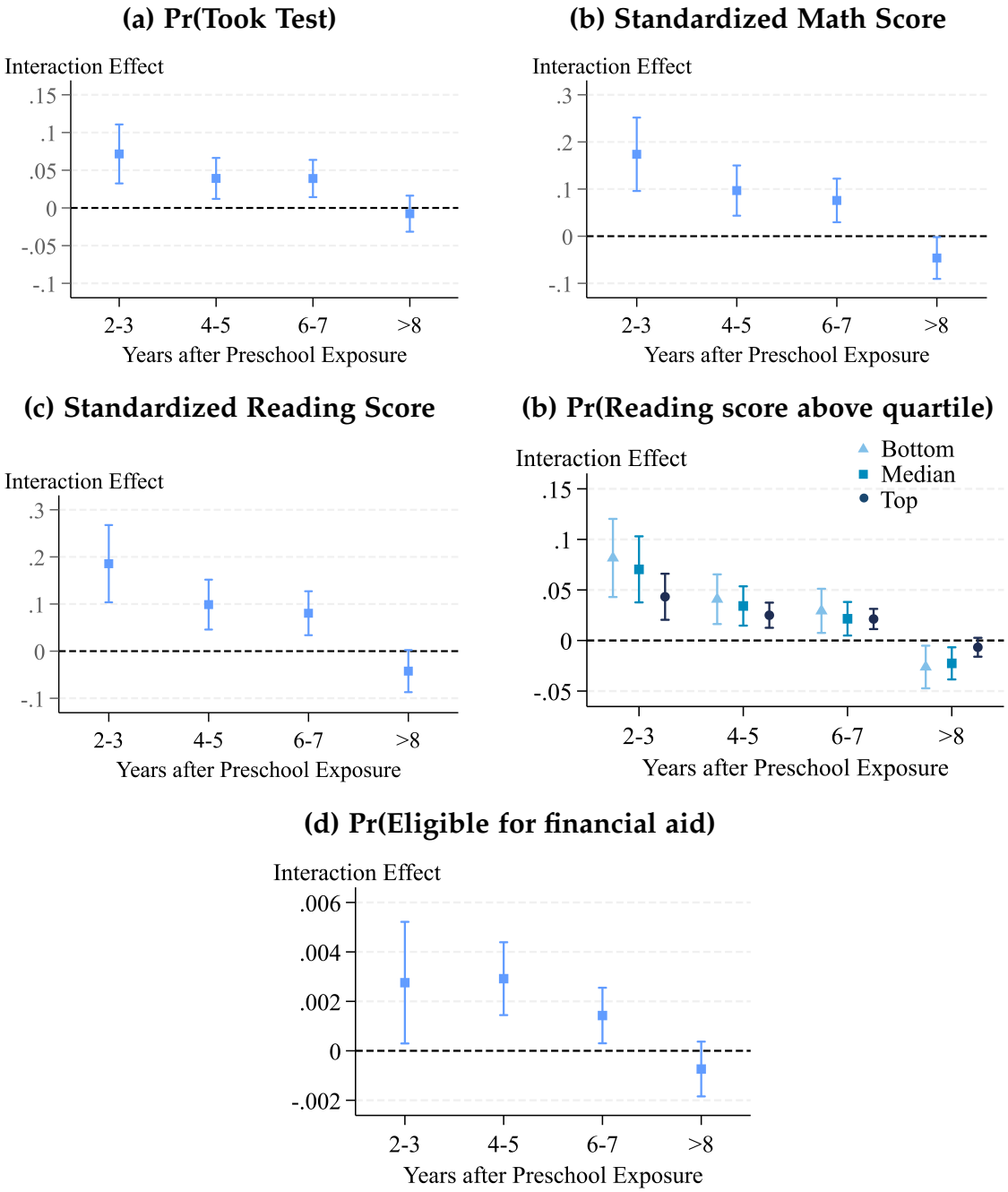
Notes: The figure shows the distribution of years between exposure to preschool and exposure to school meals.

Figure C5. Timing of School Feeding Arrival since Preschool Exposure



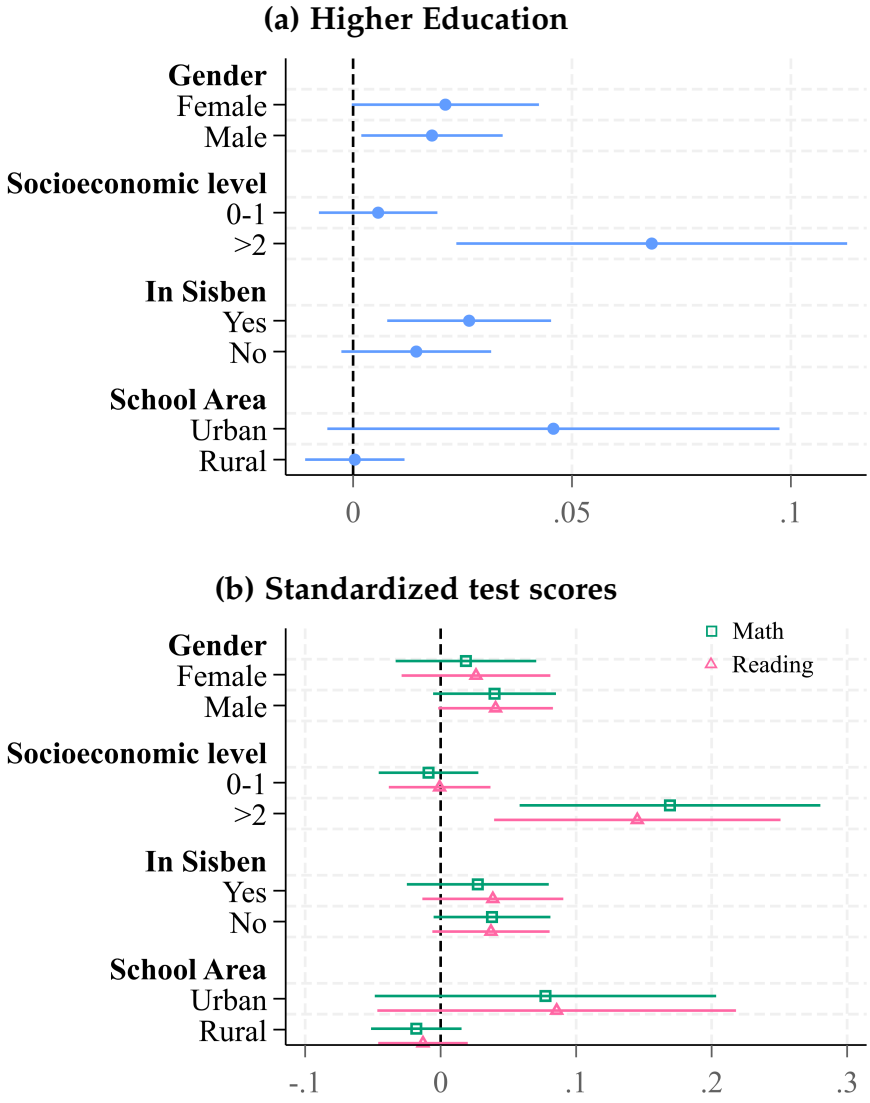
Notes: Panel (a) of the figure presents a scatter plot of years with school meals and years after preschool exposure, Panel (b) presents a scatter plot of grade when the student first received school meals and years after preschool exposure.

Figure C6. Arrival



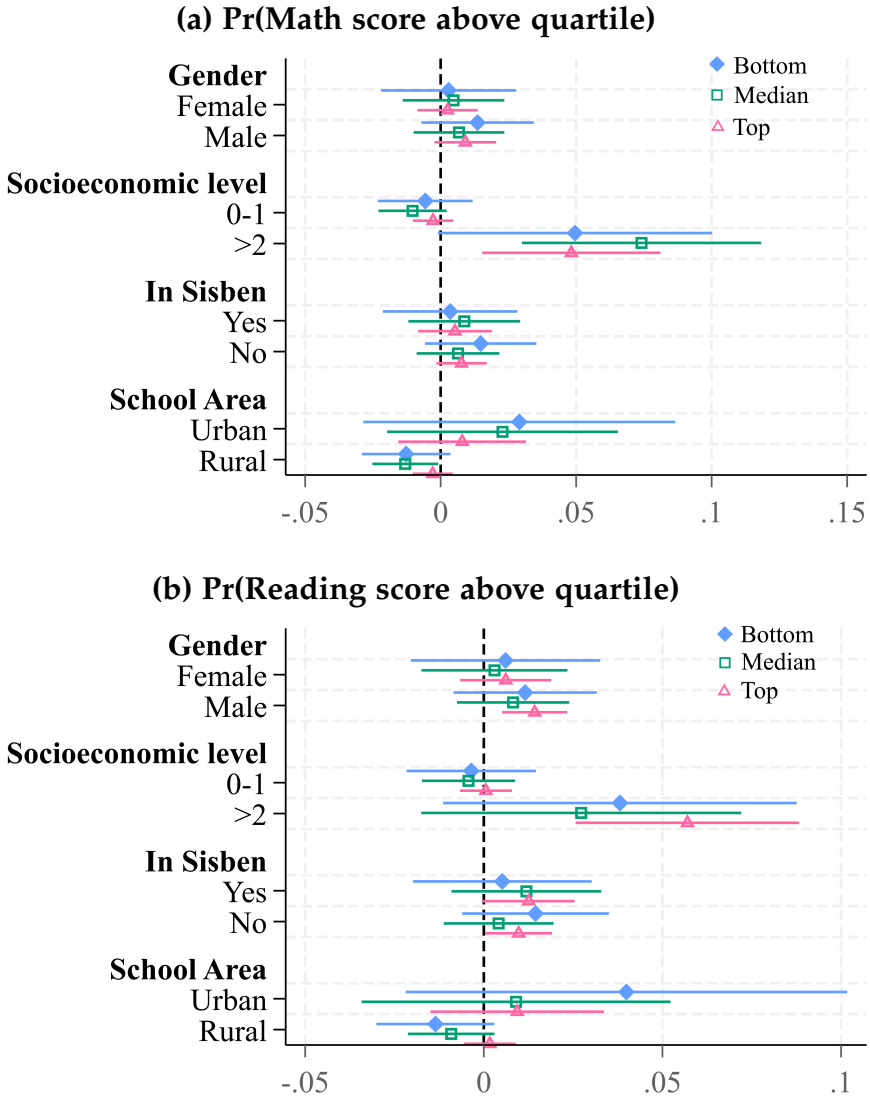
Notes: The figures present event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure C7. Interaction Effect of Preschool and School Feeding on Academic Trajectories, by student attributes



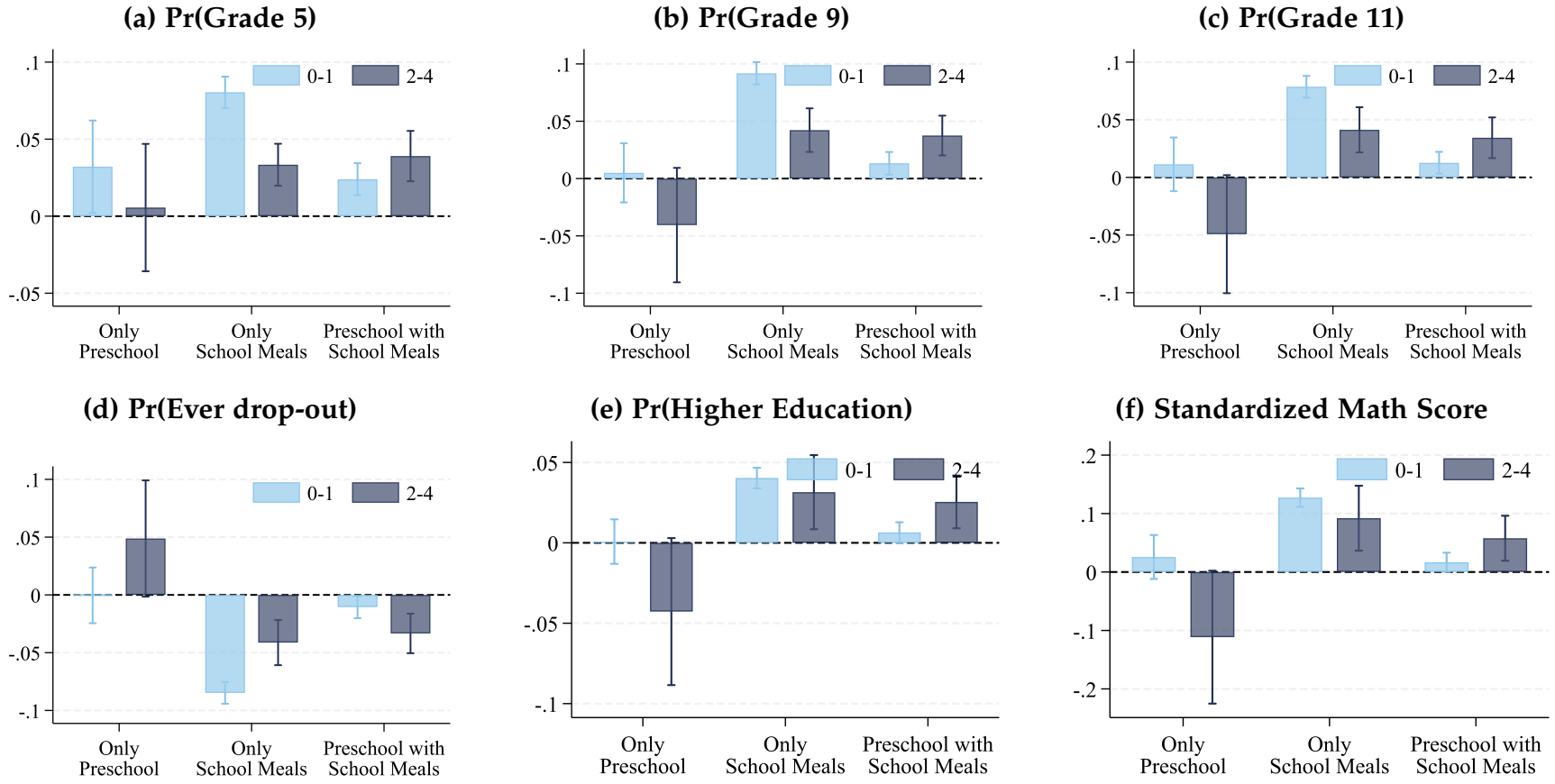
Note: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure C8. Interaction Effect of Preschool and School Feeding on Academic Trajectories, by student attributes



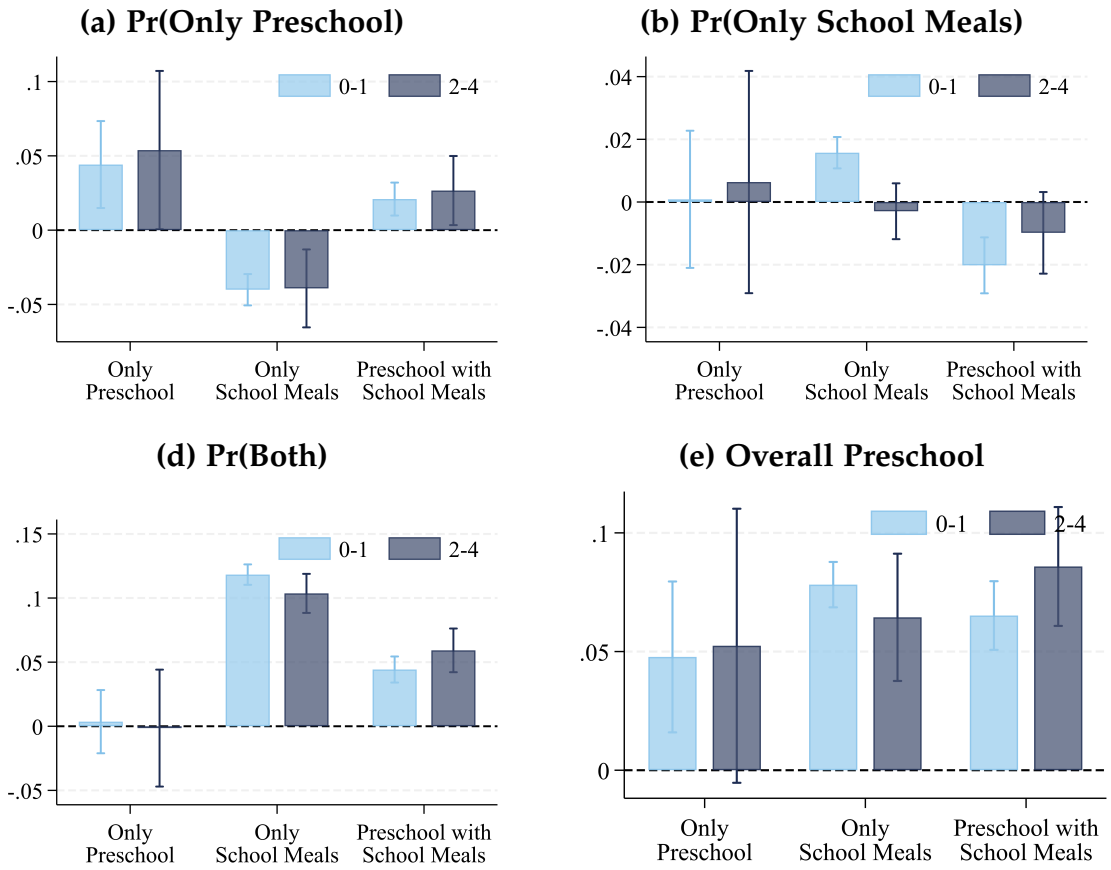
Notes: The figures present event-study estimates by subgroups. Confidence intervals at the 95% level, computed with clustered standard errors at the school level.

Figure C9. Pooled Effect of Preschool Exposure and School Feeding Exposure on Academic Outcomes, by socioeconomic level



Notes: The figures present pooled event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level. Light blue bars correspond to students in socioeconomic strata 0–1 (low-income), and dark blue bars to students in strata 2–4.

Figure C10. Pooled Effect of Preschool Exposure and School Feeding Exposure on Preschool Enrollment and School Meals Receipt, by socioeconomic level



Notes: The figures present pooled event-study estimates for relevant outcomes. Confidence intervals at the 95% level, computed with clustered standard errors at the school level. Light blue bars correspond to students in socioeconomic strata 0–1 (low-income), and dark blue bars to students in strata 2–4.

D Social Welfare

Preschool Expansion. We start with the the welfare gains from the additional policy of expanding preschool access in areas where it already existed. This expansion led to an increase in preschool enrollment of 7.1 pp We assume that the cost of providing one year of public preschool is $7,100,000$COP per child, fully subsidized to parents by the government. We estimate that the impact on the government budget of the expansion of preschools is $0.071 \times 7,100,000 = 504,100$COP. Although preschool increases the probability of reaching fifth grade, we do not detect downstream impacts on later attainment (grade 11 and higher education), or dropout. In our baseline welfare estimate, we therefore set willingness-to-pay (WTP) to zero and assume no associated fiscal externalities, so the net cost for the government equals the incremental program cost, implying an MVPF of zero.

We also compute a less conservative WTPF where we assume that beneficiaries value the subsidized preschool slot itself. Using the estimated first stage increase in preschool enrollment, and assuming that a recipient values a slot at its fiscal cost, the implied WTP is $504,100$COP. Combining the WTP and costs estimates, and ignoring the complementarities with subsequent investments, we would obtain MVPFs of 0-1.

School Feeding Expansion. We now focus on translating the effects of the school feeding program expansion into social welfare. Using the estimates of effects of this expansion on the probability of completing high school and on the probability of enrolling in higher education, we compute the additional earnings for these two types of students compared to control students. We assume that the earnings of students who were not exposed to both policies increase yearly based on estimates from a Mincer equation using household surveys. We also assume that the earnings trajectories for unexposed and exposed students differ by a constant percentage throughout the lifecycle (HS grad over HS dropout: 21,600,000/15,600,000; Higher Education graduate over HS dropout: 28,440,000/15,600,000). We compute the impact on earnings for high school graduates as the difference between their wage and the control times the effect on the probability of completing high school over 30 years. We follow the same logic for the earning gains of higher education enrollees. The present value of lifetime earnings combined for these two groups is $26,476,288$COP and we apply a 19% tax rate to compute net earnings. We also include the present value of school meals for marginal students, assuming that parents value them at their cash value, $3,449,051 * 0.136 = 469,071$COP. This results in $WTP^{\text{SFP}} = (1 - 0.19) * 26,476,288 + 469,071 = 21,914,864$COP.[30]

To quantify the impact of this expansion on the government budget, we compute: (i) the direct costs associated with meal provision, (ii) indirect costs associated to additional years in the education system because of reduced dropout and increased higher education enrollment, and (iii) the fiscal externality.

First, the cost per student-year of school meals is about $900,000$COP. We multiply the

[30]Because the cash value of school meals is small relative to net earnings, the WTP changes marginally if we do not include parents valuation of school meals per se.

unitary cost by the first stage estimate (i.e., the effect on the probability of receiving school meals) and by the average number of years students received school meals in our sample. We use a discount rate of 3%, and obtain a present value of direct costs of school meals of $3,449,051$COP. For indirect costs, we compute the additional spending that the government incurs on because of higher educational attainment in two steps. Using the estimate of the effect of the school feeding expansion on the probability of dropping out, we obtain a cost of $607,652$COP for every additional year in the public education system. In addition, we include the additional cost of higher education in the public system. We combine the average cost of a slot in a public university ($12,100,000$COP per year) with the average cost of enrollment in a two-year technical degree ($7,000,000$COP per year). We average these costs assuming 60% of public higher education enrollees go to a bachelor degree and the remaining go to a technical or technological program. We further multiply these costs by 50% (corresponding to the share of higher education enrollees that choose public institutions), and assume zero cost to the government for those who go to private universities. We multiply these costs by the effect on the probability of enrolling in higher education. The present value of these indirect costs is $6,699,764$COP. Last, we estimate that the higher earnings resulting from higher educational attainment increase tax revenue using a 19% tax rate. The present value of the fiscal externality is $0.19 * 26,476,288 = 5,030,495$COP. The net government cost sums up to $Cost^{\text{SFP}} = 3,449,051 + 6,699,764 - 5,030,495 = 5,118,320$COP.

Combining our WTP^{SFP} and $Cost^{\text{SFP}}$ estimates, we obtain an $MVPF^{\text{SFP}}$ of 4.28.

Interactions. The expansion of preschool and school feeding resulted in some students exposed to both policies, and we estimate significant interaction effects on educational attainment. These complementarities do not require additional upfront program spending beyond the direct spending on preschool slots and school meals. However, by increasing educational attainment, complementarities can raise indirect costs (e.g., additional years of schooling) and increase future earnings. Higher earnings increase beneficiaries' WTP and generate fiscal externalities that reduce the net cost to the government.

To quantify the WTP associated with the interaction effects, we follow the same logic as in the school feeding expansion case. The present value of lifetime earnings is $6,914,768$COP; with a 19% tax rate net earnings are $(1 - 0.19) * 6,914,768 = 5,600,962$. Because there are no upfront program costs, we also assume no additional cash value of the programs for the parents, which results in a $WTP^{\cap} = 5,600,962$.

For indirect costs, we multiply the effect on the probability of dropping out by the cost of each educational year and obtain a cost of $127,800$COP for every additional year in the public education system. Additional costs associated with increased provision of higher education amount to $60,000$COP. The present value of these costs combined is $1,448,180$COP. To quantify the fiscal externality, we use a tax rate of 19% and obtain $0.19 * 6,914,768 = 1,313,806$. The net government cost results in $Cost^{\cap} = 1,448,180 - 1,313,806 = 134,375$COP.

Table D1. Parameters and Assumptions

Parameters	Value	Source
Preschool Costs (millions, COP)		
Preschool slot (one time cost)	7.10	Ministry of Education and Forero et al. (2025)
First stage	0.071	Estimate in Table A4, Column (1)
Cost per exposed student	0.5041	Preschool slot×First stage
School Feeding Program Costs (millions, COP)		
School meals per student-year	0.90	Ministry of Education - SIMAT and Unidad Administrativa Especial de Alimentación Escolar – Alimentos para Aprender (2025)
First stage	0.14	Estimate in Table 3, Column (1)
Avg. years with meals	2.83	Own calculations using estimating sample, conditional on receiving meals
Cost per exposed student	0.3465	School meals×First stage×Avg. years with meals
Additional education costs (millions, COP)		
Slot per student-year (K-12)	7.10	Ministry of Education and Forero et al. (2025)
Public university, student-year	12	Ministry of Education
Public SENA slot, student-year	7	Ministry of Education
Earnings extrapolation (annual wages in millions, COP)		
School Dropout, at 22	15.6	Household Survey (*Gran Encuesta Integrada de Hogares*), DANE 2025
High School Graduate	21.6	
Higher Education Graduate	28.44	
Additional parameters		
Higher ed. enrollment rate	0.5	Ministry of Education - SNIES
University enrollment	0.6	
SENA enrollment	0.4	
Mincer equation	$0.03\text{age} - 0.0005\text{age}^2$	Household Survey (*Gran Encuesta Integrada de Hogares*), DANE 2025; López Castaño and Núñez Méndez (2007); Galvis-Aponte (2012)
Tax rate	0.19	
Discount rate	0.03	

Notes: Text and values in gray correspond to own calculations from regression estimates or descriptive statistics in the estimating sample.

Table D2. Timing between investments - lower cost *versus* higher benefits trade-off

	Years after preschool exposure				
	2–3	4–5	6–7	> 8	Overall
Interaction Effects					
Prob Grade 11	0.074	0.039	0.039	0	0.020
Prob Higher Education	0.054	0.047	0.039	0	0.012
Impact on dropout	-0.054	-0.042	-0.044	0	-0.018
Lifetime earnings	55,240,456	47,780,803	44,964,414	26,476,288	33,391,056
Willingness to Pay	45,925,480	39,761,909	37,441,923	22,425,131	28,019,926
Average number of years with PAE	4.1	3.4	3.1	2.9	2.8
Direct Costs	5,479,183	4,587,615	4,302,984	3,998,493	4,159,210
Indirect Costs	11,338,629	10,391,503	10,402,794	6,699,764	8,147,944
Fiscal Externality	10,495,687	9,078,533	8,543,239	5,030,495	6,344,301
Net Government Cost	6,322,125	5,900,765	6,162,539	5,667,762	5,756,795
MVPF	7.26	6.74	6.08	3.96	4.87

Notes: This table reports window-specific estimates of the marginal value of public funds (MVPF) of the preschool plus school feeding bundle as a function of the lag between investments. Columns correspond to the number of years between preschool and school feeding exposure; "Overall" reports pooled estimates across windows. "Interaction Effects" are the estimated DiDiD interaction terms for the probability of reaching grade 11th, enrolling in higher education, and dropping out. MVPFs estimates are computed following Section 5 and Appendix D.

Additional References

Castaño, Hugo López and Jairo Núñez Méndez, *Pobreza y desigualdad en Colombia: diagnóstico y estrategias*, Bogotá, Colombia: Departamento Nacional de Planeación (DNP) – Misión para el Diseño de una Estrategia para la Reducción de la Pobreza y la Desigualdad (MERPD), 2007. Documento MERPD/DNP (libro/informe).

Forero, David Fernando, Stephany Arenas, Nicole Puente, and Laura Rivas, "Consultoría para analizar el financiamiento de la educación en Colombia en sus diferentes niveles: Entregable 2, Capítulo 3. Sistema General de Participaciones," Technical Report, Fedesarrollo and Fundación Empresarios por la Educación (ExE), Bogotá, Colombia February 2025. Repositorio institucional Fedesarrollo (Handle: 11445/4750).

Galvis-Aponte, Luis Armando, "Informalidad laboral en las áreas urbanas de Colombia," Documentos de Trabajo sobre Economía Regional y Urbana 164, Banco de la República de Colombia, Centro de Estudios Económicos Regionales (CEER) 2012. DT SER No. 164.

Unidad Administrativa Especial de Alimentación Escolar – Alimentos para Aprender, "Informe de gestión – vigencia 2024," January 2025.